

Project Data Page:

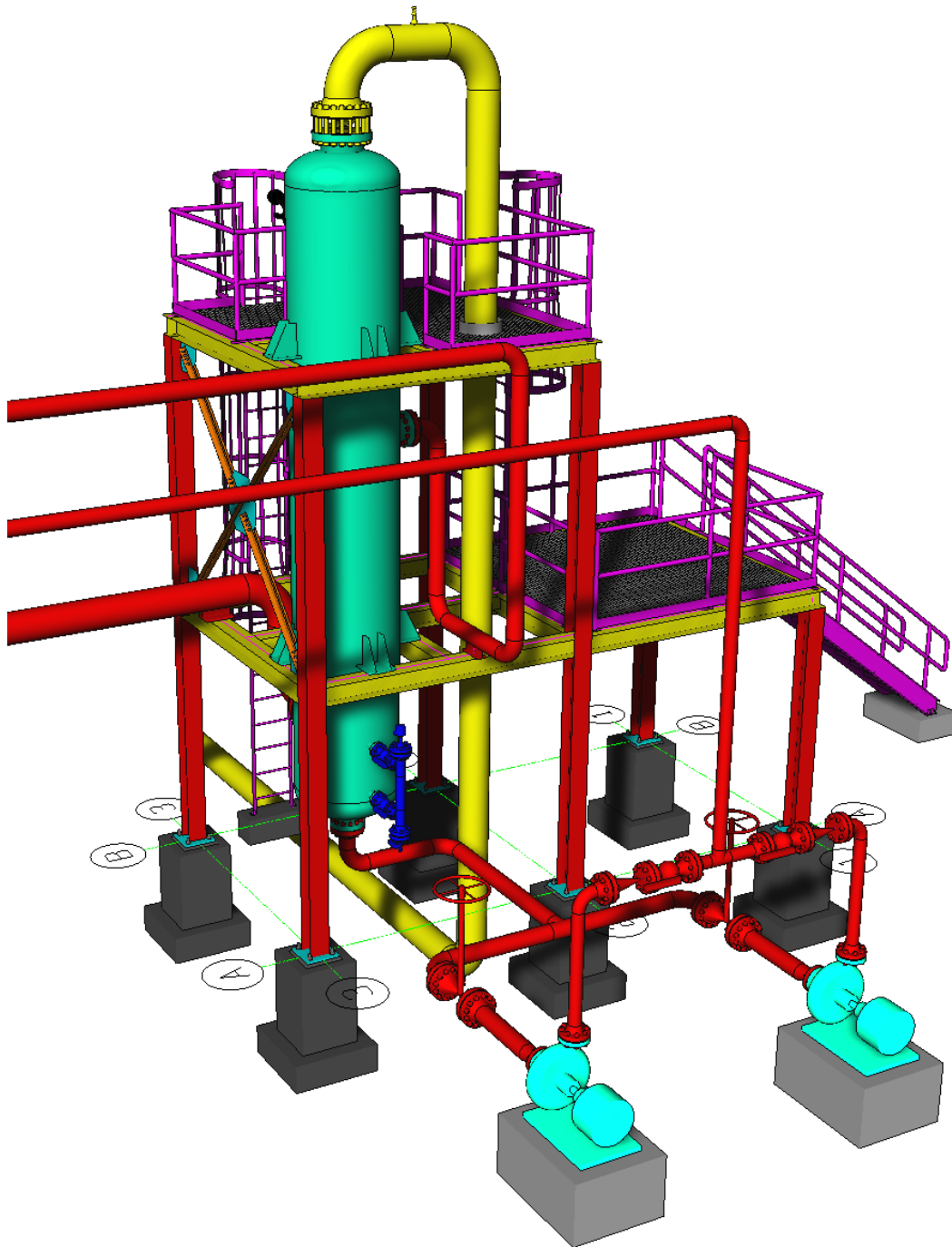




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147-D
Ammonia LetDown Drum
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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\147-D\\Ammonia Letdown Drum7%.pvdb

Date of Analysis : Sep 16,2026 11:57am

PV Elite 28 U1, March 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2150.000 mm. ID
Vessel Design Length, Tangent to Tangent 6475.00 mm.
Specified Datum Line Distance 0.00 mm.
Internal Design Temperature 65 °C
Internal Design Pressure 29.400 bars
External Design Temperature 0 °C
Maximum Allowable Working Pressure 30 bars
External Max. Allowable Working Pressure 7 bars
Shop Test Pressure 44.633 bars
Field Test Pressure 38.627 bars
Required Minimum Design Metal Temperature -33.0 °C
Warmest Computed Minimum Design Metal Temperature -33.0 °C
Wind Design Code ASCE/SEI 7-16
Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+IT
Head	SA-516 70	...	...	K02700	+IT
Nozzle	SA-266 2	...	...	K03506	+IT
Nozzle	SA-105	...	...	K03504	
Nozzle Flg	SA-105	...	...	K03504	+IT
Hrz Bolting		...	...		
Saddles	SA-516 70	...	...	K02700	+IT

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Left Dish	29.533	0.15	31.000	3.0000	No	No
S1	29.533	0.15	31.000	3.0000	N/A	No
S2	29.533	0.15	31.000	3.0000	N/A	No
S2	29.533	0.15	31.000	3.0000	N/A	No
Right Dish	29.533	0.15	31.000	3.0000	No	No

Liquid Level: 2150.00 mm. Dens.: 0.001 kg./cm3 Sp. Gr.: 0.631

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Ellipse	50.0	50.0	32.0	27.0	26.0	5.2	1.00	1.00
Cylinder	3050.0	3000.0	28.0	28.0	26.4	8.2	1.00	1.00
Cylinder	4550.0	1500.0	28.0	28.0	26.4	8.2	1.00	1.00

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Cylinder	6425.0	1875.0	28.0	28.0	26.4	8.2	1.00	1.00
Ellipse	6475.0	50.0	32.0	27.0	26.0	5.2	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Lift Lug	WRC-107/537	0.451	n/a	Passed
Lift Lug	WRC-107/537	0.524	n/a	Passed
N1_10in	WRC-297	0.928	Shell	Passed
N2-16in	WRC-297	0.928	Shell	Passed
N3-10in	WRC-297	0.928	Shell	Passed

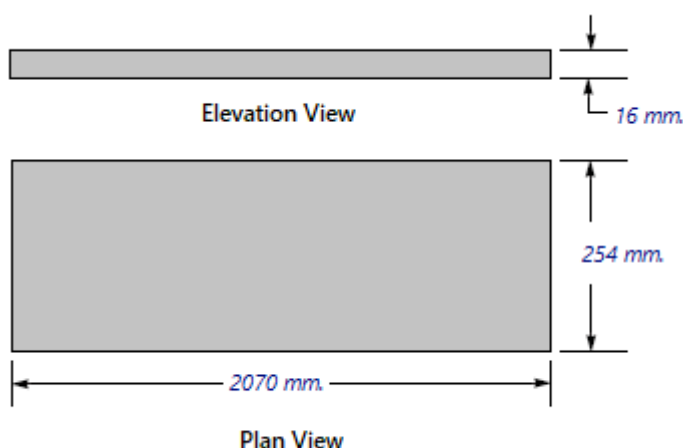
Foundation/Support Summary

Saddle Parameters:

Saddle Width	230.000 mm.
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	1750.000 mm.
Wear Pad Width	510.000 mm.
Wear Pad Thickness	28.000 mm.
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	1295.000 mm.

Baseplate Length	2070.000 mm.
Baseplate Thickness	16.000 mm.
Baseplate Width	254.000 mm.
Number of Ribs (including outside ribs)	5
Rib Thickness	10.000 mm.
Web Thickness	10.000 mm.
Height of Center Web	301.000 mm.
Number of Bolts in Baseplate	8

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	25420.75	Kgf
Maximum Transverse Saddle Shear Load	1512.08	Kgf
Maximum Longitudinal Saddle Shear Load	3024.16	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	26494.76	Kgf
Maximum Transverse Saddle Shear Load	341.88	Kgf
Maximum Longitudinal Saddle Shear Load	366.81	Kgf

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Vessel Design Summary:

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Weights:

Fabricated - Bare w/o removable internals	18705.3 kg.
Shop Test - Fabricated + water (full)	45163.5 kg.
Shipping - Fab. + rem. intls.+ shipping app.	18705.3 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	23369.9 kg.
Empty - Fab. + intls. + details + wghts.	23369.9 kg.
Operating - Empty + operating liquid (no ca)	39838.0 kg.
Field Test - Empty weight + test medium (full)	49468.0 kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
K1_3in	31	OK	...	OK	8.73	OK	Passed
K7_2in	31	...	...	OK	7.80	OK	No Calc[*]
N1_10in	31	OK	...	OK	11.10	OK	Passed
K3-3in	31	OK	...	OK	8.73	OK	Passed
K8-2in	31	...	...	OK	7.80	OK	No Calc[*]
K5-2in	31	OK	...	OK	7.80	OK	Passed
K5-2in	31	...	...	OK	7.80	OK	No Calc[*]
K2-3in	31	OK	...	OK	8.73	OK	Passed
K4-3in	31	OK	...	OK	8.73	OK	Passed
K6-2in	31	...	...	OK	7.80	OK	No Calc[*]
D1-2in	31	...	...	OK	7.80	OK	No Calc[*]
N2-16in	30	OK	...	OK	11.33	OK	Passed
N3-10in	31	OK	...	OK	11.10	OK	Passed
M2_24in	30	OK	...	...		OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 30 Nozzle : N2-16in
Minimum MAWP Shells/Flanges : 30 Element : Right Dish
Minimum MAPnc Shells/Flanges : 35 Element : Right Dish

Computed Vessel M.A.W.P. : 30 bars

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm.	Layout Angle deg	Dia. Limit mm.
20	K1_3in	1200.000	72.863	167.348
20	K7_2in	1500.000	0.000	134.400
20	N1_10in	550.000	0.000	472.000
20	K3-3in	1750.000	72.863	167.348
20	K8-2in	2000.000	0.000	134.400
20	K5-2in	2300.000	48.213	134.400
20	K2-3in	1200.000	105.243	167.348
20	K4-3in	1750.000	105.243	167.348
20	K6-2in	2300.000	105.243	134.400
20	D1-2in	2050.000	180.000	134.400
40	N2-16in	4850.000	0.000	776.800
40	N3-10in	4650.000	180.000	472.000
50	M2_24in	0.000	0.000	1183.200

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule: Step: 35 11:57am Sep 16,2026



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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
K7_2in	2.000 in	Actual	LWN	3.323	16.800	...	...	506.48	300
K8-2in	2.000 in	Actual	LWN	3.323	16.800	...	...	506.48	300
K5-2in	2.000 in	Actual	LWN	3.323	16.800	...	...	557.39	300
K6-2in	2.000 in	Actual	LWN	3.323	16.800	...	...	269.46	300
D1-2in	2.000 in	Actual	LWN	3.323	16.800	...	...	256.48	300
K1_3in	3.000 in	LWN	LWN	4.620	20.574	...	...	525.04	300
K3-3in	3.000 in	Actual	LWN	4.620	20.574	...	...	525.04	300
K2-3in	3.000 in	Actual	LWN	4.620	20.574	...	...	272.14	300
K4-3in	3.000 in	Actual	LWN	4.620	20.574	...	...	272.14	300
N1_10in	10.000 in	Actual	WNF	10.000	12.000	...	...	484.73	300
N3-10in	10.000 in	Actual	WNF	10.000	12.000	...	...	268.18	300
N2-16in	16.000 in	Actual	WNF	16.000	12.000	...	...	496.58	300
M2_24in	24.000 in	Actual	WNF	24.000	12.000	...	...	300.89	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K7_2in	SA-105	24.000	10.000	...	...	...
K8-2in	SA-105	24.000	10.000	...	...	...
K5-2in	SA-105	24.000	10.000	...	...	...
K6-2in	SA-105	24.000	10.000	...	...	...
D1-2in	SA-105	24.000	10.000	...	...	...
K1_3in	SA-266 2	24.000	10.000	...	...	...
K3-3in	SA-266 2	24.000	10.000	...	...	...
K2-3in	SA-266 2	24.000	10.000	...	...	...
K4-3in	SA-266 2	24.000	10.000	...	...	...
N1_10in	SA-266 2	24.000	10.000	...	...	...
N3-10in	SA-266 2	24.000	10.000	...	...	...
N2-16in	SA-266 2	24.000	10.000	...	...	...
M2_24in	SA-266 2	23.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
K7_2in	1500.000	0.0	477.65	0.00	S1
K8-2in	2000.000	0.0	477.65	0.00	S1
K5-2in	2300.000	48.2	477.65	0.00	S1
K6-2in	2300.000	105.2	227.65	0.00	S1
D1-2in	2050.000	180.0	227.65	0.00	S1
K1_3in	1200.000	72.9	471.55	0.00	S1
K3-3in	1750.000	72.9	471.55	0.00	S1
K2-3in	1200.000	105.2	221.55	0.00	S1



K4-3in	1750.000	105.2	221.55	0.00	S1
N1_10in	550.000	0.0	449.20	0.00	S1
N3-10in	4650.000	180.0	232.65	0.00	S2
N2-16in	4850.000	0.0	449.20	0.00	S2
M2_24in	...	0.0	250.00	0.00	Right Dish

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Bill of Material:

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 32.0mm. THK X 2150.0mm. ID X 50.0mm.	SA-516 70
1	CYLINDER: 28.0mm. THK X 2150.0mm. ID X 3000.0mm.	SA-516 70
1	CYLINDER: 28.0mm. THK X 2150.0mm. ID X 1500.0mm.	SA-516 70
1	CYLINDER: 28.0mm. THK X 2150.0mm. ID X 1875.0mm.	SA-516 70
2	INSULATION: 587mm X 80mm THK	
2	SADDLE: 230.0mm X 120 DEG	SA-516 70
1	PLATFORM: 1000mm X 2206mm WIDE	
1	LADDER: 3700.0mm	...
1	INSULATION: 3000mm X 80mm THK	
4	LIFTING LUGS:	SA-516 70
1	INSULATION: 1500mm X 80mm THK	
1	INSULATION: 1875mm X 80mm THK	
1	CLASS 300 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	NAMEPLATE	...

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Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
Left Dish	[10]	!	-33	-35		0.970	27.000	1.00	Yes
Left Dish	[7]	!	-33	-43		0.815	32.000	1.00	Yes
S1	[8]	!	-33	-36		0.945	28.000	1.00	Yes
S2	[8]	!	-33	-36		0.945	28.000	1.00	Yes
S2	[8]	!	-33	-36		0.945	28.000	1.00	Yes
Right Dish	[10]	!	-33	-35		0.970	27.000	1.00	Yes
Right Dish	[7]	!	-33	-43		0.815	32.000	1.00	Yes
K13in	[1]	!	-33	-37		0.932	28.000	1.00	Yes
Nozzle Flg	[5]	!	-33	-33					
K72in	[1]	A	7	-104		0.044	16.800	1.00	Yes
Nozzle Flg	[5]	A	-18	-42					
N110in	[1]	!	-33	-37		0.931	28.000	1.00	Yes
Nozzle Flg	[5]	!	-18	-42					
K3-3in	[1]	!	-33	-37		0.932	28.000	1.00	Yes
Nozzle Flg	[5]	!	-33	-33					
K8-2in	[1]	A	7	-104		0.044	16.800	1.00	Yes
Nozzle Flg	[5]	A	-18	-42					
K5-2in	[1]	A	7	-104		0.044	16.800	1.00	Yes
Nozzle Flg	[5]	A	-18	-42					
K2-3in	[1]	!	-33	-37		0.933	28.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-42					
K4-3in	[1]	!	-33	-37		0.933	28.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-42					
K6-2in	[1]	A	7	-104		0.045	16.800	1.00	Yes
Nozzle Flg	[5]	A	-18	-42					
D1-2in	[1]	A	7	-104		0.045	16.800	1.00	Yes
Nozzle Flg	[5]	A	-18	-42					
N2-16in	[1]	!	-33	-37		0.931	28.000	1.00	Yes
Nozzle Flg	[5]	!	-18	-42					
N3-10in	[1]	!	-33	-37		0.935	28.000	1.00	Yes
Nozzle Flg	[5]	!	-18	-42					
M224in	[1]	!	-33	-41		0.863	27.000	1.00	Yes
Nozzle Flg	[5]	!	-18	-42					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	Yes

Warmest MDMT: 7 -33

Required Minimum Design Metal Temperature -33.0 °C

Warmest Computed Minimum Design Metal Temperature -33.0 °C

Notes:

[!] - This was an impact tested material.

[1] - Governing Nozzle Weld.

[4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).

[5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).

[6] - MDMT Calculations at the Shell/Head Joint.

[7] - MDMT Calculations for the Straight Flange.

[8] - Cylinder/Cone/Flange Junction MDMT.

[9] - Calculations in the Spherical Portion of the Head.

[10] - Calculations in the Knuckle Portion of the Head.

[11] - Calculated (Body Flange) Flange MDMT.

[12] - Calculated Flat Head MDMT per UCS-66.3

[13] - Tubesheet MDMT, shell side, if applicable

[14] - Tubesheet MDMT, tube side, if applicable

[15] - Nozzle Material

[16] - Shell or Head Material

[17] - Impact Testing required

[18] - Impact Testing not required, see UCS-66(b)(3)

[20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations

[21] - Body Flange Bolting Material

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[22] - Nozzle Flange Bolting Material

[23] - Stiffening Ring to Shell Weld

[24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Input Echo:

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Units used in this Analysis (barsUniCode):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm2
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm2
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm3
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm2
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m2
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm3
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

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Design Internal Pressure (for Hydrotest)	29.4	bars
Design Internal Temperature	65.0	°C
Projection of Nozzle from Vessel Top	500	mm.
Projection of Nozzle from Vessel Bottom	500	mm.
Minimum Design Metal Temperature	-33.0	°C
Type of Construction	Welded	
Special Service	Lethal	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS



Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW
Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	0 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.
Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	1.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000
Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No



Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/5
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	Left Dish
Distance "FROM" to "TO"	50 mm.
Inside Diameter	2150 mm.
Element Thickness	27 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	32 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	29.4 bars
Design Temperature Internal Pressure	65 °C
Design External Pressure	0.15 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	246 N./mm ²
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	10
Detail Type	Liquid
Detail ID	Liquid 10
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	2150 mm.
Liquid Density	0.00063 kg./cm ³

Element From Node	10
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	-537.5 mm.
Height/Length of Insulation	587.5 mm.
Thickness of Insulation	80 mm.
Density	0.00032 kg./cm ³

Element#	2/5
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	S1
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	2150 mm.
Element Thickness	28 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	28 mm.

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External Corrosion Allowance	0 mm.
Design Internal Pressure	29.4 bars
Design Temperature Internal Pressure	65 °C
Design External Pressure	0.15 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	1245 mm.
Width of Saddle	230 mm.
Height of Saddle at Bottom	1750 mm.
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm.
Width of Wear Plate	510 mm.
Thickness of Wear Plate	28 mm.
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm.
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm.

Element From Node	20
Detail Type	Platform
Detail ID	PLAT:[1 OF 1]
Dist. from "FROM" Node / Offset dist	0 mm.
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	0.0
Platform Wind Area	64750 cm2
Platform Weight	2566.4 Kg
Platform Railing Weight	0 Kg/mm.
Platform Grating Weight	0.017 Kgs/cm2
Platform Width	2206 mm.
Platform Height	1000 mm.
Platform Clearance or End Offset	85 mm.
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm.
Ladder End Elevation	3700 mm.
Unit Weight of Ladder	0.037 Kg/mm.
Platform Length (top head platform)	6475 mm.

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	2150 mm.
Liquid Density	0.00063 kg./cm3

Element From Node	20
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	3000 mm.
Thickness of Insulation	80 mm.
Density	0.00032 kg./cm3

Element From Node	20
Detail Type	Nozzle
Detail ID	K13in

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Dist. from "FROM" Node / Offset dist	1150	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	72.863403	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	34.981	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2 +IT	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	K72in	
Dist. from "FROM" Node / Offset dist	1450	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	18.522	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	N110in	
Dist. from "FROM" Node / Offset dist	500	mm.
Nozzle Diameter	10	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	117.43	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2 +IT	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	K3-3in	
Dist. from "FROM" Node / Offset dist	1700	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	72.863403	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	34.981	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2 +IT	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	K8-2in	
Dist. from "FROM" Node / Offset dist	1950	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	18.522	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	K5-2in	
Dist. from "FROM" Node / Offset dist	2250	mm.
Nozzle Diameter	2	in.

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Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	48.212898	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	18.522	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	K2-3in	
Dist. from "FROM" Node / Offset dist	1150	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	105.243	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	22.861	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2 +IT	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	K4-3in	
Dist. from "FROM" Node / Offset dist	1700	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	105.243	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	22.861	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2 +IT	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	K6-2in	
Dist. from "FROM" Node / Offset dist	2250	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	105.243	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	11.609	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	D1-2in	
Dist. from "FROM" Node / Offset dist	2000	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	11.609	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	20	
Detail Type	Weight	
Detail ID	Fp wt	
Dist. from "FROM" Node / Offset dist	1245	mm.
Miscellaneous Weight	180	Kgf
Offset from Element Centerline	1500	mm.

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Element#	3/5
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	S2
Distance "FROM" to "TO"	1500 mm.
Inside Diameter	2150 mm.
Element Thickness	28 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	28 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	29.4 bars
Design Temperature Internal Pressure	65 °C
Design External Pressure	0.15 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	30
Detail Type	Liquid
Detail ID	Liquid 30
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	2150 mm.
Liquid Density	0.00063 kg./cm3
Element From Node	30
Detail Type	Insulation
Detail ID	Ins: 30
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	1500 mm.
Thickness of Insulation	80 mm.
Density	0.00032 kg./cm3
Element From Node	30
Detail Type	Weight
Detail ID	Baffles
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	2250 Kg
Offset from Element Centerline	0 mm.
Element From Node	30
Detail Type	Weight
Detail ID	Clip Wt
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	150 Kg
Offset from Element Centerline	0 mm.

Element#	4/5
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	S2
Distance "FROM" to "TO"	1875 mm.
Inside Diameter	2150 mm.
Element Thickness	28 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	28 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	29.4 bars
Design Temperature Internal Pressure	65 °C

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Design External Pressure	0.15	bars
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	40	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	630	mm.
Width of Saddle	230	mm.
Height of Saddle at Bottom	1750	mm.
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm.
Width of Wear Plate	510	mm.
Thickness of Wear Plate	28	mm.
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.30000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm.
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm.
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid 40	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	2150	mm.
Liquid Density	0.00063	kg./cm3
Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 40	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	1875	mm.
Thickness of Insulation	80	mm.
Density	0.00032	kg./cm3
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N2-16in	
Dist. from "FROM" Node / Offset dist	300	mm.
Nozzle Diameter	16	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	288.3	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	+IT
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N3-10in	
Dist. from "FROM" Node / Offset dist	100	mm.
Nozzle Diameter	10	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	102.12	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	+IT

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Element From Node	40
Detail Type	Weight
Detail ID	Fp
Dist. from "FROM" Node / Offset dist	630 mm.
Miscellaneous Weight	180 Kgf
Offset from Element Centerline	1500 mm.

Element#	5/5
Element From Node	50
Element To Node	60
Element Type	Elliptical
Description	Right Dish
Distance "FROM" to "TO"	50 mm.
Inside Diameter	2150 mm.
Element Thickness	27 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	32 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	29.4 bars
Design Temperature Internal Pressure	65 °C
Design External Pressure	0.15 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Liquid
Detail ID	Liquid 50
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	2150 mm.
Liquid Density	0.00063 kg./cm ³

Element From Node	50
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	587.5 mm.
Thickness of Insulation	80 mm.
Density	0.00032 kg./cm ³

Element From Node	50
Detail Type	Nozzle
Detail ID	M224in
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2 +IT

Element From Node	50
Detail Type	Weight
Detail ID	M2David24"
Dist. from "FROM" Node / Offset dist	500 mm.
Miscellaneous Weight	50 Kgf
Offset from Element Centerline	300 mm.



XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
Left Dish		50	...	50	...
S1		3050	...	3000	...
S2		4550	...	1500	...
S2		6425	...	1875	...
Right Dish		6475	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
Left Dish		29.533	32	3	2150	138	1
S1		29.533	28	3	2150	138	2
S2		29.533	28	3	2150	138	3
S2		29.533	28	3	2150	138	4
Right Dish		29.533	32	3	2150	138	5

Element Required Thickness and MAWP :

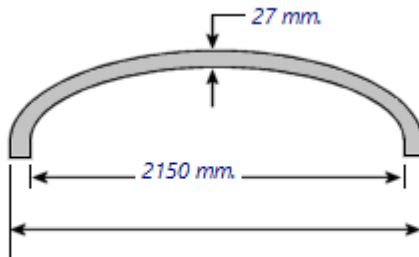
From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
Left Dish		29.4	30.6331	34.5709	27	26.036	1
S1		29.4	31.4288	35.3883	28	26.3721	2
S2		29.4	31.4288	35.3883	28	26.3721	3
S2		29.4	31.4288	35.3883	28	26.3721	4
Right Dish		29.4	30.6331	34.5709	27	26.036	5

Minimum 30 35

MAWP: 30 bars, limited by: Nozzle Reinforcement.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:**ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: Left Dish**

Elliptical Head Design Information:					
Design Pressure	P	29.533 bars	Int. Design Temperature		65 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	2150.000 mm.	Actual Stress		132.5 N./mm^2
Spec. Min. Thk t _e , or s _a		27.0000 mm.	Spec. Nominal Thk.		32.000 mm.
Surface Area		56612.5 cm^2	SG of Contents		0.631
Element Volume		1482720.9 cm3	Weight of Contents		935.0 Kg
Empty Weight		1498.3 Kg	Operating Weight		2433.3 Kg
			Comp. All. Ope. Stress		132.2 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 26.875 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 13.438 mm.

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

= $(P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P)$ Appendix 1-4(c)= $(29.533 \cdot 2156.0 \cdot 0.996) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 29.533)$ = $23.0360 + 3.0000 = 26.0360$ mm.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.133 bars

= $(2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t)$ per Appendix 1-4 (c)= $(2 \cdot 138.0 \cdot 1.0 \cdot 24.0) / (0.996 \cdot 2156.0 + 0.2 \cdot 24.0)$ = $31 - 0 = 31$ bars

Maximum Allowable Pressure, New and Cold [MAPNC]:

= $(2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t)$ per Appendix 1-4 (c)= $(2 \cdot 138.0 \cdot 1.0 \cdot 27.0) / (1.0 \cdot 2150.0 + 0.2 \cdot 27.0)$

= 35 bars

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Element: Left Dish

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Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (29.533 \cdot (0.996 \cdot 2156.0 + 0.2 \cdot 24.0)) / (2 \cdot 1.0 \cdot 24.0)$$

$$= 132.466 \text{ N/mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (29.533 \cdot 1078.0) / (138.0 \cdot 1.0 - 0.6 \cdot 29.533) + 3.0$$

$$= 26.372 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.133 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 29.0) / (1078.0 + 0.6 \cdot 29.0)$$

$$= 37 - 0 = 36 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (2156.0 / (2 \cdot 540.5)))^2 / 6$$

$$= 0.996305$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 6.291 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 26.875 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 13.438 mm.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Left Dish**Elliptical Head From 10 to 20 Ext. Chart: CS-2 at 0 °C**

Elastic Modulus from Chart: CS-2 at 0 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
24.000	2204.00	91.83	0.0015124	97.00

$$MAEP = B / (K_0 \cdot Do/t) = 97.0023 / (0.9 \cdot 91.8333) = 11.7358 \text{ bars}$$

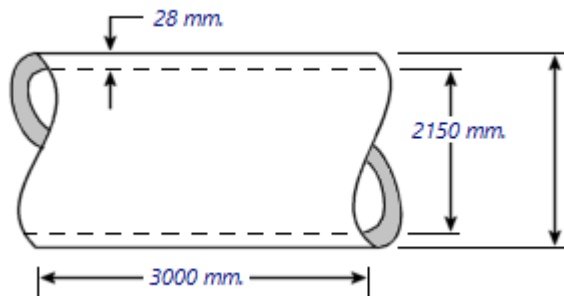
Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
2.173	2204.00	1014.13	0.0001370	13.69

$$MAEP = B / (K_0 \cdot Do/t) = 13.6923 / (0.9 \cdot 1014.1263) = 0.15 \text{ bars}$$



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: S1

Cylindrical Shell Design Information:					
Design Pressure	P	29.533 bars	Int. Design Temperature		65 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	2150.000 mm.	Actual Stress		129.1 N./mm^2
Spec. Min. Thk t _e , or s _a		28.0000 mm.	Spec. Nominal Thk.		28.000 mm.
Surface Area		207910.6 cm^2	SG of Contents		0.631
Element Volume		10893460.0 cm3	Weight of Contents		6869.7 Kg
Empty Weight		4454.7 Kg	Operating Weight		11324.4 Kg
			Comp. All. Ope. Stress		133.0 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (29.533 \cdot 1078.0) / (138.0 \cdot 1.0 - 0.6 \cdot 29.533)$$

$$= 23.3721 + 3.0000 = 26.3721 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.133 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 25.0) / (1078.0 + 0.6 \cdot 25.0)$$

$$= 32 - 0 = 31 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 28.0) / (1075.0 + 0.6 \cdot 28.0)$$

$$= 35 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (29.533 \cdot (1078.0 + 0.6 \cdot 25.0)) / (1.0 \cdot 25.0)$$

$$= 129.126 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.286 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: S1

Cylindrical Shell From 20 to 30 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	2206.00	6833.33	88.24	3.0976	0.0004883	48.82

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 48.8188) / (3 \cdot 88.24) = 7.3762 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.218	2206.00	6833.33	422.81	3.0976	0.0000476	4.76

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 4.7572) / (3 \cdot 422.8051) = 0.15 \text{ bars}$$

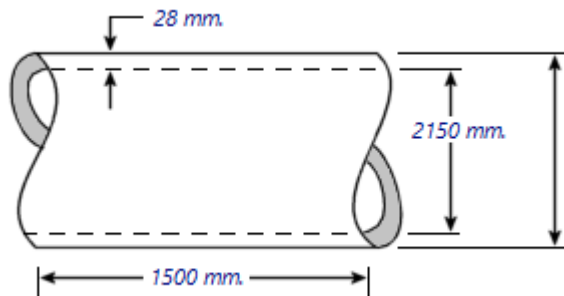
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	2206.00	183266.17	88.24	50.0000	0.0001439	14.39

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 14.3861) / (3 \cdot 88.24) = 2.1737 \text{ bars}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: S2

Cylindrical Shell Design Information:					
Design Pressure	P	29.533 bars	Int. Design Temperature		65 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		1500.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	2150.000 mm.	Actual Stress		129.1 N./mm^2
Spec. Min. Thk t _e , or s _a		28.0000 mm.	Spec. Nominal Thk.		28.000 mm.
Surface Area		103955.3 cm^2	SG of Contents		0.631
Element Volume		5446730.0 cm3	Weight of Contents		3434.8 Kg
Empty Weight		2227.3 Kg	Operating Weight		5662.2 Kg
			Comp. All. Ope. Stress		133.0 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (29.533 \cdot 1078.0) / (138.0 \cdot 1.0 - 0.6 \cdot 29.533)$$

$$= 23.3721 + 3.0000 = 26.3721 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.133 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 25.0) / (1078.0 + 0.6 \cdot 25.0)$$

$$= 32 - 0 = 31 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 28.0) / (1075.0 + 0.6 \cdot 28.0)$$

$$= 35 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (29.533 \cdot (1078.0 + 0.6 \cdot 25.0)) / (1.0 \cdot 25.0)$$

$$= 129.126 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.286 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: S2

Cylindrical Shell From 30 to 40 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	2206.00	6833.33	88.24	3.0976	0.0004883	48.82

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 48.8188) / (3 \cdot 88.24) = 7.3762 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.218	2206.00	6833.33	422.81	3.0976	0.0000476	4.76

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 4.7572) / (3 \cdot 422.8051) = 0.15 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	2206.00	183266.17	88.24	50.0000	0.0001439	14.39

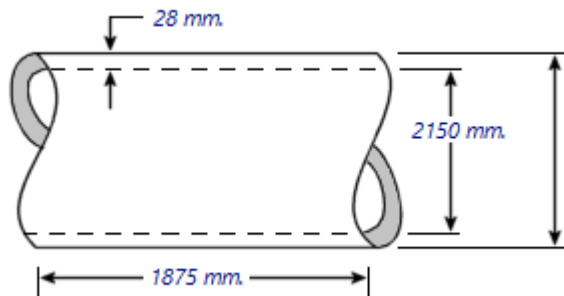
$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 14.3861) / (3 \cdot 88.24) = 2.1737 \text{ bars}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: S2

Cylindrical Shell Design Information:					
Design Pressure	P	29.533 bars	Int. Design Temperature		65 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		1875.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	2150.000 mm.	Actual Stress		129.1 N./mm^2
Spec. Min. Thk t _e , or s _a		28.0000 mm.	Spec. Nominal Thk.		28.000 mm.
Surface Area		129944.1 cm^2	SG of Contents		0.631
Element Volume		6808413.0 cm3	Weight of Contents		4293.6 Kg
Empty Weight		2784.2 Kg	Operating Weight		7077.7 Kg
			Comp. All. Ope. Stress		133.0 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (29.533 \cdot 1078.0) / (138.0 \cdot 1.0 - 0.6 \cdot 29.533)$$

$$= 23.3721 + 3.0000 = 26.3721 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.133 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 25.0) / (1078.0 + 0.6 \cdot 25.0)$$

$$= 32 - 0 = 31 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 28.0) / (1075.0 + 0.6 \cdot 28.0)$$

$$= 35 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (29.533 \cdot (1078.0 + 0.6 \cdot 25.0)) / (1.0 \cdot 25.0)$$

$$= 129.126 \text{ N./mm}^2$$

147-D

Ammonia LetDown Drum

QUEP

PV Elite 28 U1

Elite® v28

Licensee: SL User

FileName : 147-D_Ammonia Letdown Drum_7%

Element: S2

Step: 7 11:57am Sep 16,2026

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HEXAGON

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.286 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: S2

Cylindrical Shell From 40 to 50 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	2206.00	6833.33	88.24	3.0976	0.0004883	48.82

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 48.8188) / (3 \cdot 88.24) = 7.3762 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.218	2206.00	6833.33	422.81	3.0976	0.000476	4.76

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 4.7572) / (3 \cdot 422.8051) = 0.15 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	2206.00	183266.17	88.24	50.0000	0.0001439	14.39

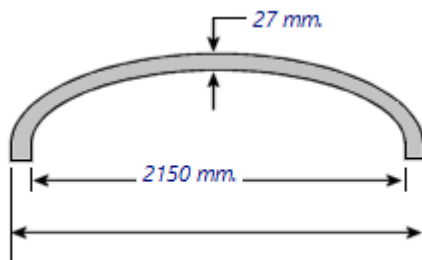
$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 14.3861) / (3 \cdot 88.24) = 2.1737 \text{ bars}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Right Dish

Elliptical Head Design Information:					
Design Pressure	P	29.533 bars	Int. Design Temperature	65 °C	
Material		SA-516 70	Ext. Design Temperature	0 °C	
Length		50.000 mm.	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	2150.000 mm.	Actual Stress	132.5 N./mm^2	
Spec. Min. Thk t _e , or s _a		27.0000 mm.	Spec. Nominal Thk.	32.000 mm.	
Surface Area		56612.5 cm^2	SG of Contents	0.631	
Element Volume		1482720.9 cm3	Weight of Contents	935.0 Kg	
Empty Weight		1498.3 Kg	Operating Weight	2433.3 Kg	
			Comp. All. Ope. Stress	133.0 N./mm^2	
			Aspect Ratio	2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 26.875 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 13.438 mm.

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (29.533 \cdot 2156.0 \cdot 0.996) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 29.533)$$

$$= 23.0360 + 3.0000 = 26.0360 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.133 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 24.0) / (0.996 \cdot 2156.0 + 0.2 \cdot 24.0)$$

$$= 31 - 0 = 31 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 27.0) / (1.0 \cdot 2150.0 + 0.2 \cdot 27.0)$$

$$= 35 \text{ bars}$$

147-D

Ammonia LetDown Drum

QUEP

PV Elite 28 U1

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FileName : 147-D Ammonia Letdown Drum_7%

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Element: Right Dish

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HEXAGON

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (29.533 \cdot (0.996 \cdot 2156.0 + 0.2 \cdot 24.0)) / (2 \cdot 1.0 \cdot 24.0)$$

$$= 132.466 \text{ N/mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (29.533 \cdot 1078.0) / (138.0 \cdot 1.0 - 0.6 \cdot 29.533) + 3.0$$

$$= 26.372 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.133 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 29.0) / (1078.0 + 0.6 \cdot 29.0)$$

$$= 37 - 0 = 36 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (2156.0 / (2 \cdot 540.5)))^2 / 6$$

$$= 0.996305$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 6.291 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 26.875 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 13.438 mm.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Right Dish**Elliptical Head From 50 to 60 Ext. Chart: CS-2 at 0 °C**

Elastic Modulus from Chart: CS-2 at 0 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
24.000	2204.00	91.83	0.0015124	97.00

$$MAEP = B / (K_0 \cdot Do/t) = 97.0023 / (0.9 \cdot 91.8333) = 11.7358 \text{ bars}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
2.173	2204.00	1014.13	0.0001370	13.69

$$MAEP = B / (K_0 \cdot Do/t) = 13.6923 / (0.9 \cdot 1014.1263) = 0.15 \text{ bars}$$



Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	38.627 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	38.220 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	44.633 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	32.685 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	42.042 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	42.042 bars
App 27-4	= M.A.W.P.	29.713 bars

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Left Dish	179.0	234.0	0.765	44.84
S1	174.9	234.0	0.747	44.84
S2	174.9	234.0	0.747	44.84
S2	174.9	234.0	0.747	44.84
Right Dish	179.0	234.0	0.765	44.84

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
K1_3in	Nozzle	138.00	138.00	1.000
K7_2in	Nozzle	138.00	138.00	1.000
N1_10in	Nozzle	138.00	138.00	1.000
K3-3in	Nozzle	138.00	138.00	1.000
K8-2in	Nozzle	138.00	138.00	1.000
K5-2in	Nozzle	138.00	138.00	1.000
K2-3in	Nozzle	138.00	138.00	1.000
K4-3in	Nozzle	138.00	138.00	1.000
K6-2in	Nozzle	138.00	138.00	1.000
D1-2in	Nozzle	138.00	138.00	1.000
N2-16in	Nozzle	138.00	138.00	1.000
N3-10in	Nozzle	138.00	138.00	1.000
M2_24in	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Left Dish	138.00	138.00	1.000
S1	138.00	138.00	1.000
S2	138.00	138.00	1.000
S2	138.00	138.00	1.000
Right Dish	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
K1_3in	11.00	225.00	0.049
K7_2in	9.47	225.00	0.042
N1_10in	45.67	225.00	0.203

K3-3in	11.00	225.00	0.049
K8-2in	9.47	225.00	0.042
K5-2in	9.47	225.00	0.042
K2-3in	11.00	225.00	0.049
K4-3in	11.00	225.00	0.049
K6-2in	9.47	225.00	0.042
D1-2in	9.47	225.00	0.042
N2-16in	74.15	225.00	0.330
N3-10in	45.67	225.00	0.203
M2_24in	112.12	225.00	0.498

Elements Suitable for Test Pressure.

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**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	38.627 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	38.220 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	44.633 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	32.685 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	42.042 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	42.042 bars
App 27-4	= M.A.W.P.	29.713 bars

UG-99(b), Test Pressure Calculation:

= Test Factor • MAWP • Stress Ratio

= 1.3 • 29.713 • 1.0

= 38.627 bars

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Left Dish	174.4	234.0	0.745	38.89
S1	170.0	234.0	0.727	38.89
S2	170.0	234.0	0.727	38.89
S2	170.0	234.0	0.727	38.89
Right Dish	174.4	234.0	0.745	38.89

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
K1_3in	Nozzle	138.00	138.00	1.000
K7_2in	Nozzle	138.00	138.00	1.000
N1_10in	Nozzle	138.00	138.00	1.000
K3-3in	Nozzle	138.00	138.00	1.000
K8-2in	Nozzle	138.00	138.00	1.000
K5-2in	Nozzle	138.00	138.00	1.000
K2-3in	Nozzle	138.00	138.00	1.000
K4-3in	Nozzle	138.00	138.00	1.000
K6-2in	Nozzle	138.00	138.00	1.000
D1-2in	Nozzle	138.00	138.00	1.000
N2-16in	Nozzle	138.00	138.00	1.000
N3-10in	Nozzle	138.00	138.00	1.000
M2_24in	Nozzle	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Left Dish	138.00	138.00	1.000
S1	138.00	138.00	1.000
S2	138.00	138.00	1.000
S2	138.00	138.00	1.000
Right Dish	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

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Field Pressure Test Results:

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Description	Ambient	Operating	Ratio
K1_3in	9.54	225.00	0.042
K7_2in	8.21	225.00	0.037
N1_10in	39.60	225.00	0.176
K3-3in	9.54	225.00	0.042
K8-2in	8.21	225.00	0.037
K5-2in	8.21	225.00	0.037
K2-3in	9.54	225.00	0.042
K4-3in	9.54	225.00	0.042
K6-2in	8.21	225.00	0.037
D1-2in	8.21	225.00	0.037
N2-16in	64.30	225.00	0.286
N3-10in	39.60	225.00	0.176
M2_24in	97.22	225.00	0.432

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	2204	24	0.0015124	97.0023
20	30	6833.33	2206	25	0.0004883	48.8188
30	40	6833.33	2206	25	0.0004883	48.8188
40	50	6833.33	2206	25	0.0004883	48.8188
50	60	No Calc	2204	24	0.0015124	97.0023

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	27	5.1733	0.15	11.7358
20	30	28	8.21753	0.15	7.37624
30	40	28	8.21753	0.15	7.37624
40	50	28	8.21753	0.15	7.37624
50	60	27	5.1733	0.15	11.7358

Minimum

7

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	6833.33	183266	No Calc	No Calc
30	40	6833.33	183266	No Calc	No Calc
40	50	6833.33	183266	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm3	Corroded Metal Wgt. kg.	Corroded ID Volume cm3	Extra due Misc % kg.
10	20	1498.29	1482720	1357.83	1494659	104.881
20	30	4454.69	10893460	3982.88	10954345	311.828
30	40	2227.34	5446730	1991.44	5477172	155.914
40	50	2784.18	6808413	2489.3	6846466	194.892
50	60	1498.29	1482720	1357.83	1494659	104.881
Total		12462	26114046.00	11179	26267302.00	872

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
10	Liqd	935.04	-179.167	...	...	Liquid 10
10	Insl	192.526	-243.75	...	...	Ins: 10
20	Sadl	632.283	1245	1384.5	...	Left Saddle
20	Plat	2746.05	3237.5	1179.2	...	PLAT:[1 OF 1]
20	Liqd	6869.68	1500	...	...	Liquid 20
20	Insl	552.199	1500	...	...	Ins: 20
20	Nozl	37.4301	1150	327.976	1063.68	K1_3in
20	Nozl	19.8186	1450	1100.4	...	K7_2in
20	Nozl	125.65	500	1202	...	N1_10in
20	Nozl	37.4301	1700	327.976	1063.68	K3-3in
20	Nozl	19.8186	1950	1100.4	...	K8-2in
20	Nozl	19.8186	2250	733.268	820.487	K5-2in
20	Nozl	24.4618	1150	-292.649	1073.94	K2-3in
20	Nozl	24.4618	1700	-292.649	1073.94	K4-3in
20	Nozl	12.4215	2250	-289.31	1061.69	K6-2in
20	Nozl	12.4215	2000	-1100.4	...	D1-2in
20	Wght	180.003	1245	-1500	...	Fp wt
30	Liqd	3434.84	750	...	...	Liquid 30
30	Insl	276.099	750	...	...	Ins: 30
30	Wght	2250	...	...	...	Baffles
30	Wght	150.003	...	...	...	Clip Wt
40	Sadl	632.283	630	1384.5	...	Left Saddle
40	Liqd	4293.55	937.5	...	...	Liquid 40
40	Insl	345.124	937.5	...	...	Ins: 40
40	Nozl	308.484	300	1278.2	...	N2-16in
40	Nozl	109.266	100	-1202	...	N3-10in
40	Wght	180.003	630	-1500	...	Fp
50	Liqd	935.04	229.167	...	...	Liquid 50
50	Insl	192.526	293.75	...	...	Ins: 50
50	Nozl	904.093	587.5	...	...	M2_24in
50	Wght	50	587.5	...	300	M2_Davit_24"

Total Weight of Each Detail Type:

Saddles	1264.6
Platforms	2746.1
Liquid	16468.2
Insulation	1558.5
Nozzles	1655.6
Weights	2810.0

Sum of the Detail Weights 26502.8 kg.

Weight Summation Results: (kg.)

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Element and Detail Weights:

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	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	13335.2	13335.2	13335.2	13335.2	13335.2	13335.2
Saddles	1264.6	1264.6	1264.6	1264.6	1264.6	1264.6
Nozzles	1655.6	1655.6	1655.6	1655.6	1655.6	1655.6
Wld Weights	2450.0	2450.0	2450.0	2450.0	2450.0	2450.0
Platforms	...	...	...	2746.1	2746.1	2746.1
Insulation	...	...	...	1558.5	1558.5	1558.5
Empty Weights	...	...	...	360.0	360.0	...
Test Weights	...	360.0	...	...	...	...
Ope Weights	...	...	...	...	...	360.0
Ope. Liquid	...	...	...	...	...	16468.2
Test Liquid	...	26098.1	...	...	...	...
Totals	18705.3	45163.5	18705.3	23369.9	23369.9	39838.0

Field Installation Options:

Platforms installed before lifting.

Insulation installed before lifting.

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	18705.3 kg.
Shop Test Wt.	- Fabricated weight + water (full)	45163.5 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	18705.3 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	23369.9 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	23369.9 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	39838.0 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	38464.7 kg.
Field Test Wt.	- Empty weight + test medium (full)	49468.0 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm^2	Inside Surface Area cm^2
10	20	56612.5	53484.4
20	30	207911	202633
30	40	103955	101316
40	50	129944	126645
50	60	56612.5	53484.4

Total	555035.1	537563.4 cm2
Total	597.4	578.6 ft²

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Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
K1_3in	48.87	51.00	65	300	GR 1.1	...	...	...	...
K7_2in	48.87	51.00	65	300	GR 1.1	...	...	...	...
N1_10in	48.87	51.00	65	300	GR 1.1	22.786	48.871	48.871	48.871
K3-3in	48.87	51.00	65	300	GR 1.1	...	...	...	...
K8-2in	48.87	51.00	65	300	GR 1.1	...	...	...	...
K5-2in	48.87	51.00	65	300	GR 1.1	...	...	...	...
K2-3in	48.87	51.00	65	300	GR 1.1	...	...	...	...
K4-3in	48.87	51.00	65	300	GR 1.1	...	...	...	...
K6-2in	48.87	51.00	65	300	GR 1.1	...	...	...	...
D1-2in	48.87	51.00	65	300	GR 1.1	...	...	...	...
N2-16in	48.87	51.00	65	300	GR 1.1	18.554	48.871	48.871	48.871
N3-10in	48.87	51.00	65	300	GR 1.1	22.786	48.871	48.871	48.871
M2_24in	48.87	51.00	65	300	GR 1.1	...	...	...	...
Min Rating	48.871	51.000, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
N1_10in	302.387	0.50	1261	0	29.713	48.871	Pass
N2-16in	442.976	0.50	3229	0	29.713	48.871	Pass
N3-10in	302.387	0.50	1261	0	29.713	48.871	Pass

Evaluating Flange on Nozzle [N110in]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(1261) + 4 \cdot \text{abs}(0) \cdot 302.387 \leq \pi \cdot 302.387^3 \cdot [(48.871 - 29.713) + 500 \cdot 48.871]$$

$$= 0.20184E+05 \leq 0.38615E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N110in [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(48.871, 0.500 \cdot 48.871 + 48.871 - 20184.0 / (\pi \cdot 302.387^3))$$

$$= \min(48.871, 50.520)$$

$$= 48.871 \text{ bars}$$

Evaluating Flange on Nozzle [N2-16in]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(3229) + 4 \cdot \text{abs}(0) \cdot 442.976 \leq \pi \cdot 442.976^3 \cdot [(48.871 - 29.713) + 500 \cdot 48.871]$$

$$= 0.51671E+05 \leq 0.12140E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2-16in [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(48.871, 0.500 \cdot 48.871 + 48.871 - 51670.7 / (\pi \cdot 442.976^3))$$

$$= \min(48.871, 54.752)$$

$$= 48.871 \text{ bars}$$

Evaluating Flange on Nozzle [N3-10in]:

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Nozzle Flange MAWP:

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$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(1261) + 4 \cdot \text{abs}(0) 302.387 \leq \pi \cdot 302.387^3 [(48.871 - 29.713) + 500 \cdot 48.871] \\ &= 0.20184E+05 \leq 0.38615E+05 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N3-10in [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(48.871, 0.500 \cdot 48.871 + 48.871 - 20184.0 / (\pi \cdot 302.387^3)) \\ &= \min(48.871, 50.520) \\ &= 48.871 \text{ bars} \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	0 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1+1.7 \cdot 3.4 \cdot 0.228 \cdot 0.976)/(1+1.7 \cdot 3.4 \cdot 0.228)))$
 $= \min(0.85, 0.912)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.534
Structure Height to Diameter ratio	3.069

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$; $Zg = 274320.0$ mm.

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1750.0 + 0.0 = 1750.0$ mm.
 $= 5.741$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (5.741 ft.) < 15 ft.
 $= 2.01(15/Zg)^{2/\alpha}$
 $= 2.01(15/900.0)^{2/9.5}$
 $= 0.849$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1



As there is No Hill Present: [Kzt]:

$K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$= (1 + K1 \cdot K2 \cdot K3)^2$

$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$

$= 1.0$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$

$= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$

$= 16.0 \text{ psf } [78.12] \text{ Kgs/m}^2$

Force on the first element [F]:

$= qz \cdot G \cdot Cf \cdot \text{WindArea}$

$= 16.0 \cdot 0.85 \cdot 0.534 \cdot 14.407$

$= 104.7 \text{ lbs. } [47.5] \text{ Kg}$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Left Dish	1750.0	0.000	0.000	0.000	0.849	1.000	78.120
S1	1750.0	0.000	0.000	0.000	0.849	1.000	78.120
S2	1750.0	0.000	0.000	0.000	0.849	1.000	78.120
S2	1750.0	0.000	0.000	0.000	0.849	1.000	78.120
Right Dish	1750.0	0.000	0.000	0.000	0.849	1.000	78.120

Platform Load Calculations

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf	Cf
PLAT:[1 OF 1]	64750.00	1750.00	78.12	505.83	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf
Fp wt	0.00	0.00	78.12	0.00
Baffles	0.00	1750.00	78.12	0.00
Clip Wt	0.00	1750.00	78.12	0.00
Fp	0.00	0.00	78.12	0.00
M2Davit24"	0.00	1750.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1750	2836.8	13384.7	78.12	52.2537
20	30	1750	2839.2	85176	78.12	888.936
30	40	1750	2839.2	42588	78.12	166.263
40	50	1750	2839.2	53235	78.12	207.829
50	60	1750	2836.8	13384.7	78.12	52.2537

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	1.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}
 SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\
 SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\
 SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\
 SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104
 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot hn^x$ where $Ct = 0.020$, $x = 0.75$ and hn = Structural Height (ft.)
 $= 0.020(9.26840.75)$
 $= 0.106$ seconds

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/33.0)$
 $= 0.030$

Check the Value of T which is the smaller of $Cu \cdot Ta$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.106, 0.03) \text{ per 12.8.2}$
 $= 0.030$

Loads on Building/Structure Supported Equipment:

$$\begin{aligned}
 &\text{Seismic Force per equation 13.3-1, [Fp]:} \\
 &= 0.4 \cdot Ap \cdot Sds \cdot W(1 + 2 \cdot (z/h)) / (R / Ie) \\
 &= 0.4 \cdot 1.0 \cdot 0.157 \cdot 38573(1 + 2 \cdot 1.0) / (3.0/1.25) \\
 &= 3024.159 \text{ Kgf}
 \end{aligned}$$

Maximum value of Fp per equation 13.3-2:
 $= 1.6 \cdot Sds \cdot I \cdot W$
 $= 1.6 \cdot 0.157 \cdot 1.25 \cdot 38573 = 12096.64 \text{ Kgf}$

Minimum value of Fp per equation 13.3-3:
 $= 0.3 \cdot 0.16 \cdot 1.25 \cdot 38573$
 $= 2268.120 \text{ Kgf}$

Total Base Shear $V = Fp$, [V]:
 $= 3024.159 \text{ Kgf}$

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.157 \cdot 38573$

= 1209.7 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	1075	5510.49	432.023
20	30	1075	5510.49	432.023
20	30	1075	5510.49	432.023
30	40	1075	5510.49	432.023
40	0	1075	5510.49	432.023
40	50	1075	5510.49	432.023
50	60	1075	5510.49	432.023

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Platform(s) installed in the Field before being lifted.
Insulation is installed in the Field before being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	3237.500 mm.
Center of Gravity of the Platforms	3287.500 mm.
Center of Gravity of the Liquid	3237.500 mm.
Center of Gravity of the Insulation	3237.500 mm.
Center of Gravity of the Nozzles	5294.143 mm.
Center of Gravity of the Added Weights (Operating)	3144.528 mm.
Center of Gravity of the Added Weights (Empty)	3144.528 mm.
Center of Gravity of Bare Shell New and Cold	3237.499 mm.
Center of Gravity of Bare Shell Corroded	3237.500 mm.
Vessel CG in the Operating Condition	3322.799 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	3402.389 mm.
Vessel CG in the Test Condition	3307.106 mm.

Rigging Analysis Results:

- Lifting weight based on the user selected calculated erected weight

Total Effective Length of Vessel for this analysis	6475.00 mm.
Total vessel weight (No Liquid) Twt	23369.87 Kg
Impact weight multiplication factor Imp	1.50
Design lifting weight, DWT = Imp * Twt	35054.80 Kg
Distance to the Lug on the Left End	1000.00 mm.
Distance to the Lug on the Right End	5470.00 mm.
Design Reaction force at the Left End Lug	16214.70 Kg
Design Reaction force at the Right End Lug	18840.11 Kg
CG distance during field lifting operation	3402.39 mm.
CG distance from the left end lug/trunnion	2402.39 mm.
CG distance from the nearer lifting lug/trunnion	2067.61 mm.
Compressive Allowable Increase Factor	1.00

Critical Values: (N./mm², mm.)

	Max Stress	Elevation	Allowables
Bending	0.70	1250.00	112.88 (UG-23)
Shear	0.58	650.00	96.60 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm.	Bending Moment Kg-m.	Bending Stress N./mm ²	Shear Force Kg	Shear Stress N./mm ²
0.00	0.0	0.0	8680.2	0.5
1850.00	3572.1	0.3	2700.3	0.1
3650.00	687.6	0.1	3309.1	0.2
4925.00	2980.6	0.3	8316.9	0.4
6425.00	1226.8	0.1	667.7	0.0

Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.0062 at elevation 1250.0 mm. - Must be <=1



Note: The rigging analysis is performed using a uniformly distributed load.

--- Plot data successfully generated ...----

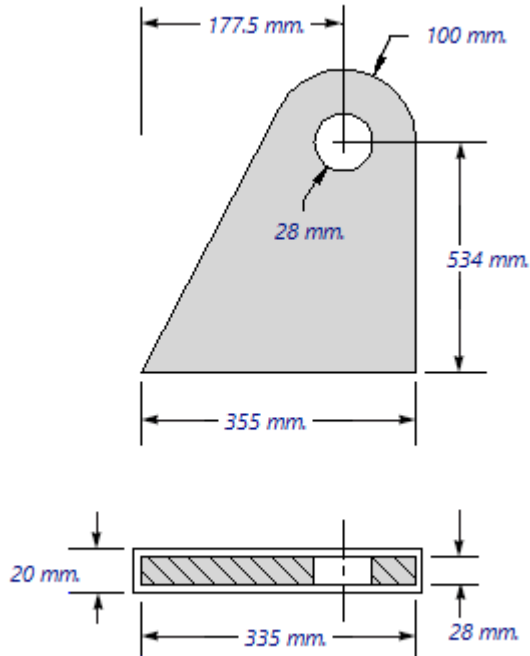
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Lifting Lug Calculations: Lug(s) on Left End of Vessel

Input Values:

Lifting Lug Material		SA-516 70
Lifting Lug Yield Stress	Yield	262.00 N./mm ²
Total Height of Lifting Lug	w	355.0000 mm.
Thickness of Lifting Lug	t	28.0000 mm.
Diameter of Hole in Lifting Lug	dh	28.0000 mm.
Radius of Semi-Circular Arc of Lifting Lug	r	100.0000 mm.
Height of Lug from Edge to Center of Hole	h	177.5000 mm.
Offset from Vessel Surface to Center of Hole	off	534.0000 mm.
Lug Fillet Weld Size	tw	20.0000 mm.
Length of weld along side of Lifting Lug	wl	335.0000 mm.
Length of Weld along Bottom of Lifting Lug	wb	20.0000 mm.
Thickness of Collar (if any)	tc	0.0000 mm.
Diameter of Collar (if any)	dc	0.0000 mm.
Impact Factor	Impfac	1.50
Sling Angle from Horizontal		90.0 degrees
Number of Lugs in Group		2

Lifting Lug Orientation to Vessel: Perpendicular
Lift Orientation : Horizontal Lift



PV Elite does not compute weak axis bending forces on the lugs. It is assumed that a spreader bar is used.

Computed Results:

Total vessel weight (no liquid)		23369.87 Kg
Design reaction force at the left end lug		16214.70 Kg
Design reaction force at the right end lug		18840.11 Kg
Force along vessel axis	Fax	0.00 Kg
Force normal to vessel	Fn	8107.35 Kg



Force tangential to vessel Ft 0.00 Kgf

Converting the weld leg dimension (tw) to the weld throat dimension.

Weld Group Inertia Calculations:

Weld group inertia about the longitudinal axis	I11	10584.556 cm ⁴
Weld group centroid distance in the long. direction	Y11	181.640 mm.
Dist. of weld group centroid from lug bottom	Y11b	167.500 mm.
Weld group inertia about the circumferential axis	I1c	438.254 cm ⁴
Weld group centroid distance in the circ. direction	Y1c	28.140 mm.

Note: The Impact Factor is applied to the Forces acting on the Lug.

Primary Shear Stress in the Welds due to Shear Loads [Ssll]:

$$= \sqrt{(Fax^2 + Ft^2 + Fn^2) / ((2 \cdot (wl + wb)) \cdot tw)}$$

$$= \sqrt{(0^2 + 0^2 + 8107^2) / ((2 \cdot (335.0 + 20.0)) \cdot 14.14)}$$

$$= 7.92 \text{ N./mm}^2$$

Shear Stress in the Welds due to Bending Loads [Sblf]:

$$= (Fax \cdot \text{off} \cdot Y11/I11) + (Ft \cdot \text{off} \cdot Y1c/I1c)$$

$$= (0 \cdot 534.0 \cdot 181.64/10584.556) +$$

$$(0 \cdot 534.0 \cdot 28.14/438.254)$$

$$= 0.00 \text{ N./mm}^2$$

Total Shear Stress for Combined Loads [St]:

$$= Ssll + Sblf$$

$$= 7.92 + 0.0$$

$$= 7.92 \text{ N./mm}^2$$

Allowable Shear Stress for Combined Loads [Sta]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac (AISC Shear Allowable)}$$

$$= 0.4 \cdot 262 \cdot 1.0$$

$$= 104.80 \text{ N./mm}^2$$

Shear Stress in Lug above Hole [Shs]:

$$= \sqrt{(Pl^2 + Fax^2) / Sha}$$

$$= \sqrt{(8107^2 + 0^2) / 48.16}$$

$$= 16.51 \text{ N./mm}^2$$

Allowable Shear Stress in Lug above Hole [Sas]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac}$$

$$= 0.4 \cdot 262 \cdot 1.0$$

$$= 104.80 \text{ N./mm}^2$$

Pin Hole Bearing Width [Pbw]:

$$= \text{Lug Thickness} + 2 \cdot \text{Collar Thickness}$$

$$= 28.0 + 2 \cdot 0.0$$

$$= 28.000 \text{ mm.}$$

Pin Hole Bearing Stress [Pbs]:

$$= \sqrt{(Pl^2 + Faxial^2) / (Pbw \cdot dh)}$$

$$= \sqrt{(8107^2 + 0^2) / (28.0 \cdot 28.0)}$$

$$= 101.41 \text{ N./mm}^2$$

Allowable Bearing Stress [Pba]:

$$= \min(0.75 \cdot \text{Yield} \cdot \text{Occfac}, 0.9 \cdot \text{Yield}) \text{ AISC Bearing All.}$$

$$= \min(0.75 \cdot 262 \cdot 1.0, 235.8)$$

$$= 196.50 \text{ N./mm}^2$$

Bending Stress at the Base of the Lug [Fbs]:

$$= Ft \cdot \text{off} / (w \cdot t^2/6) + Fax \cdot \text{off} / (w^2 \cdot t/6)$$

$$= 0 \cdot 534.0 / (355.0 \cdot 28.0^2/6) +$$

$$0 \cdot 534.0 / (355.0^2 \cdot 28.0/6)$$

$$= 0.00 \text{ N./mm}^2$$

Tensile Stress at the Base of the Lug [Fa]:

$$= F_n / (w \cdot t)$$

$$= 8107 / (355.0 \cdot 28.0)$$

$$= 8.00 \text{ N./mm}^2$$

Total Combined Stress at the Base of the Lug:

$$= F_{bs} + F_a$$

$$= 0.0 + 8.0$$

$$= 8.00 \text{ N./mm}^2$$

Lug Allowable Stress for Bending and Tension:

$$= \min(0.66 \cdot \text{Yield} \cdot \text{Occfac}, 0.75 \cdot \text{Yield})$$

$$= \min(0.66 \cdot 262 \cdot 1.0, 196.5)$$

$$= 172.92 \text{ N./mm}^2$$

Required Shackle Pin Diameter [Spd]:

$$= \sqrt{((2(F_n^2 + F_{ax}^2)^{1/2} / (\pi \cdot \text{Sta})))}$$

$$= \sqrt{((2(8107^2 + 0^2)^{1/2} / (\pi \cdot 105)))}$$

$$= 21.9769 \text{ mm.}$$

WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).



The ratio Beta1/Beta2 for rectangular attachments was outside the range of values in tables 7 & 8 and was set to be an acceptable (maximum or minimum) value. This generally results in one dimension being altered so that the aspect ratio does not exceed 4 to 1.

Attachment dimensions (C11P, C22P, etc.) may be modified accounting for attachment welds where applicable.

Input Echo, WRC107/537 Item 1, Description: Lift Lug

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	0.0000 mm.
Vessel Diameter	Dv	2150.000 mm.
Vessel Thickness	Tv	28.000 mm.

Note: Lifting lug calculations are performed at ambient temperature conditions.

Attachment Type	Type	Rectangular
Parameter C11	C11	20.00 mm.
Parameter C22	C22	80.00 mm.
Design Internal Pressure	Dp	0.000 bars
Include Pressure Thrust		No

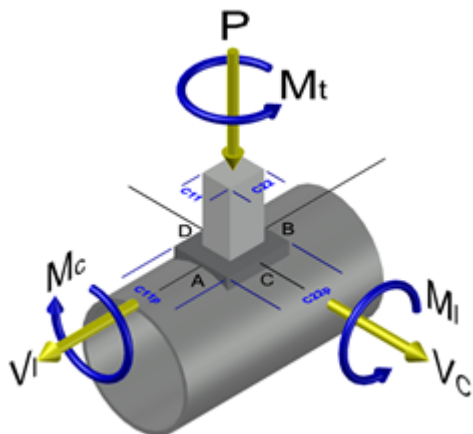
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-8107.3	Kgf
Longitudinal Shear (SUS)	Vl	-0.0	Kgf
Circumferential Shear (SUS)	Vc	0.0	Kgf
Circumferential Moment (SUS)	Mc	0.0	Kg-m.
Longitudinal Moment (SUS)	Ml	-0.0	Kg-m.
Torsional Moment (SUS)	Mt	0.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUSained loads:

Radial Load	P	-8107.3	Kgf
Circumferential Shear	VC	0.0	Kgf
Longitudinal Shear	VL	-0.0	Kgf
Circumferential Moment	MC	0.0	Kg-m.
Longitudinal Moment	ML	-0.0	Kg-m.
Torsional Moment	MT	0.0	Kg-m.

Dimensionless Parameters used : Gamma = 38.89

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.027	4C	7.499	(A,B)
N(PHI) / (P/Rm)	0.027	3C	7.687	(C,D)
M(PHI) / (P)	0.016	2C1	0.238	(A,B)
M(PHI) / (P)	0.016	1C !	0.274	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.015	3A	0.087	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.018	1A !	0.104	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.023	3B	0.741	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.020	1B !	0.063	(A,B,C,D)
N(x) / (P/Rm)	0.022	3C	7.775	(A,B)
N(x) / (P/Rm)	0.022	4C	7.539	(C,D)
M(x) / (P)	0.023	1C1	0.270	(A,B)
M(x) / (P)	0.023	2C	0.217	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.015	4A	0.102	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.025	2A !	0.063	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.023	4B	0.180	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.028	2B !	0.106	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb.	P	19.6	19.6	19.6	19.6	20.0	20.0	20.0	20.0

147-D

Ammonia LetDown Drum

QUEP

PV Elite 28 U1 Licensee: SL User

FileName : 147-D Ammonia Letdown Drum_7%

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Lifting Lug Calcs: Left End

Step: 17 11:57am Sep 16,2026



HEXAGON

Circ. Bend. P	144.6	-144.6	144.6	-144.6	166.7	-166.7	166.7	-166.7
Circ. Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Circ. Bend. MC	0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Circ. Memb. ML	0.0	0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML	0.0	-0.0	-0.0	0.0	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.	164.2	-125.1	164.2	-125.1	186.8	-146.7	186.8	-146.7
Long. Memb. P	20.3	20.3	20.3	20.3	19.7	19.7	19.7	19.7
Long. Bend. P	164.5	-164.5	164.5	-164.5	132.3	-132.3	132.3	-132.3
Long. Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Long. Bend. MC	0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Long. Memb. ML	0.0	0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML	0.0	-0.0	-0.0	0.0	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.	184.8	-144.3	184.8	-144.3	152.0	-112.7	152.0	-112.7
Shear VC	0.0	0.0	-0.0	-0.0	0.0	0.0	0.0	0.0
Shear VL	-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Shear MT	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Str. Int.	184.8	144.3	184.8	144.3	186.8	146.7	186.8	146.7

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		19.6	19.6	19.6	19.6	20.0	20.0	20.0	20.0
Circ. Q (SUS)		144.6	-144.6	144.6	-144.6	166.7	-166.7	166.7	-166.7
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		20.3	20.3	20.3	20.3	19.7	19.7	19.7	19.7
Long. Q (SUS)		164.5	-164.5	164.5	-164.5	132.3	-132.3	132.3	-132.3
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.0	0.0	-0.0	-0.0	0.0	0.0	0.0	0.0
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		20.3	20.3	20.3	20.3	20.0	20.0	20.0	20.0
Pm+Pl+Q (Total)		184.8	144.3	184.8	144.3	186.8	146.7	186.8	146.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	20.27	207.00	Passed
Pm+Pl+Q (TOTAL)	186.79	413.99	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.



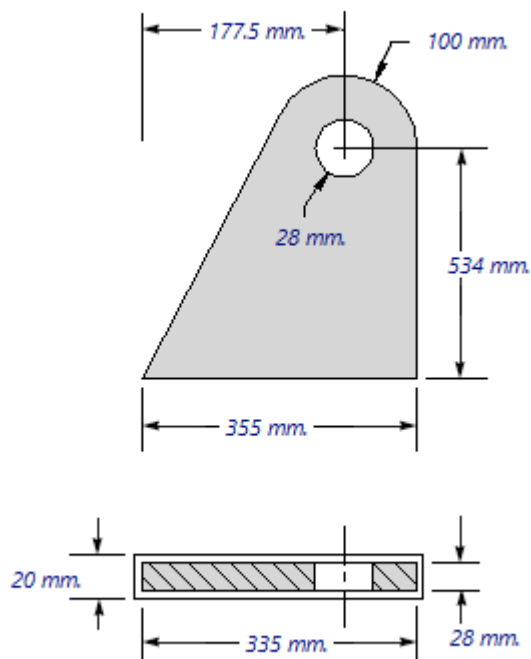


Lifting Lug Calculations: Lug(s) on Right End of Vessel

Input Values:

Lifting Lug Material		SA-516 70
Lifting Lug Yield Stress	Yield	262.00 N./mm ²
Total Height of Lifting Lug	w	355.0000 mm.
Thickness of Lifting Lug	t	28.0000 mm.
Diameter of Hole in Lifting Lug	dh	28.0000 mm.
Radius of Semi-Circular Arc of Lifting Lug	r	100.0000 mm.
Height of Lug from Edge to Center of Hole	h	177.5000 mm.
Offset from Vessel Surface to Center of Hole	off	534.0000 mm.
Lug Fillet Weld Size	tw	20.0000 mm.
Length of weld along side of Lifting Lug	wl	335.0000 mm.
Length of Weld along Bottom of Lifting Lug	wb	20.0000 mm.
Thickness of Collar (if any)	tc	0.0000 mm.
Diameter of Collar (if any)	dc	0.0000 mm.
Impact Factor	Impfac	1.50
Number of Lugs in Group		2

Lifting Lug Orientation to Vessel: Perpendicular
Lift Orientation : Horizontal Lift



PV Elite does not compute weak axis bending forces on the lugs. It is assumed that a spreader bar is used.

Computed Results:

Total vessel weight (no liquid)		23369.87 Kg
Design reaction force at the left end lug		16214.70 Kg
Design reaction force at the right end lug		18840.11 Kg
Force along vessel axis	Fax	0.00 Kg
Force normal to vessel	Fn	9420.06 Kg
Force tangential to vessel	Ft	0.00 Kg



Sling angle from horizontal

0.0 deg

Converting the weld leg dimension (tw) to the weld throat dimension.

Weld Group Inertia Calculations:

Weld group inertia about the longitudinal axis	I11	10584.556 cm4
Weld group centroid distance in the long. direction	Y11	181.640 mm.
Dist. of weld group centroid from lug bottom	Y11b	167.500 mm.
Weld group inertia about the circumferential axis	I1c	438.254 cm4
Weld group centroid distance in the circ. direction	Y1c	28.140 mm.

Note: The Impact Factor is applied to the Forces acting on the Lug.

Primary Shear Stress in the Welds due to Shear Loads [Ssll]:

$$= \sqrt{(F_{ax}^2 + F_t^2 + F_n^2)} / ((2 \cdot (w_l + w_b)) \cdot t_w)$$

$$= \sqrt{(0^2 + 0^2 + 9420^2)} / ((2 \cdot (335.0 + 20.0)) \cdot 14.14)$$

$$= 9.20 \text{ N./mm}^2$$

Shear Stress in the Welds due to Bending Loads [Sblf]:

$$= (F_{ax} \cdot \text{off} \cdot Y_{11}/I_{11}) + (F_t \cdot \text{off} \cdot Y_{1c}/I_{1c})$$

$$= (0 \cdot 534.0 \cdot 181.64/10584.556) +$$

$$(0 \cdot 534.0 \cdot 28.14/438.254)$$

$$= 0.00 \text{ N./mm}^2$$

Total Shear Stress for Combined Loads [St]:

$$= S_{sll} + S_{blf}$$

$$= 9.202 + 0.0$$

$$= 9.20 \text{ N./mm}^2$$

Allowable Shear Stress for Combined Loads [Sta]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac} \text{ (AISC Shear Allowable)}$$

$$= 0.4 \cdot 262 \cdot 1.0$$

$$= 104.80 \text{ N./mm}^2$$

Shear Stress in Lug above Hole [Shs]:

$$= \sqrt{(P_l^2 + F_{ax}^2)} / S_{ha}$$

$$= \sqrt{(9420^2 + 0^2)} / 48.16$$

$$= 19.18 \text{ N./mm}^2$$

Allowable Shear Stress in Lug above Hole [Sas]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac}$$

$$= 0.4 \cdot 262 \cdot 1.0$$

$$= 104.80 \text{ N./mm}^2$$

Pin Hole Bearing Width [Pbw]:

$$= \text{Lug Thickness} + 2 \cdot \text{Collar Thickness}$$

$$= 28.0 + 2 \cdot 0.0$$

$$= 28.000 \text{ mm.}$$

Pin Hole Bearing Stress [Pbs]:

$$= \sqrt{(P_l^2 + F_{axial}^2)} / (P_{bw} \cdot d_h)$$

$$= \sqrt{(9420^2 + 0^2)} / (28.0 \cdot 28.0)$$

$$= 117.83 \text{ N./mm}^2$$

Allowable Bearing Stress [Pba]:

$$= \min(0.75 \cdot \text{Yield} \cdot \text{Occfac}, 0.9 \cdot \text{Yield}) \text{ AISC Bearing All.}$$

$$= \min(0.75 \cdot 262 \cdot 1.0, 235.8)$$

$$= 196.50 \text{ N./mm}^2$$

Bending Stress at the Base of the Lug [Fbs]:

$$= F_t \cdot \text{off} / (w \cdot t^2/6) + F_{ax} \cdot \text{off} / (w^2 \cdot t/6)$$

$$= 0 \cdot 534.0 / (355.0 \cdot 28.0^2/6) +$$

$$0 \cdot 534.0 / (355.0^2 \cdot 28.0/6)$$

$$= 0.00 \text{ N./mm}^2$$



Tensile Stress at the Base of the Lug [Fa]:

$$= F_n / (w \cdot t)$$

$$= 9420 / (355.0 \cdot 28.0)$$

$$= 9.29 \text{ N./mm}^2$$

Total Combined Stress at the Base of the Lug:

$$= F_{bs} + F_a$$

$$= 0.0 + 9.3$$

$$= 9.29 \text{ N./mm}^2$$

Lug Allowable Stress for Bending and Tension:

$$= \min(0.66 \cdot \text{Yield} \cdot \text{Occfac}, 0.75 \cdot \text{Yield})$$

$$= \min(0.66 \cdot 262 \cdot 1.0, 196.5)$$

$$= 172.92 \text{ N./mm}^2$$

Required Shackle Pin Diameter [Spd]:

$$= \sqrt{((2(F_n^2 + F_{ax}^2)^{1/2} / (\pi \cdot \text{Sta})))}$$

$$= \sqrt{(2(9420^2 + 0^2)^{1/2} / (\pi \cdot 105))}$$

$$= 23.6894 \text{ mm.}$$

WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).



The ratio Beta1/Beta2 for rectangular attachments was outside the range of values in tables 7 & 8 and was set to be an acceptable (maximum or minimum) value. This generally results in one dimension being altered so that the aspect ratio does not exceed 4 to 1.

Attachment dimensions (C11P, C22P, etc.) may be modified accounting for attachment welds where applicable.

Input Echo, WRC107/537 Item 1, Description: Lift Lug

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	0.0000 mm.
Vessel Diameter	Dv	2150.000 mm.
Vessel Thickness	Tv	28.000 mm.

Note: Lifting lug calculations are performed at ambient temperature conditions.

Attachment Type	Type	Rectangular
Parameter C11	C11	20.00 mm.
Parameter C22	C22	80.00 mm.
Design Internal Pressure	Dp	0.000 bars
Include Pressure Thrust		No

External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-9420.1	Kgf
Longitudinal Shear (SUS)	Vl	-0.0	Kgf
Circumferential Shear (SUS)	Vc	0.0	Kgf
Circumferential Moment (SUS)	Mc	0.0	Kg-m.
Longitudinal Moment (SUS)	Ml	-0.0	Kg-m.
Torsional Moment (SUS)	Mt	0.0	Kg-m.

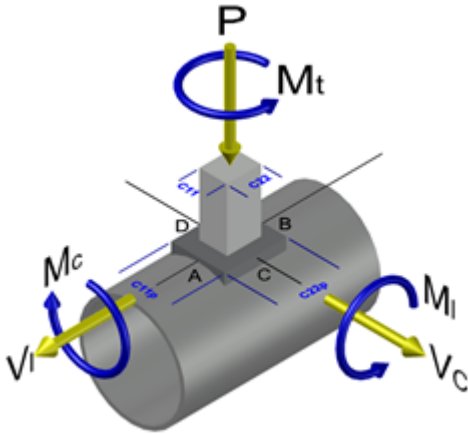
Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the



results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	-9420.1	Kgf
Circumferential Shear	VC	0.0	Kgf
Longitudinal Shear	VL	-0.0	Kgf
Circumferential Moment	MC	0.0	Kg-m.
Longitudinal Moment	ML	-0.0	Kg-m.
Torsional Moment	MT	0.0	Kg-m.

Dimensionless Parameters used : Gamma = 38.89

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
$N(\Phi) / (P/R_m)$	0.027	4C	7.499	(A,B)
$N(\Phi) / (P/R_m)$	0.027	3C	7.687	(C,D)
$M(\Phi) / (P)$	0.016	2C1	0.238	(A,B)
$M(\Phi) / (P)$	0.016	1C !	0.274	(C,D)
$N(\Phi) / (MC/(R_m^{*2} * Beta))$	0.015	3A	0.087	(A,B,C,D)
$M(\Phi) / (MC/(R_m * Beta))$	0.018	1A !	0.104	(A,B,C,D)
$N(\Phi) / (ML/(R_m^{*2} * Beta))$	0.023	3B	0.741	(A,B,C,D)
$M(\Phi) / (ML/(R_m * Beta))$	0.020	1B !	0.063	(A,B,C,D)
$N(x) / (P/R_m)$	0.022	3C	7.775	(A,B)
$N(x) / (P/R_m)$	0.022	4C	7.539	(C,D)
$M(x) / (P)$	0.023	1C1	0.270	(A,B)
$M(x) / (P)$	0.023	2C	0.217	(C,D)
$N(x) / (MC/(R_m^{*2} * Beta))$	0.015	4A	0.102	(A,B,C,D)
$M(x) / (MC/(R_m * Beta))$	0.025	2A !	0.063	(A,B,C,D)
$N(x) / (ML/(R_m^{*2} * Beta))$	0.023	4B	0.180	(A,B,C,D)
$M(x) / (ML/(R_m * Beta))$	0.028	2B !	0.106	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl

147-D

Ammonia LetDown Drum

QUEP

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FileName : 147-D Ammonia Letdown Drum_7%

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Lifting Lug Calcs: Right End

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Circ. Memb. P	22.7	22.7	22.7	22.7	23.3	23.3	23.3	23.3
Circ. Bend. P	168.0	-168.0	168.0	-168.0	193.7	-193.7	193.7	-193.7
Circ. Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Circ. Bend. MC	0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Circ. Memb. ML	0.0	0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML	0.0	-0.0	-0.0	0.0	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.	190.7	-145.3	190.7	-145.3	217.0	-170.5	217.0	-170.5
Long. Memb. P	23.6	23.6	23.6	23.6	22.8	22.8	22.8	22.8
Long. Bend. P	191.2	-191.2	191.2	-191.2	153.7	-153.7	153.7	-153.7
Long. Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Long. Bend. MC	0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Long. Memb. ML	0.0	0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML	0.0	-0.0	-0.0	0.0	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.	214.7	-167.6	214.7	-167.6	176.6	-130.9	176.6	-130.9
Shear VC	0.0	0.0	-0.0	-0.0	0.0	0.0	0.0	0.0
Shear VL	-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Shear MT	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Str. Int.	214.7	167.6	214.7	167.6	217.0	170.5	217.0	170.5

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		22.7	22.7	22.7	22.7	23.3	23.3	23.3	23.3
Circ. Q (SUS)		168.0	-168.0	168.0	-168.0	193.7	-193.7	193.7	-193.7
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		23.6	23.6	23.6	23.6	22.8	22.8	22.8	22.8
Long. Q (SUS)		191.2	-191.2	191.2	-191.2	153.7	-153.7	153.7	-153.7
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.0	0.0	-0.0	-0.0	0.0	0.0	0.0	0.0
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		23.6	23.6	23.6	23.6	23.3	23.3	23.3	23.3
Pm+Pl+Q (Total)		214.7	167.6	214.7	167.6	217.0	170.5	217.0	170.5

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	23.55	207.00	Passed
Pm+Pl+Q (TOTAL)	217.04	413.99	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

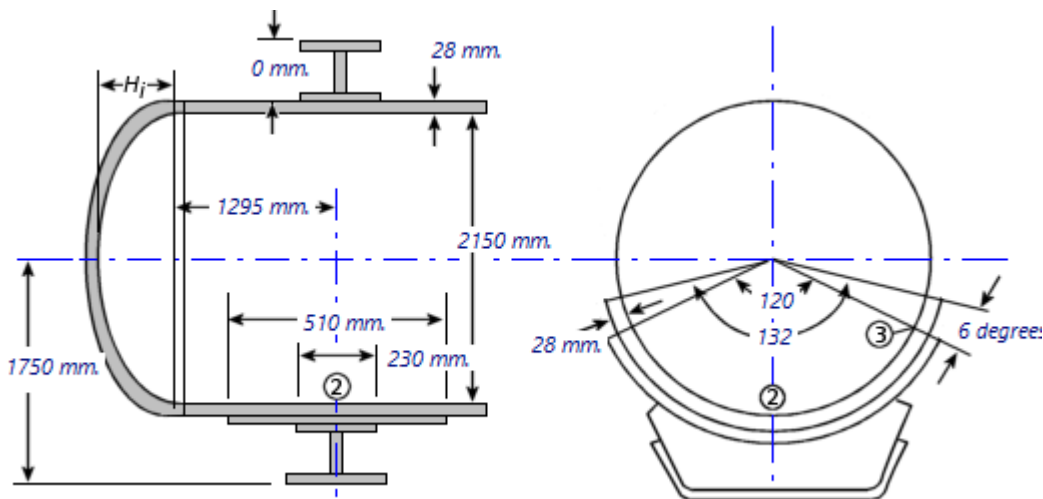


ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1090.50	mm.
Shell Thickness used in this Case	t	28.000	mm.
Stiffened Vessel Length per 4.15.6	L	6475.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	1295.00	mm.
Saddle Width	b1 or b	230.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	510.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	28.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm2
Inside or Mean Depth of Head	Hi or hm	552.50	mm.
Shell Allowable Stress used in Calculation		138.00	N./mm2
Head Allowable Stress used in Calculation		138.00	N./mm2
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1750.00	mm.
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		23099.42	Kgf
Pressure used in Analysis	P	29.466	bars

Horizontal Vessel Analysis Results:	Actual N./mm^2	Allowable N./mm^2
Long. Stress at Top of Midspan	63.74	138.00
Long. Stress at Bottom of Midspan	64.80	138.00
Long. Stress at Top of Saddles	71.18	138.00
Long. Stress at Bottom of Saddles	60.44	138.00
Tangential Shear in Shell	5.24	110.40
Circ. Stress at Horn of Saddle	10.81	172.50
Circ. Compressive Stress in Shell	0.67	138.00



Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(1367.5/2 + 0) \cdot 1750.0/1958.9496$$

$$= 1832.5 \text{ Kg}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(733.61, 0) \cdot 1750.0/3885.0$$

$$= 330.5 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(3024.16, 0) \cdot 1750.0/3885.0$$

$$= 1362.2 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(3024/2 + 0) \cdot 1750.0/1958.9496$$

$$= 4052.4 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 19047 + \max(330, 1833, 1362, 4052)$$

$$= 23099.4 \text{ Kg}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08(2.206 + \pi/4(2.366 \cdot 1.2)^2)$$

$$= 7194.198 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	23731.71	Kgf
Transverse Shear Load	1512.08	Kgf
Longitudinal Shear Load	3024.16	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(2150.0/2 + 28.0 + 28.0)\sin(120.0/2), 2 \cdot 1090.5)$$

$$= 1958.950 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -23099 \cdot 1295.0 [1 - (1 - 1295.0/6475.0 + (1090.5^2 - 552.5^2)/(2 \cdot 1295.0 \cdot 6475.0))/(1 + (4 \cdot 552.5)/(3 \cdot 6475.0))]$$

$$= -7011.8 \text{ Kg-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 23099 \cdot 6475/4((1 + 2(1091^2 - 553^2)/(6475^2))/(1 + (4 \cdot 553)/(3 \cdot 6475)) - 4 \cdot 1295/6475)$$

$$= 5074.6 \text{ Kg-m}$$



Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 29.466 \cdot 1090.5 / (2 \cdot 25.0) - 5074.6 / (\pi \cdot 1090.5^2 \cdot 25.0)$$

$$= 63.74 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 29.466 \cdot 1090.5 / (2 \cdot 25.0) + 5074.6 / (\pi \cdot 1090.5^2 \cdot 25.0)$$

$$= 64.80 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 29.466 \cdot 1090.5 / (2 \cdot 25.0) - 7011.8 / (0.1066 \cdot \pi \cdot 1090.5^2 \cdot 25.0)$$

$$= 71.18 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 29.466 \cdot 1090.5 / (2 \cdot 25.0) + 7011.8 / (0.1923 \cdot \pi \cdot 1090.5^2 \cdot 25.0)$$

$$= 60.44 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 23099(6475.0 - 2 \cdot 1295.0) / (6475.0 + (4 \cdot 552.5 / 3))$$

$$= 12443.9 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 12443.9 / (1090.5 \cdot 25.0)$$

$$= 5.24 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1090.5 \cdot 25.0)}$$

$$= 128.789 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(230.0 + 1.56 \cdot \sqrt{(1090.5 \cdot 25.0)}, 2 \cdot 1295.0)$$

$$= 487.58 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(137.997 / 137.997, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot t_r))$$

$$= -0.7603 \cdot 23099 \cdot 0.1 / (487.577(25.0 + 1.0 \cdot 28.0))$$

$$= -0.67 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r_{*}]:

$$= -Q / (4(t + \eta \cdot t_r) b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot t_r)^2)$$

$$= -23099 / (4(25.0 + 1.0 \cdot 28.0) 487.577) -$$

$$12 \cdot 0.053 \cdot 23099 \cdot 1090.5 / (6475.0(25.0 + 1.0 \cdot 28.0)^2)$$

$$= -10.81 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 3885.0 \text{ mm.}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000012 \cdot 3885.0(65.0 - 21.1)$$

$$= 2.025 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length

Bplen

2070.0000 mm.

147-D

Ammonia LetDown Drum

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FileName : 147-D_Ammonia Letdown Drum_7%

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Saddle Calcs: Operating Case: Step: 19 11:57am Sep 16,2026



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Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	254.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm.
Web Thickness	Webtk	10.0000	mm.
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	246.0	N./mm2
Height of Web at Center	Hw,c	301.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1078.0 \cdot 25.0)} + 510.0)^3) / 12$$

$$= 99.752 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 99.752 + 191.524(129.403 - 12.5)^2$$

$$= 26274.078 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	766.1	25.0	12.5	191.5	239405.2	0.263E+05
Wearplate	510.0	28.0	39.0	142.8	556920.0	0.118E+05
Web	10.0	603.0	354.5	60.3	2137634.2	0.488E+05
BasePlate	254.0	16.0	664.0	40.6	2698495.2	0.116E+06
Totals	...	...	...	435.3	5632454.5	0.203E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 5632455 / 435.264$$

$$= 129.403 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 23099.424$$

$$= 4701.2334 \text{ Kgf}$$

$$\text{Tension Stress, St} = (Fh / A_s) = 1.8915 \text{ N./mm2}$$

$$\text{Allowed Stress, Sa} = 0.6 \cdot \text{Yield Str} = 147.6000 \text{ N./mm2}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1750.0 - 1078.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 858.501 \text{ mm.}$$

$$\text{Bending Moment, M} = Fh \cdot d = 4036.0942 \text{ Kg-m.}$$

$$\text{Bending Stress, Sb} = (M \cdot C1 / I) = 2.5229 \text{ N./mm2}$$

$$\text{Allowed Stress, Sa} = 2/3 \cdot \text{Yield Str} = 164.0000 \text{ N./mm2}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$



$$= \sqrt{3(23099 + 632)254.0 / (4 \cdot 2070.0 \cdot 164.0)}$$

$$= 11.428 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1075.0 + 28.0 + 28.0)$$

$$= 1958.950 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1958.9496 / (5 - 1)$$

$$= 489.737 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 489.7374 \cdot 254.0/2$$

$$= 621.966 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 23731.707 / (2070.0 \cdot 254.0)$$

$$= 4.514 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 622.0 \cdot 4.51$$

$$= 2807.320 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 22.0 + 24.487$$

$$= 46.487 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/\text{Ar}$$

$$= 2807.3/46.4869$$

$$= 5.922 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	230.0	...	23.0	...	0.101E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 230.0 - 10.0$$

$$= 220.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0.0/46.487$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 230.0/2$$

$$= 115.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{1013.9/46.487}$$

$$= 46.702 \text{ mm.}$$



Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1750.0 - \cos(120.0/2) (1075.0 + 28.0 + 28.0) - 16.0$$

$$= 1168.500 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 246.0)}$$

$$= 126.667$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1168.5 / 46.702$$

$$= 25.020$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{\text{plen}}) \cdot e \cdot L / 2$$

$$= 3024.2 / (2 \cdot 2070.0) \cdot 489.737 \cdot 1168.5 / 2$$

$$= 209.014 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($25.0203 < 126.6669$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (25.02)^2 / (2 \cdot 126.67^2)) 246 /$$

$$(5/3 + 3 \cdot (25.02) / (8 \cdot 126.67) - (25.02^3) / (8 \cdot 126.67^3))$$

$$= 138.6 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 5.92 / 138.64 + (209.01 \cdot 115.0 / 10139168) / 164.0$$

$$= 0.184$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	220.0	0.0	22.0	0.0	0.101E+04
Web	489.7	10.0	0.0	49.0	0.0	4.08
Totals	...	...	...	71.0	...	0.102E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 70.974$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 230.0 / 2$$

$$= 115.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1750.0 - \cos(1131.0/2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 603.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1017.9 / 70.974)}$$

$$= 37.871 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 603.0 / 37.871$$

$$= 15.922$$

Unit Force [Force,u]:



$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 3024.159 / (2 \cdot 2070.0)$$

$$= 0.730 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.73 \cdot 489.737 \cdot 603.0$$

$$= 215.722 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 215.722 / 88514.297$$

$$= 23.900 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ (15.9225 < 126.6669) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (15.92)^2 / (2 \cdot 126.67^2)) 246 / (5/3 + 3 \cdot (15.92) / (8 \cdot 126.67) - (15.92^3) / (8 \cdot 126.67^3))$$

$$= 142.4 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 7.77/142.43 + (215.72 \cdot 115.0 / 1017.914) / 164.0$$

$$= 0.200$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm2
Bolt Corrosion Allowance	Bca	0.0 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	1.5500 cm2
BasePlate Allowable Stress	S	138.00 N./mm2
Area Available in a Single Bolt	BlArea	1.5500 cm2
Saddle Load Q0 (Weight)	Q0	19679.3 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	1362.2 Kg
Maximum Transverse Force	Ft	1512.1 Kg
Maximum Longitudinal Force	Fl	3024.2 Kg
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

$$= 6410.5 \text{ Kg}$$

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= \max(Ff, Fl) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 6410.5 / (1.55 \cdot 8)$$

$$= 50.7 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1512.1 / (1.55 \cdot 8)$$

$$= 12.0 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= FI / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 3024.16 / (82.8 \cdot 8.0)$$

$$= 0.4477 \text{ cm}^2$$

Bolt Area due to Transverse Load:

147-D

Ammonia LetDown Drum

QUEP

PV Elite 28 U1 Licensee: SL User

FileName : 147-D Ammonia Letdown Drum_7%

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Saddle Calcs: Operating Case: Step: 19 11:57am Sep 16,2026



HEXAGON

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1750.0 \cdot 1512.08 + 0.0$$

$$= 2646.19 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 2646.19/19679.32$$

$$= 134.46 \text{ mm.} < B_{plen}/6 \rightarrow \text{No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0.4477, 0.0]$$

$$= 0.4477 \text{ cm}^2$$

Note: Saddle and wear plate material are both specified as impact tested.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:**Input and Calculated Values:**

Vessel Mean Radius	Rm	1090.50	mm.
Shell Thickness used in this Case	t	28.000	mm.
Stiffened Vessel Length per 4.15.6	L	6475.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	1295.00	mm.
Saddle Width	b1 or b	230.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	510.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	28.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	552.50	mm.
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1750.00	mm.
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		24788.47	Kgf
Pressure used in Analysis	P	29.466	bars

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		63.70	138.00
Long. Stress at Bottom of Midspan		64.84	138.00
Long. Stress at Top of Saddles		71.68	138.00
Long. Stress at Bottom of Saddles		60.16	138.00
Tangential Shear in Shell		5.62	110.40
Circ. Stress at Horn of Saddle		11.60	172.50
Circ. Compressive Stress in Shell		0.72	138.00



Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(1367.5/2 + 0) \cdot 1750.0/1958.9496$$

$$= 1832.5 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(733.61, 0) \cdot 1750.0/3885.0$$

$$= 330.5 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(3024.16, 0) \cdot 1750.0/3885.0$$

$$= 1362.2 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(3024/2 + 0) \cdot 1750.0/1958.9496$$

$$= 4052.4 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 20736 + \max(330, 1833, 1362, 4052)$$

$$= 24788.5 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08(2.206 + \pi/4(2.366 \cdot 1.2)^2)$$

$$= 7194.198 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	25420.75	Kgf
Transverse Shear Load	1512.08	Kgf
Longitudinal Shear Load	3024.16	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(2150.0/2 + 28.0 + 28.0)\sin(120.0/2), 2 \cdot 1090.5)$$

$$= 1958.950 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(2a \cdot L))/(1 + (4\text{hm})/3L)]$$

$$= -24788 \cdot 1295.0 [1 - (1 - 1295.0/6475.0 + (1090.5^2 - 552.5^2)/(2 \cdot 1295.0 \cdot 6475.0))/(1 + (4 \cdot 552.5)/(3 \cdot 6475.0))]$$

$$= -7524.5 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(\text{Rm}^2 - \text{hm}^2)/(L^2))/(1 + (4\text{hm}^2)/(3L)) - 4a/L)$$

$$= 24788 \cdot 6475/4((1 + 2(1091^2 - 553^2)/(6475^2))/(1 + (4 \cdot 553)/(3 \cdot 6475)) - 4 \cdot 1295/6475)$$

$$= 5445.7 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:



$$\begin{aligned}
 &= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t) \\
 &= 29.466 \cdot 1090.5 / (2 \cdot 25.0) - 5445.7 / (\pi \cdot 1090.5^2 \cdot 25.0) \\
 &= 63.70 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= 29.466 \cdot 1090.5 / (2 \cdot 25.0) + 5445.7 / (\pi \cdot 1090.5^2 \cdot 25.0) \\
 &= 64.84 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t) \\
 &= 29.466 \cdot 1090.5 / (2 \cdot 25.0) - 7524.5 / (0.1066 \cdot \pi \cdot 1090.5^2 \cdot 25.0) \\
 &= 71.68 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 29.466 \cdot 1090.5 / (2 \cdot 25.0) + 7524.5 / (0.1923 \cdot \pi \cdot 1090.5^2 \cdot 25.0) \\
 &= 60.16 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L-2a) / (L + (4 \cdot hm/3)) \\
 &= 24788(6475.0 - 2 \cdot 1295.0) / (6475.0 + (4 \cdot 552.5/3)) \\
 &= 13353.8 \text{ Kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 1.1707 \cdot 13353.81 / (1090.5 \cdot 25.0) \\
 &= 5.62 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(1090.5 \cdot 25.0)} \\
 &= 128.789 \text{ mm.}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\
 &= \min(230.0 + 1.56 \cdot \sqrt{(1090.5 \cdot 25.0)}, 2 \cdot 1295.0) \\
 &= 487.58 \text{ mm.}
 \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$\begin{aligned}
 &= \min(Sr/S, 1) \\
 &= \min(137.997/137.997, 1) \\
 &= 1.0000
 \end{aligned}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\
 &= -0.7603 \cdot 24788 \cdot 0.1 / (487.577(25.0 + 1.0 \cdot 28.0)) \\
 &= -0.72 \text{ N./mm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [σ_7, r_{*}]:

$$\begin{aligned}
 &= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2) \\
 &= -24788 / (4(25.0 + 1.0 \cdot 28.0)487.577) - \\
 &\quad 12 \cdot 0.053 \cdot 24788 \cdot 1090.5 / (6475.0(25.0 + 1.0 \cdot 28.0)^2) \\
 &= -11.60 \text{ N./mm}^2
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2070.0000	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	254.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm.
Web Thickness	Webtk	10.0000	mm.
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	246.0	N./mm ²
Height of Web at Center	Hw,c	301.0	mm.

Friction Coefficient mu 0.300

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1078.0 \cdot 25.0) + 510.0})^3) / 12$$

$$= 99.752 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 99.752 + 191.524 (129.403 - 12.5)^2$$

$$= 26274.078 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	766.1	25.0	12.5	191.5	239405.2	0.263E+05
Wearplate	510.0	28.0	39.0	142.8	556920.0	0.118E+05
Web	10.0	603.0	354.5	60.3	2137634.2	0.488E+05
BasePlate	254.0	16.0	664.0	40.6	2698495.2	0.116E+06
Totals	...	...	...	435.3	5632454.5	0.203E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 5632455 / 435.264$$

$$= 129.403 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 24788.471$$

$$= 5044.9912 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 2.0298 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 147.6000 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1750.0 - 1078.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 858.501 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 4331.2168 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 2.7073 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 164.0000 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(24788 + 632) \cdot 254.0) / (4 \cdot 2070.0 \cdot 164.0)}$$

$$= 11.828 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1075.0 + 28.0 + 28.0)$$

$$= 1958.950 \text{ mm.}$$



Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nribs} - 1)$
 $= 1958.9496 / (5 - 1)$
 $= 489.737 \text{ mm.}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 489.7374 \cdot 254.0 / 2$
 $= 621.966 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 25420.754 / (2070.0 \cdot 254.0)$
 $= 4.835 \text{ Kg/cm}^2$

Axial Load [P]:
 $= \text{Ap} \cdot \text{Bp}$
 $= 622.0 \cdot 4.83$
 $= 3007.124 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 22.0 + 24.487$
 $= 46.487 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / \text{Ar}$
 $= 3007.1 / 46.4869$
 $= 6.344 \text{ N/mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	230.0	...	23.0	...	0.101E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 230.0 - 10.0$
 $= 220.000 \text{ mm.}$

Distance to Centroid from Datum [ytot]:
 $= \text{AY} / \text{A}$
 $= 0.0 / 46.487$
 $= 0.000 \text{ mm.}$

Distance to Centroid [C1]:
 $= \text{Saddle Width} / 2$
 $= 230.0 / 2$
 $= 115.000 \text{ mm.}$

Radius of Gyration [r]:
 $= \sqrt{(\text{Total Inertia} / \text{Total Area})}$
 $= \sqrt{(1013.9 / 46.487)}$
 $= 46.702 \text{ mm.}$

Length of Outer Rib [L]:
 $= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 $= 1750.0 - \cos(120.0/2) (1075.0 + 28.0 + 28.0) - 16.0$
 $= 1168.500 \text{ mm.}$

Intermediate Term [Cc]:
 $= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 $= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 246.0)}$



$$= 126.667$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1168.5/46.702$$

$$= 25.020$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 3024.2 / (2 \cdot 2070.0) \cdot 489.737 \cdot 1168.5 / 2$$

$$= 209.014 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($25.0203 < 126.6669$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (25.02)^2 / (2 \cdot 126.67^2)) 246 /$$

$$(5/3 + 3 \cdot (25.02) / (8 \cdot 126.67) - (25.02^3) / (8 \cdot 126.67^3))$$

$$= 138.6 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 6.34/138.64 + (209.01 \cdot 115.0/10139168) / 164.0$$

$$= 0.188$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	220.0	0.0	22.0	0.0	0.101E+04
Web	489.7	10.0	0.0	49.0	0.0	4.08
Totals	...	...	...	71.0	...	0.102E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/70.974$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 230.0/2$$

$$= 115.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1750.0 - \cos(1131.0/2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 603.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{1017.9/70.974}$$

$$= 37.871 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 603.0/37.871$$

$$= 15.922$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 3024.159 / (2 \cdot 2070.0)$$

$$= 0.730 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.73 \cdot 489.737 \cdot 603.0$$

$$= 215.722 \text{ Kg-m.}$$



Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 215.722/88514.297$$

$$= 23.900 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (15.9225 < 126.6669) per AISC E2-1 [Sca]:

$$= (1 - (K/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (K/r) / (8 \cdot C_c) - (K/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (15.92)^2 / (2 \cdot 126.67^2)) 246 /$$

$$(5/3 + 3 \cdot (15.92) / (8 \cdot 126.67) - (15.92^3) / (8 \cdot 126.67^3))$$

$$= 142.4 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 8.33/142.43 + (215.72 \cdot 115.0 / 1017.914) / 164.0$$

$$= 0.204$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm2
Bolt Corrosion Allowance	Bca	0.0 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	1.5500 cm2
BasePlate Allowable Stress	S	138.00 N./mm2
Area Available in a Single Bolt	BltArea	1.5500 cm2
Saddle Load Q0 (Weight)	Q0	21368.4 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	1362.2 Kg
Maximum Transverse Force	Ft	1512.1 Kg
Maximum Longitudinal Force	Fl	3024.2 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b, l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 3024.2 / (1.55 \cdot 8)$$

$$= 23.9 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b, t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1512.1 / (1.55 \cdot 8)$$

$$= 12.0 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q_0 > Q_L \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 3024.16 / (82.8 \cdot 8.0)$$

$$= 0.4477 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 1750.0 \cdot 1512.08 + 0.0$$

$$= 2646.19 \text{ Kg-m.}$$

Eccentricity (e):

$$= R_{mom} / Q_0$$

$$= 2646.19 / 21368.37$$

$$= 123.83 \text{ mm.} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0.0, 0.4477, 0.0]

= 0.4477 cm²

Note: Saddle and wear plate material are both specified as impact tested.

Review notes about nozzle loadings and supports in the Warnings report.

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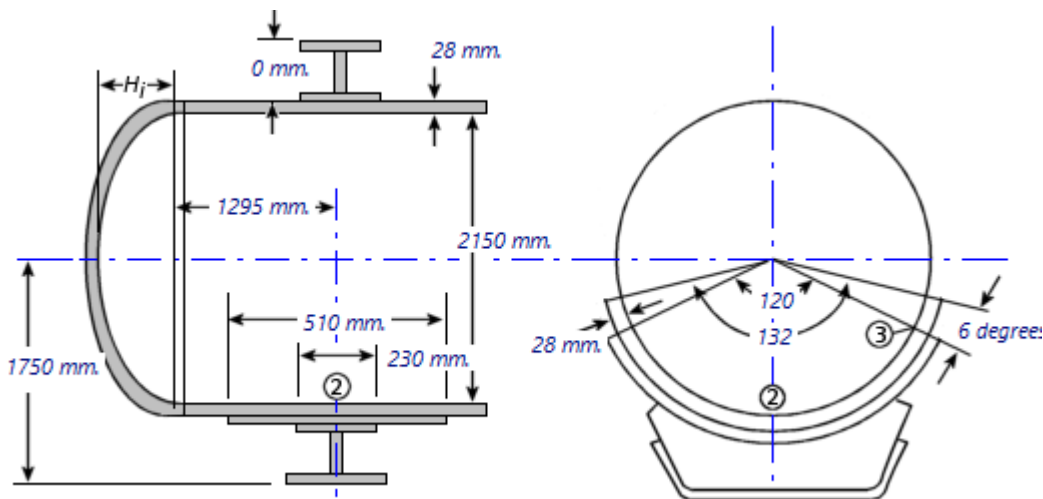


ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1089.00	mm.
Shell Thickness used in this Case	t	28.000	mm.
Stiffened Vessel Length per 4.15.6	L	6475.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	1295.00	mm.
Saddle Width	b1 or b	230.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	510.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	28.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	551.00	mm.
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1750.00	mm.
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	P	24173.43	Kgf
Pressure used in Analysis		38.734	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	74.83	234.00
Long. Stress at Bottom of Midspan	75.83	234.00
Long. Stress at Top of Saddles	81.80	234.00
Long. Stress at Bottom of Saddles	71.74	234.00
Tangential Shear in Shell	4.90	138.58
Circ. Stress at Horn of Saddle	15.41	350.99
Circ. Compressive Stress in Shell	0.81	234.00



Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (683.8/2 + 0) \cdot 1750.0/1958.9496$$

$$= 916.3 \text{ Kg}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(366.81, 0) \cdot 1750.0/3885.0$$

$$= 165.2 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 23257 + \max(165, 916, 0, 0)$$

$$= 24173.4 \text{ Kg}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08 (2.206 + \pi/4 (2.366 \cdot 1.2)^2)$$

$$= 7194.198 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	24805.72	Kgf
Transverse Shear Load	341.88	Kgf
Longitudinal Shear Load	366.81	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(2150.0/2 + 28.0 + 28.0) \sin(120.0/2), 2 \cdot 1089.0)$$

$$= 1958.950 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -24173 \cdot 1295.0 [1 - (1 - 1295.0/6475.0 + (1089.0^2 - 551.0^2) / (2 \cdot 1295.0 \cdot 6475.0)) / (1 + (4 \cdot 551.0)/(3 \cdot 6475.0))]$$

$$= -7333.8 \text{ Kg-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 24173 \cdot 6475/4 ((1 + 2(1089^2 - 551^2)/(6475^2)) / (1 + (4 \cdot 551)/(3 \cdot 6475)) - 4 \cdot 1295/6475)$$

$$= 5318.0 \text{ Kg-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 \cdot t)$$

$$= 38.734 \cdot 1089.0/(2 \cdot 28.0) - 5318.0/(\pi \cdot 1089.0^2 \cdot 28.0)$$

$$= 74.83 \text{ N/mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$$

$$= 38.734 \cdot 1089.0/(2 \cdot 28.0) + 5318.0/(\pi \cdot 1089.0^2 \cdot 28.0)$$

$$= 75.83 \text{ N/mm}^2$$



Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 38.734 \cdot 1089.0 / (2 \cdot 28.0) - 7333.8 / (0.1066 \cdot \pi \cdot 1089.0^2 \cdot 28.0)$$

$$= 81.80 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 38.734 \cdot 1089.0 / (2 \cdot 28.0) + 7333.8 / (0.1923 \cdot \pi \cdot 1089.0^2 \cdot 28.0)$$

$$= 71.74 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 24173(6475.0 - 2 \cdot 1295.0) / (6475.0 + (4 \cdot 551.0 / 3))$$

$$= 13026.1 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 13026.09 / (1089.0 \cdot 28.0)$$

$$= 4.90 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1089.0 \cdot 28.0)}$$

$$= 136.203 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(230.0 + 1.56 \cdot \sqrt{(1089.0 \cdot 28.0)}, 2 \cdot 1295.0)$$

$$= 502.41 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(S_r / S, 1)$$

$$= \min(137.997 / 233.996, 1)$$

$$= 0.5897$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 24173 \cdot 0.1 / (502.407(28.0 + 0.59 \cdot 28.0))$$

$$= -0.81 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -24173 / (4(28.0 + 0.59 \cdot 28.0)502.407) -$$

$$12 \cdot 0.053 \cdot 24173 \cdot 1089.0 / (6475.0(28.0 + 0.59 \cdot 28.0)^2)$$

$$= -15.41 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2070.0000	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	254.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm.
Web Thickness	Webtk	10.0000	mm.
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm2
Height of Web at Center	Hw,c	301.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1075.0 \cdot 28.0)} + 510.0)^3) / 12$$

$$= 142.807 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:



$$\begin{aligned}
 &= \text{ShellInertia} + A\text{Shell}(C1 - Y)^2 \\
 &= 142.807 + 218.582(124.852 - 14.0)^2 \\
 &= 27002.547 \text{ cm}^4
 \end{aligned}$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ² cm ⁴
Shell	780.6	28.0	14.0	218.6	306014.8	0.270E+05
Wearplate	510.0	28.0	42.0	142.8	599760.1	0.990E+04
Web	10.0	603.0	357.5	60.3	2155724.2	0.509E+05
BasePlate	254.0	16.0	667.0	40.6	2710687.2	0.119E+06
Totals	...	...	...	462.3	5772186.5	0.207E+06

Distance to Centroid [C1]:

$$\begin{aligned}
 &= AY / A \\
 &= 5772187/462.322 \\
 &= 124.852 \text{ mm.}
 \end{aligned}$$

Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 120.0/2 \\
 &= 120.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta)) \\
 &= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0)) \\
 &= 0.2035
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.204 \cdot 24173.434 \\
 &= 4919.8179 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh/As) = 1.9795 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2
 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 1750.0 - 1075.0 \cdot \sin(120.0/2) / 1.0472 \\
 &= 860.982 \text{ mm.}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 4235.9580 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 2.5023 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt{3(24173 + 632) \cdot 254.0 / (4 \cdot 2070.0 \cdot 173.33)} \\
 &= 11.365 \text{ mm.}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120.0/2) (1075.0 + 28.0 + 28.0) \\
 &= 1958.950 \text{ mm.}
 \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (\text{Nr ribs} - 1) \\
 &= 1958.9496 / (5 - 1) \\
 &= 489.737 \text{ mm.}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e \cdot \text{Bpwid} / 2 \\
 &= 489.7374 \cdot 254.0/2 \\
 &= 621.966 \text{ cm}^2
 \end{aligned}$$



Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 24805.717 / (2070.0 \cdot 254.0)$$

$$= 4.718 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 622.0 \cdot 4.72$$

$$= 2934.369 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 22.0 + 24.487$$

$$= 46.487 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 2934.4 / 46.4869$$

$$= 6.190 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	230.0	...	23.0	...	0.101E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 230.0 - 10.0$$

$$= 220.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 46.487$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 230.0 / 2$$

$$= 115.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1013.9 / 46.487)}$$

$$= 46.702 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1750.0 - \cos(120.0/2) (1075.0 + 28.0 + 28.0) - 16.0$$

$$= 1168.500 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1168.5 / 46.702$$

$$= 25.020$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 366.8 / (2 \cdot 2070.0) \cdot 489.737 \cdot 1168.5 / 2$$



= 25.352 Kg-m.

Compressive Allowable, $KL/r < C_c$ (25.0203 < 123.2106) per AISC E2-1 [Sca]:

$$= (1 - (K/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (K/r) / (8 \cdot C_c) - (K/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (25.02)^2 / (2 \cdot 123.21^2)) 260 / (5/3 + 3 \cdot (25.02) / (8 \cdot 123.21) - (25.02^3) / (8 \cdot 123.21^3))$$

$$= 146.2 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 6.19 / 146.19 + (25.35 \cdot 115.0 / 10139168) / 173.33$$

$$= 0.059$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	220.0	0.0	22.0	0.0	0.101E+04
Web	489.7	10.0	0.0	49.0	0.0	4.08
Totals	...	...	...	71.0	...	0.102E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 70.974$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 230.0 / 2$$

$$= 115.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{phk}$$

$$= 1750.0 - \cos(1131.0 / 2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 603.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (1017.9 / 70.974) }$$

$$= 37.871 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 603.0 / 37.871$$

$$= 15.922$$

Unit Force [Force,u]:

$$= F / (2 \cdot \text{Baseplate Length})$$

$$= 366.807 / (2 \cdot 2070.0)$$

$$= 0.089 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.089 \cdot 489.737 \cdot 603.0$$

$$= 26.165 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 26.165 / 88514.297$$

$$= 2.899 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (15.9225 < 123.2106) per AISC E2-1 [Sca]:

$$= (1 - (K/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (K/r) / (8 \cdot C_c) - (K/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (15.92)^2 / (2 \cdot 123.21^2)) 260 / (5/3 + 3 \cdot (15.92) / (8 \cdot 123.21) - (15.92^3) / (8 \cdot 123.21^3))$$

$$= 150.3 \text{ N./mm}^2$$

**AISC Unity Check of Inside Ribs (must be ≤ 1)**

$$= Sc/Sca + (Mbase,c \cdot C1/I)/Sba$$

$$= 8.13/150.35 + (26.17 \cdot 115.0/1017.914)/173.33$$

$$= 0.071$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm2
Bolt Corrosion Allowance	Bca	0.0 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	1.5500 cm2
BasePlate Allowable Stress	S	138.00 N./mm2
Area Available in a Single Bolt	BltArea	1.5500 cm2
Saddle Load Q0 (Weight)	Q0	23889.5 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	165.2 Kgf
Maximum Transverse Force	Ft	341.9 Kgf
Maximum Longitudinal Force	F1	366.8 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 366.8 / (1.55 \cdot 8)$$

$$= 2.9 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 341.9 / (1.55 \cdot 8)$$

$$= 2.7 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 366.81 / (82.8 \cdot 8.0)$$

$$= 0.0543 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1750.0 \cdot 341.88 + 0.0$$

$$= 598.31 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 598.31 / 23889.46$$

$$= 25.04 \text{ mm.} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0.0543, 0.0]$$

$$= 0.0543 \text{ cm}^2$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:**Input and Calculated Values:**

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Saddle Calcs: Test Case:

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Vessel Mean Radius	Rm	1089.00	mm.
Shell Thickness used in this Case	t	28.000	mm.
Stiffened Vessel Length per 4.15.6	L	6475.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	1295.00	mm.
Saddle Width	b1 or b	230.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	510.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	28.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm2
Inside or Mean Depth of Head	Hi or hm	551.00	mm.
Shell Allowable Stress used in Calculation		234.00	N./mm2
Head Allowable Stress used in Calculation		234.00	N./mm2
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1750.00	mm.
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		25862.48	Kgf
Pressure used in Analysis	P	38.734	bars

Horizontal Vessel Analysis Results:	Actual N./mm^2	Allowable N./mm^2
Long. Stress at Top of Midspan	74.79	234.00
Long. Stress at Bottom of Midspan	75.86	234.00
Long. Stress at Top of Saddles	82.25	234.00
Long. Stress at Bottom of Saddles	71.49	234.00
Tangential Shear in Shell	5.25	138.58
Circ. Stress at Horn of Saddle	16.49	350.99
Circ. Compressive Stress in Shell	0.86	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (683.8/2 + 0) \cdot 1750.0/1958.9496$
 $= 916.3 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(366.81, 0) \cdot 1750.0/3885.0$
 $= 165.2 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 24946 + \max(165, 916, 0, 0)$
 $= 25862.5 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 \cdot OD \cdot \text{WindDiaMult})^2)$
 $= 1.1 \cdot 766.08 (2.206 + \pi/4 (2.366 \cdot 1.2)^2)$
 $= 7194.198 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	26494.76	Kgf
Transverse Shear Load	341.88	Kgf



Longitudinal Shear Load 366.81 Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(2150.0/2 + 28.0 + 28.0)\sin(120.0/2), 2 \cdot 1089.0)$$

$$= 1958.950 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -25862 \cdot 1295.0 [1 - (1 - 1295.0/6475.0 + (1089.0^2 - 551.0^2)/(2 \cdot 1295.0 \cdot 6475.0))/(1 + (4 \cdot 551.0)/(3 \cdot 6475.0))]$$

$$= -7846.3 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 \{ (1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L \}$$

$$= 25862 \cdot 6475.0 / 4 \{ (1 + 2(1089^2 - 551^2)/(6475^2)) / (1 + (4 \cdot 551)/(3 \cdot 6475)) - 4 \cdot 1295/6475 \}$$

$$= 5689.6 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 38.734 \cdot 1089.0/(2 \cdot 28.0) - 5689.6/(\pi \cdot 1089.0^2 \cdot 28.0)$$

$$= 74.79 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 38.734 \cdot 1089.0/(2 \cdot 28.0) + 5689.6/(\pi \cdot 1089.0^2 \cdot 28.0)$$

$$= 75.86 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 38.734 \cdot 1089.0/(2 \cdot 28.0) - 7846.3/(0.1066 \cdot \pi \cdot 1089.0^2 \cdot 28.0)$$

$$= 82.25 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 38.734 \cdot 1089.0/(2 \cdot 28.0) + 7846.3/(0.1923 \cdot \pi \cdot 1089.0^2 \cdot 28.0)$$

$$= 71.49 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 25862(6475.0 - 2 \cdot 1295.0)/(6475.0 + (4 \cdot 551.0/3))$$

$$= 13936.3 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 13936.25 / (1089.0 \cdot 28.0)$$

$$= 5.25 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1089.0 \cdot 28.0)}$$

$$= 136.203 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

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$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(230.0 + 1.56 \cdot \sqrt{(1089.0 \cdot 28.0)}, 2 \cdot 1295.0)$$

$$= 502.41 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(Sr/S, 1)$$

$$= \min(137.997/233.996, 1)$$

$$= 0.5897$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 25862 \cdot 0.1 / (502.407(28.0 + 0.59 \cdot 28.0))$$

$$= -0.86 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -25862 / (4(28.0 + 0.59 \cdot 28.0)502.407) -$$

$$12 \cdot 0.053 \cdot 25862 \cdot 1089.0 / (6475.0(28.0 + 0.59 \cdot 28.0)^2)$$

$$= -16.49 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2070.0000	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	254.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm.
Web Thickness	Webtk	10.0000	mm.
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	301.0	mm.
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1075.0 \cdot 28.0)} + 510.0) 28.0^3) / 12$$

$$= 142.807 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 142.807 + 218.582(124.852 - 14.0)^2$$

$$= 27002.547 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ² cm ⁴
Shell	780.6	28.0	14.0	218.6	306014.8	0.270E+05
Wearplate	510.0	28.0	42.0	142.8	599760.1	0.990E+04
Web	10.0	603.0	357.5	60.3	2155724.2	0.509E+05
BasePlate	254.0	16.0	667.0	40.6	2710687.2	0.119E+06
Totals	...	...	...	462.3	5772186.5	0.207E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 5772187/462.322$$

$$= 124.852 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$



$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 25862.48$$

$$= 5263.5757 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) =$$

$$2.1178 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} =$$

$$155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1750.0 - 1075.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 860.982 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d =$$

$$4531.9341 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) =$$

$$2.6771 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} =$$

$$173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(25862 + 632) \cdot 254.0) / (4 \cdot 2070.0 \cdot 173.33)}$$

$$= 11.745 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1075.0 + 28.0 + 28.0)$$

$$= 1958.950 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1958.9496 / (5 - 1)$$

$$= 489.737 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 489.7374 \cdot 254.0 / 2$$

$$= 621.966 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 26494.766 / (2070.0 \cdot 254.0)$$

$$= 5.039 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 622.0 \cdot 5.04$$

$$= 3134.173 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 22.0 + 24.487$$

$$= 46.487 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 3134.2 / 46.4869$$

$$= 6.612 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

B

D

Y

A

AY

Io

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Rib+Web	10.0	230.0	...	23.0	...	0.101E+04
---------	------	-------	-----	------	-----	-----------

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 230.0 - 10.0
 = 220.000 mm.

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0.0/46.487
 = 0.000 mm.

Distance to Centroid [C1]:

= Saddle Width / 2
 = 230.0/2
 = 115.000 mm.

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{1013.9/46.487}$
 = 46.702 mm.

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1750.0 - cos(120.0/2)(1075.0 + 28.0 + 28.0) - 16.0
 = 1168.500 mm.

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09/260.0}$
 = 123.211

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 1168.5/46.702
 = 25.020

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 366.8/ (2 • 2070.0) • 489.737 • 1168.5/2
 = 25.352 Kg-m.

Compressive Allowable, KL/r < Cc (25.0203 < 123.2106) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (25.02)^2 / (2 \cdot 123.21^2)) 260 / (5/3 + 3 \cdot (25.02) / (8 \cdot 123.21) - (25.02^3) / (8 \cdot 123.21^3))$
 = 146.2 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I)/Sba
 = 6.61/146.19 + (25.35 • 115.0/10139168)/173.33
 = 0.061

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	220.0	0.0	22.0	0.0	0.101E+04
Web	489.7	10.0	0.0	49.0	0.0	4.08
Totals	...	...	...	71.0	...	0.102E+04

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0.0/70.974
 = 0.000 mm.



Distance to Centroid [C1]:

= Saddle Width / 2

= 230.0/2

= 115.000 mm.

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= 1750.0 - cos(1131.0/2)(16.0 + 0.0 + 0.0) - 0.0

= 603.000 mm.

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{1017.9/70.974}$

= 37.871 mm.

Slenderness ratio [KL/r]:

= KL/r

= 1 • 603.0/37.871

= 15.922

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 366.807/(2 • 2070.0)

= 0.089 Kg/mm.

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 0.089 • 489.737 • 603.0

= 26.165 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 26.165/88514.297

= 2.899 N./mm²

Compressive Allowable, KL/r < Cc (15.9225 < 123.2106) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (15.92)^2 / (2 \cdot 123.21^2)) 260 /$

$(5/3 + 3 \cdot (15.92) / (8 \cdot 123.21) - (15.92^3) / (8 \cdot 123.21^3))$

= 150.3 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 8.68/150.35 + (26.17 • 115.0/1017.914)/173.33

= 0.074

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm2
Bolt Corrosion Allowance	Bca	0.0 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	1.5500 cm2
BasePlate Allowable Stress	S	138.00 N./mm2
Area Available in a Single Bolt	BltArea	1.5500 cm2
Saddle Load Q0 (Weight)	Q0	25578.5 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	165.2 Kg
Maximum Transverse Force	Ft	341.9 Kg
Maximum Longitudinal Force	F1	366.8 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,I]:

= FI / (Bolt Area • Number of Bolts)

= 366.8/(1.55 • 8)



= 2.9 N./mm². Must be less than 82.8 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $341.9 / (1.55 \cdot 8)$

= 2.7 N./mm². Must be less than 82.8 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F_t / (\text{BoltShearAllowable} \cdot \text{Nbols})$

= $366.81 / (82.8 \cdot 8.0)$

= 0.0543 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot F_t + \text{Sum of X Moments}$

= $1750.0 \cdot 341.88 + 0.0$

= 598.31 Kg-m.

Eccentricity (e):

= $Rmom / QO$

= $598.31 / 25578.51$

= 23.39 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$

= $\max[0.0, 0.0543, 0.0]$

= 0.0543 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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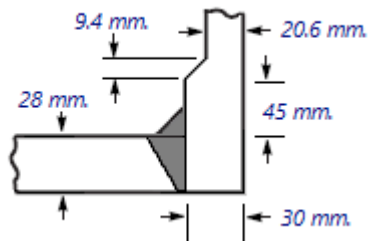
Input, Nozzle Desc: K13in**From: 20**

Pressure for Reinforcement Calculations	P	29.447	bars
Temperature for Internal Pressure	Temp	65	°C
Design External Pressure	Pext	0.15	bars
Temperature for External Pressure	Tempex	0	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2150.00	mm.
Design Length of Section	L	6833.3330	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Cylinder/Cone Centerline	L1	325.0000	mm.
Distance from Bottom/Left Tangent		1200.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		72.86	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm.
Flange Material		SA-105	+IT
Flange Type		N/A	
Hub Height of Integral Nozzle	h	45.0000	mm.
Height of Beveled Transition	L'	9.4260	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm.
Style of FVC Nozzle		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	471.5520	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K13in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.000 in.
Actual Thickness Used in Calculation	0.810 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 29.45 \cdot 1078.0 / (138 \cdot 1.0 - 0.6 \cdot 29.45)$
 $= 23.3029 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 29.45 \cdot 41.1 / (138 \cdot 1.0 - 0.6 \cdot 29.45)$
 $= 0.8885 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3089 mm.

Intermediate Hub Nozzle Calculations:

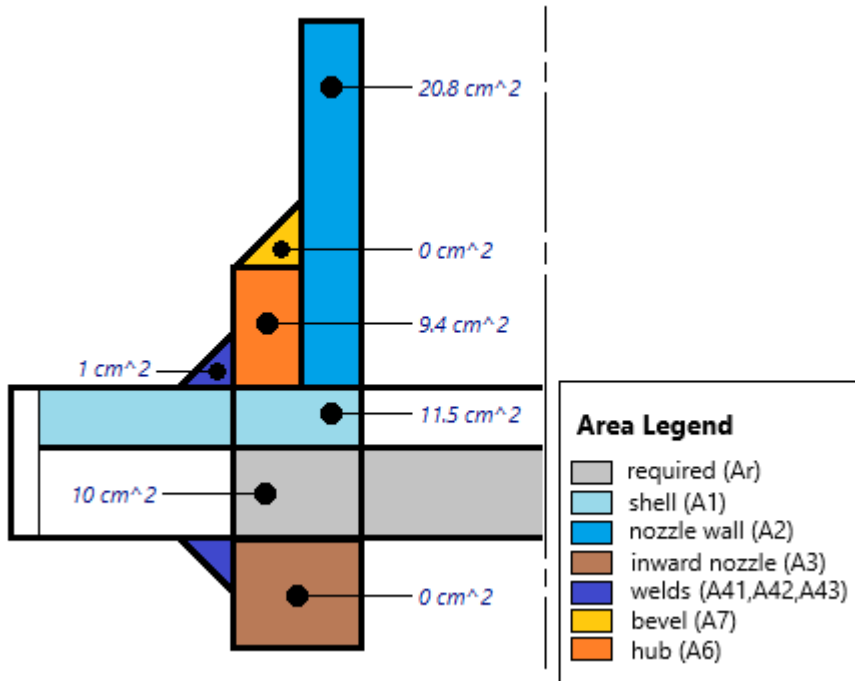
Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 9.426 / 0.5773$
 $= 16.3263 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	172.2974 mm.
Parallel to Vessel Wall, opening length	d	86.1487 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	60.2613 mm.



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	10.038	2.247	NA
Area in Shell	A1	11.500	17.042	NA
Area in Nozzle Wall	A2	20.811	21.534	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	9.372	9.372	NA
TOTAL AREA AVAILABLE	Atot	42.683	48.948	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 72.59 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (86.1487 \cdot 23.3029 \cdot 0.5 + 2 \cdot 27.0 \cdot 23.3029 \cdot 0.5 \cdot (1 - 1.0))$$

$$= 10.038 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 86.149 (1.0 \cdot 25.0 - 0.5 \cdot 23.303) - 2 \cdot 27.0$$

$$(1.0 \cdot 25.0 - 0.5 \cdot 23.3029) \cdot (1 - 1.0)$$

$$= 11.500 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2 / \sin(\alpha3)$$

$$= (2 \cdot 60.26) (17.57 - 0.89) 1.0 / \sin(75.1)$$

$$= 20.811 \text{ cm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.



Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i\text{-can}/0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{hp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(60.3, 471.6, 45.0)) \cdot (30.0 - 20.6) \cdot 1.0$$

$$= 9.372 \text{ cm}^2 \text{ Includes Bevel Area of } 0.888 \text{ cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Weld Size Calculations, Description: K13in

Intermediate Calc. for nozzle/shell Welds T_{min} 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (10.0376 - 11.4996 + 2 \cdot 27.0 \cdot 1.0 \cdot (1.0 \cdot 25.0 - 11.6515)) \cdot 138)$$

$$= 8085.78 \text{ Kgf}$$

For hub type nozzles, A₂ includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i\text{-Can}/0.707)^2 \cdot fr_2) \cdot S_v$$

$$= (30.1829 + 1.0 - 0.0 \cdot 1.0) \cdot 138$$

$$= 43879.11 \text{ Kgf}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$$

$$= (30.1829 + 0.0 + 1.0 + (13.5)) \cdot 138$$

$$= 62875.66 \text{ Kgf}$$

Weld Load [W3]:

$$= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$$

$$= (30.1829 + 0.0 + 1.0 + 0.0 + (13.5)) \cdot 138$$

$$= 62875.66 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [S_{onw}]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 142.7428 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 15460. \text{ Kgf}$$

Shear, Nozzle Wall [S_{nw}]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 57.2229) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 47810. \text{ Kgf}$$

Tension, Shell Groove Weld [T_{ngw}]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 142.7428 \cdot (24.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 56035. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (15460 + 47810) = 63270 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + T_{pgw} + T_{ngw} + Sinw)$$

$$= (15460 + 0 + 56035 + 0) = 71495 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + T_{ngw} + Sinw)$$

$$= (15460 + 56035 + 0) = 71495 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 63270	Kgf, must exceed W = 8085 Kgf	or W1 = 43879 Kgf
Path 2-2 = 71494	Kgf, must exceed W = 8085 Kgf	or W2 = 62875 Kgf
Path 3-3 = 71494	Kgf, must exceed W = 8085 Kgf	or W3 = 62875 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 31 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 bars

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K13in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.000 in.
Actual Thickness Used in Calculation	0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 29.45 \cdot 1078.0 / (138 \cdot 1.0 - 0.6 \cdot 29.45)$$

$$= 23.3029 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 29.45 \cdot 41.1 / (138 \cdot 1.0 - 0.6 \cdot 29.45)$$

$$= 0.8885 \text{ mm.}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.3089 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 9.426 / 0.5773$$

$$= 16.3263 \text{ mm.}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 167.3480 mm.

147-D

Ammonia Letdown Drum

QUEP

PV Elite 28 U1 Licensee: SL User

FileName : 147-D_Ammonia Letdown Drum_7%

Nozzle Calcs.: K1_3in

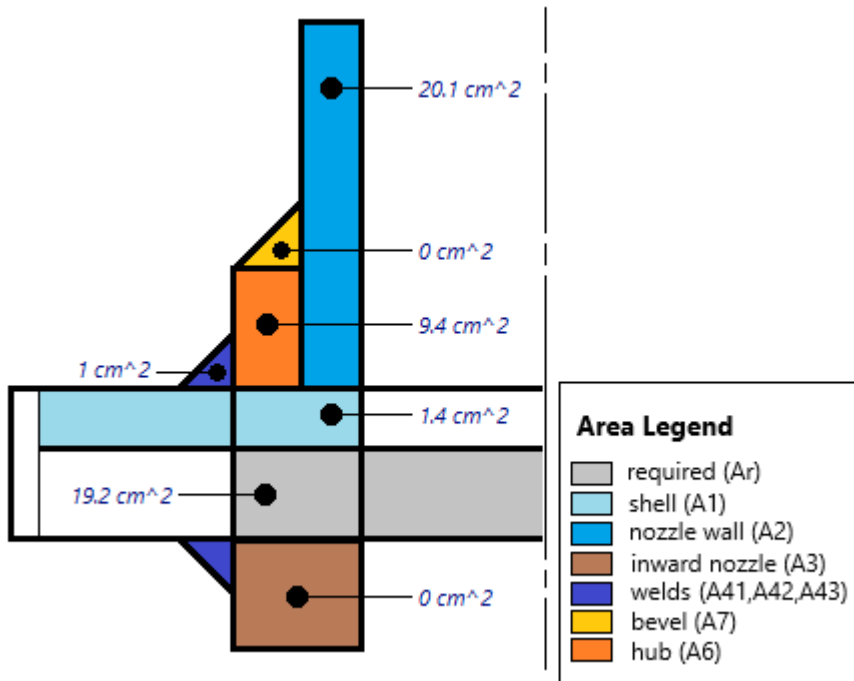
Noz1: 27 11:57am Sep 16,2026

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HEXAGON

Parallel to Vessel Wall Rn+tn+t 83.6740 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 60.2613 mm.



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	19.155	2.144	NA
Area in Shell	A1	1.445	16.844	NA
Area in Nozzle Wall	A2	20.110	20.808	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	9.372	9.372	NA
TOTAL AREA AVAILABLE	Atot	31.927	48.025	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (82.2 \cdot 23.3029 \cdot 1.0 + 2 \cdot 27.0 \cdot 23.3029 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 19.155 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 85.148(1.0 \cdot 25.0 - 1.0 \cdot 23.303) - 2 \cdot 27.0$$

$$(1.0 \cdot 25.0 - 1.0 \cdot 23.3029) \cdot (1 - 1.0)$$

$$= 1.445 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 60.26)(17.57 - 0.89)1.0$$

$$= 20.110 \text{ cm}^2$$



Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i\text{-can}/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot fr2$$

$$= (2 \cdot \min(60.3, 471.6, 45.0)) \cdot (30.0 - 20.6) \cdot 1.0$$

$$= 9.372 \text{ cm}^2 \text{ Includes Bevel Area of } 0.888 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.8885 \text{ mm.}$
Wall Thickness per UG16(b),	$tr16b = 4.5000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$trb1 = 26.3029 \text{ mm.}$
Wall Thickness	$tb1 = \max(trb1, tr16b) = 26.3029 \text{ mm.}$
Wall Thickness	$tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$
Wall Thickness per table UG-45	$tb3 = 8.7300 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.73, \max(26.3029, 4.5)]$$

$$= 8.7300 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.8885, 8.73)$$

$$= 8.7300 \text{ mm.}$$

Available Nozzle Neck Thickness = 20.5740 mm. -> OK

Weld Size Calculations, Description: K13in

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm.}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (19.155 - 1.445 + 2 \cdot 27.0 \cdot 1.0 \cdot (1.0 \cdot 25.0 - 23.3029)) \cdot 138)$$

$$= 26210.18 \text{ Kg}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i\text{-Can}/0.707)^2 \cdot fr2) \cdot S_v$$

$$= (29.4818 + 1. - 0.0 \cdot 1.0) \cdot 138$$

$$= 42892.46 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S_v$$

$$= (29.4818 + 0.0 + 1. + (13.5)) \cdot 138$$

$$= 61889.00 \text{ Kg}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (29.4818 + 0.0 + 1. + 0.0 + (13.5)) \cdot 138$$

$$= 61889.00 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 136.2 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 14751. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 54.6) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 45619. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 136.2 \cdot (24.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 53466. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (14751 + 45619) = 60370 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (14751 + 0 + 53466 + 0) = 68218 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (14751 + 53466 + 0) = 68218 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 60370	Kgf, must exceed W = 26210 Kg	or W1 = 42892 Kg
Path 2-2 = 68217	Kgf, must exceed W = 26210 Kg	or W2 = 61889 Kg
Path 3-3 = 68217	Kgf, must exceed W = 26210 Kg	or W3 = 61889 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 31 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 bars

The Drop for this Nozzle is : 24.1465 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 525.0358 mm.

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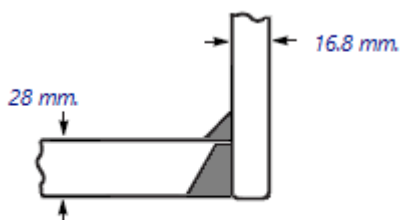
**Input, Nozzle Desc: K72in****From: 20**

Pressure for Reinforcement Calculations	P	29.400	bars
Temperature for Internal Pressure	Temp	65	°C
Design External Pressure	Pext	0.15	bars
Temperature for External Pressure	Tempex	0	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2150.00	mm.
Design Length of Section	L	6833.3330	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1500.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	477.6480	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K72in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 29.4 \cdot 1078.0 / (138 \cdot 1.0 - 0.6 \cdot 29.4)$
 $= 23.2653 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 29.4 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 29.4)$
 $= 0.6129 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.2565 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	134.4000 mm.
Parallel to Vessel Wall	Rn+tn+t	67.2000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000 mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K72in.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.6129 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 26.2653 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 26.2653 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 7.8000 mm.

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(26.2653, 4.5)]$
 $= 7.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(3.6129, 7.8)$
 $= 7.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: A

Govrn. thk, tg = 16.8, tr = 0.613, c = 3.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.044$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -42 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 29.40/51.00 = 0.576

Weld Size Calculations, Description: K72in

Intermediate Calc. for nozzle/shell Welds Tmin 13.8000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.8286 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 506.4767 mm.

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 67.20 mm.

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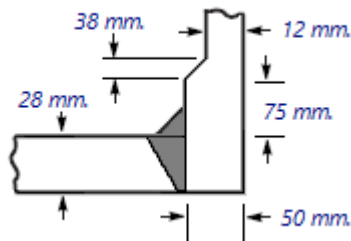
**Input, Nozzle Desc: N110in****From: 20**

Pressure for Reinforcement Calculations	P	29.400	bars
Temperature for Internal Pressure	Temp	65	°C
Design External Pressure	Pext	0.15	bars
Temperature for External Pressure	Tempex	0	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2150.00	mm.
Design Length of Section	L	6833.3330	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		550.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	+IT
Hub Height of Integral Nozzle	h	75.0000	mm.
Height of Beveled Transition	L'	38.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	449.2000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N110in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	10.000 in.
Actual Thickness Used in Calculation	0.472 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 29.4 \cdot 1078.0 / (138 \cdot 1.0 - 0.6 \cdot 29.4)$
 $= 23.2653 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 29.4 \cdot 127.0 / (138 \cdot 1.0 + 0.4 \cdot 29.4)$
 $= 2.6830 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4764 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 38.0 / 0.5773$
 $= 65.8180 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

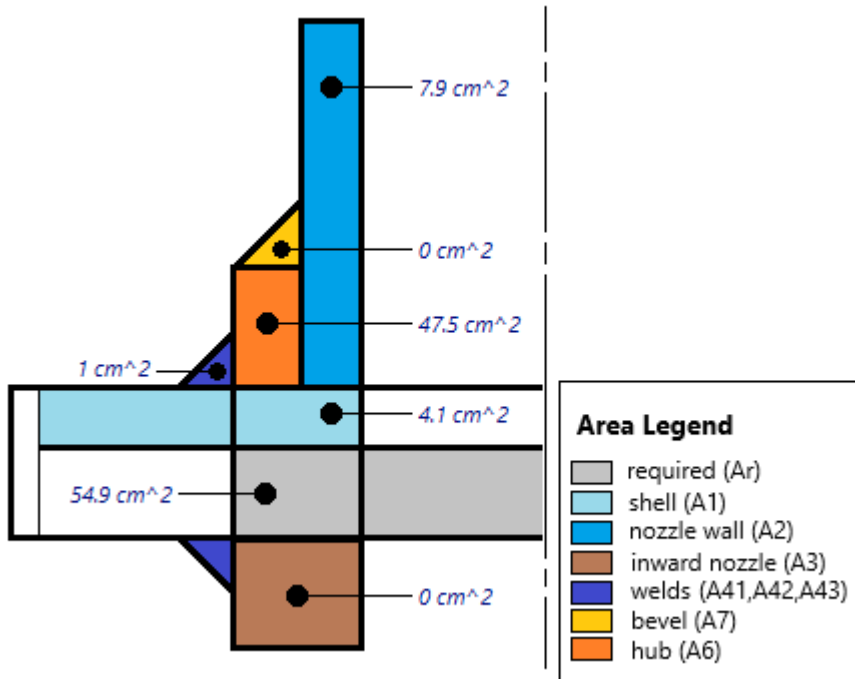
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	472.0000 mm.
Parallel to Vessel Wall, opening length	d	236.0000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	62.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	54.906	6.157	NA
Area in Shell	A1	4.094	46.687	NA
Area in Nozzle Wall	A2	7.896	10.655	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	47.500	47.500	NA
TOTAL AREA AVAILABLE	Atot	60.490	105.841	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (236.0 \cdot 23.2653 \cdot 1.0 + 2 \cdot 47.0 \cdot 23.2653 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 54.906 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 236.0(1.0 \cdot 25.0 - 1.0 \cdot 23.265) - 2 \cdot 47.0$$

$$(1.0 \cdot 25.0 - 1.0 \cdot 23.2653) \cdot (1 - 1.0)$$

$$= 4.094 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 62.5)(9.0 - 2.68)1.0$$

$$= 7.896 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

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$$\begin{aligned}
 &= W_o^2 \cdot fr2 + (W_i - can/0.707)^2 \cdot fr2 \\
 &= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 \\
 &= 1.000 \text{ cm}^2
 \end{aligned}$$

Area Available in the Hub Section [A6]:

$$\begin{aligned}
 &= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2 \\
 &= (2 \cdot \min(62.5, 449.2, 75.0)) \cdot (50.0 - 12.0) \cdot 1.0 \\
 &= 47.500 \text{ cm}^2
 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.6830 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 26.2653 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 26.2653 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 11.1026 mm.

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned}
 &= \min[tb3, \max(tb1, tb2)] \\
 &= \min[11.103, \max(26.2653, 4.5)] \\
 &= 11.1026 \text{ mm.}
 \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned}
 &= \max(ta, tb) \\
 &= \max(5.683, 11.1026) \\
 &= 11.1026 \text{ mm.}
 \end{aligned}$$

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 51.5,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 293.5 N./mm ²	Passed
Occasional	: 18.6,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 21.2,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-18 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-42 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 29.40/51.00 = 0.576$$

Weld Size Calculations, Description: N110in

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

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Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (54.9062 - 4.0938 + 2 \cdot 47.0 \cdot 1.0 \cdot (1.0 \cdot 25.0 - 23.2653)) \cdot 138)$$

$$= 73795.20 \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (55.3963 + 1. - 0.0 \cdot 1.0) \cdot 138$$

$$= 79358.09 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (55.3963 + 0.0 + 1. + (23.5)) \cdot 138$$

$$= 112426.16 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (55.3963 + 0.0 + 1. + 0.0 + (23.5)) \cdot 138$$

$$= 112426.16 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 254.0 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 27510. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 122.5) \cdot (50.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 178165. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnv} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 254.0 \cdot (24.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 99710. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (27510 + 178165) = 205675 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (27510 + 0 + 99710 + 0) = 127220 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (27510 + 99710 + 0) = 127220 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 205675 Kgf, must exceed W = 73795 Kgf	or W1 = 79358 Kgf
Path 2-2 = 127219 Kgf, must exceed W = 73795 Kgf	or W2 = 112426 Kgf
Path 3-3 = 127219 Kgf, must exceed W = 73795 Kgf	or W3 = 112426 Kgf

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Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 31 bars

Nozzle is O.K. for the External Pressure: 0 bars

The Drop for this Nozzle is : 7.5282 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 484.7282 mm.

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 236.00 mm.

Input Echo, WRC297 Item 1, Description: N110in :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	3.0000	mm.	
Shell External Corrosion Allowance	0.0000	mm.	
Shell Diameter	2150.000	mm.	
Shell Thickness	28.0000	mm.	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD		
Nozzle Corrosion Allowance	3.0000	mm.	
Nozzle Diameter	330.000	mm.	
Nozzle Thickness	50.0000	mm.	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-266 2		
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:*External Forces and Moments in WRC 107/537 Convention:**These loads are assumed to be SUStained loads.*

Design Internal Pressure	Dp	29.40	bars
Radial Load	P	1911.36	Kgf
Circumferential Shear	Vc	1274.24	Kgf
Longitudinal Shear	Vl	1274.24	Kgf
Circumferential Moment	Mc	892.01	Kg-m.
Longitudinal Moment	Ml	892.01	Kg-m.
Torsional Moment	Mt	1274.31	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (6.600) is < 10
check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 330.0 + 2 \cdot 1.65 \cdot \sqrt{1090.5(28.0 - 3.0)}$$

$$= 874.875 \text{ mm.}$$

WRC 297 Curve Access Parameters:

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Vessel Mean Diameter (D) = 2181.000 mm.
 Nozzle Outside Diameter (d) = 330.000 mm.
 Vessel Thickness used (T) = 25.000 mm.
 Nozzle Thickness used (t) = 47.000 mm.
 T / t = 0.532
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.413$$

Nr/P = 0.086
 Mr/P = 0.177
 M0/P = 0.052
 N0/P = 0.115
 MrD/Mc = 0.297
 NrDL/Mc = 0.100
 M0d/Mc = 0.093
 N0DL/Mc = 0.055
 MrD/M1 = 0.152
 NrDL/M1 = 0.080
 M0D/M1 = 0.050
 N0DL/M1 = 0.132

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-3	-3	-3	-3
Outplane Bending (P)	-9	9	-9	9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-12	12	12	-12
Normal Pressure Stress	125	128	125	128
Outplane Stress Summary	94	141	130	127

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-31	31	-31	31
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-38	38	38	-38
Inplane Pressure Stress	62	62	62	62
Inplane Stress Summary	-13	127	70	56

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Nozzle Calcs.: N1_10in

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Two * Max Shear Stress	108	142	130	127
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	-3	-3	-3	-3
Outplane Bending (P)	-9	9	-9	9
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-23	23	23	-23
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	62	62	62	62

Outplane Stress Summary	23	89	75	47
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-31	31	-31	31
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-75	75	75	-75
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	125	128	125	128

Inplane Stress Summary	10	229	170	86
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	24	229	170	86
------------------------	----	-----	-----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	-3	-3	-3	-3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-5	-5	5	5
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	10	7	10

Hoop Stress Summary	-1	1	9	13
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Nozzle Stresses (N./mm²):

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LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-6	6	-6	6
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-8	8	8	-8
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-15	14	7	5

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	2	2	1	1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	15	15	10	13

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-3	-3	-3	-3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-2	-2	2	2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	10	7	10
Hoop Stress Summary	1	5	6	9

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-6	6	-6	6
Axial Membrane (Mc)	-3	-3	3	3
Axial Bending (Mc)	-17	17	17	-17
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-24	24	17	-4

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

147-D

Ammonia LetDown Drum

QUEP

PV Elite 28 U1

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FileName : 147-D Ammonia Letdown Drum_7%

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Nozzle Calcs.: N1_10in

Nozl: 29 11:57am Sep 16,2026



HEXAGON

Shear Stress Summary 1 1 2 2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	26	24	17	14

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	125	128	125	128	125	128	125	128
Circ. Pl (SUS)	-8	-8	2	2	-6	-6	2	2
Circ. Q (SUS)	-21	21	3	-3	-106	106	44	-44
Long. Pm (SUS)	62	62	62	62	62	62	62	62
Long. Pl (SUS)	-5	-5	1	1	-5	-5	-1	-1
Long. Q (SUS)	-69	69	7	-7	-32	32	14	-14
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	125.0	128.0	125.0	128.0	125.0	128.0	125.0	128.0
Pm+Pl (SUS)	117.0	120.0	127.0	130.0	119.0	122.0	127.0	130.0
Pm+Pl+Q (Total)	108.1	141.3	130.1	127.1	25.3	228.0	171.0	86.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	128.00	138.00	Passed
Pm+Pl (SUS)	130.00	207.00	Passed
Pm+Pl+Q (TOTAL)	228.03	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	7	10	7	10	7	10	7	10
Circ. Pl (SUS)	-8	-8	2	2	-5	-5	-1	-1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-3	-3	3	3	-3	-3	3	3
Long. Q (SUS)	-14	14	2	-2	-23	23	11	-11
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	7.0	10.0	7.0	10.0	7.0	10.0	7.0	10.0

Pm+Pl (SUS)	1.0	2.0	9.0	12.0	2.0	5.0	6.0	9.0
Pm+Pl+Q (Total)	14.1	14.1	9.6	12.1	25.1	23.1	17.1	14.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	10.00	138.00	Passed
Pm+Pl (SUS)	12.00	207.00	Passed
Pm+Pl+Q (TOTAL)	25.08	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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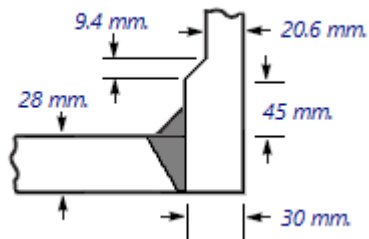
**Input, Nozzle Desc: K3-3in****From: 20**

Pressure for Reinforcement Calculations	P	29.447	bars
Temperature for Internal Pressure	Temp	65	°C
Design External Pressure	Pext	0.15	bars
Temperature for External Pressure	Tempex	0	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2150.00	mm.
Design Length of Section	L	6833.3330	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Cylinder/Cone Centerline	L1	325.0000	mm.
Distance from Bottom/Left Tangent		1750.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		72.86	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm.
Flange Material		SA-105	+IT
Hub Height of Integral Nozzle	h	45.0000	mm.
Height of Beveled Transition	L'	9.4260	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	471.5520	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K3-3in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.000 in.
Actual Thickness Used in Calculation	0.810 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 29.45 \cdot 1078.0 / (138 \cdot 1.0 - 0.6 \cdot 29.45)$
 $= 23.3029 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 29.45 \cdot 41.1 / (138 \cdot 1.0 - 0.6 \cdot 29.45)$
 $= 0.8885 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3089 mm.

Intermediate Hub Nozzle Calculations:

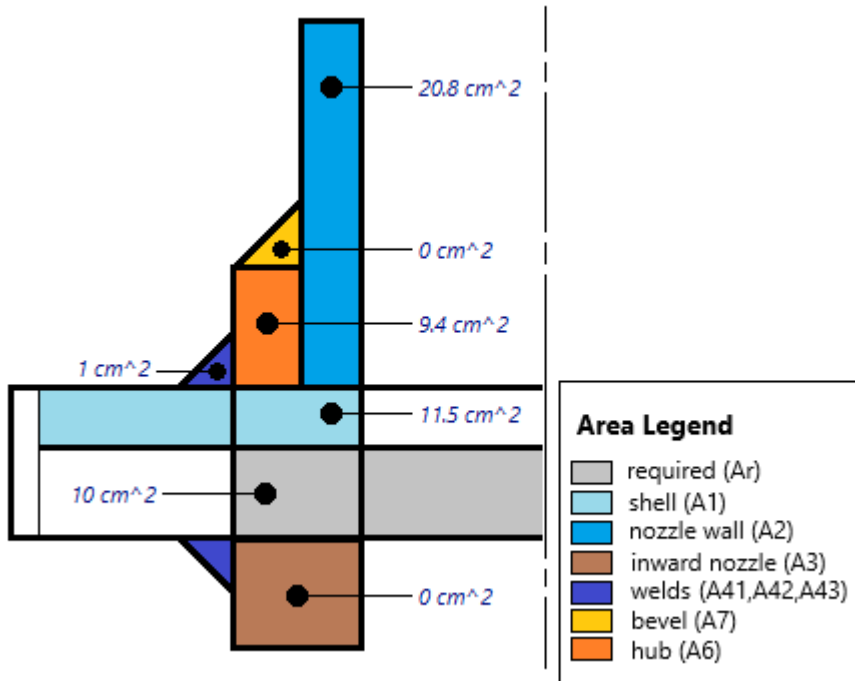
Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 9.426 / 0.5773$
 $= 16.3263 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	172.2974 mm.
Parallel to Vessel Wall, opening length	d	86.1487 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	60.2613 mm.



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	10.038	2.247	NA
Area in Shell	A1	11.500	17.042	NA
Area in Nozzle Wall	A2	20.811	21.534	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	9.372	9.372	NA
TOTAL AREA AVAILABLE	Atot	42.683	48.948	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

72.59 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (86.1487 \cdot 23.3029 \cdot 0.5 + 2 \cdot 27.0 \cdot 23.3029 \cdot 0.5 \cdot (1 - 1.0))$$

$$= 10.038 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 86.149 (1.0 \cdot 25.0 - 0.5 \cdot 23.303) - 2 \cdot 27.0$$

$$(1.0 \cdot 25.0 - 0.5 \cdot 23.3029) \cdot (1 - 1.0)$$

$$= 11.500 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2 / \sin(\alpha3)$$

$$= (2 \cdot 60.26) (17.57 - 0.89) 1.0 / \sin(75.1)$$

$$= 20.811 \text{ cm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.



Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i\text{-can}/0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(60.3, 471.6, 45.0)) \cdot (30.0 - 20.6) \cdot 1.0$$

$$= 9.372 \text{ cm}^2 \text{ Includes Bevel Area of } 0.888 \text{ cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Weld Size Calculations, Description: K3-3in

Intermediate Calc. for nozzle/shell Welds T_{min} 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W_o mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (10.0376 - 11.4996 + 2 \cdot 27.0 \cdot 1.0 \cdot (1.0 \cdot 25.0 - 11.6515)) \cdot 138)$$

$$= 8085.78 \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i\text{-Can}/0.707)^2 \cdot fr_2) \cdot S_v$$

$$= (30.1829 + 1. - 0.0 \cdot 1.0) \cdot 138$$

$$= 43879.11 \text{ Kgf}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$$

$$= (30.1829 + 0.0 + 1. + (13.5)) \cdot 138$$

$$= 62875.66 \text{ Kgf}$$

Weld Load [W3]:

$$= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$$

$$= (30.1829 + 0.0 + 1. + 0.0 + (13.5)) \cdot 138$$

$$= 62875.66 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 142.7428 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 15460. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 57.2229) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 47810. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:



$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\
 &= (3.1416/2.0) \cdot 142.7428 \cdot (24.0 - 0.0) \cdot 0.74 \cdot 138 \\
 &= 56035. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (15460 + 47810) = 63270 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + T_{pgw} + T_{ngw} + Sinw) \\
 &= (15460 + 0 + 56035 + 0) = 71495 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + T_{ngw} + Sinw) \\
 &= (15460 + 56035 + 0) = 71495 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 63270	Kgf, must exceed W = 8085 Kgf	or W1 = 43879 Kgf
Path 2-2 = 71494	Kgf, must exceed W = 8085 Kgf	or W2 = 62875 Kgf
Path 3-3 = 71494	Kgf, must exceed W = 8085 Kgf	or W3 = 62875 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 31 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 bars

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K3-3in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.000 in.
Actual Thickness Used in Calculation	0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$\begin{aligned}
 &= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= 29.45 \cdot 1078.0 / (138 \cdot 1.0 - 0.6 \cdot 29.45) \\
 &= 23.3029 \text{ mm.}
 \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned}
 &= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= 29.45 \cdot 41.1 / (138 \cdot 1.0 - 0.6 \cdot 29.45) \\
 &= 0.8885 \text{ mm.}
 \end{aligned}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.3089 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$\begin{aligned}
 &= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30) \\
 &= 9.426 / 0.5773 \\
 &= 16.3263 \text{ mm.}
 \end{aligned}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 167.3480 mm.

147-D

Ammonia Letdown Drum

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FileName : 147-D_Ammonia Letdown Drum_7%

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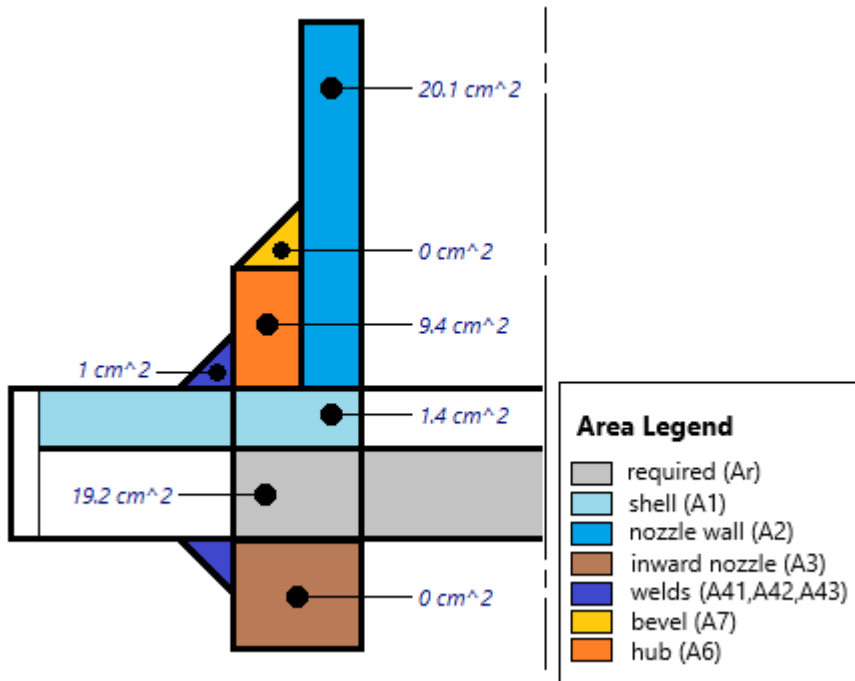
Nozzle Calcs.: K3-3in

Noz1: 30 11:57am Sep 16,2026



HEXAGON

Parallel to Vessel Wall Rn+tn+t 83.6740 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 60.2613 mm.



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	19.155	2.144	NA
Area in Shell	A1	1.445	16.844	NA
Area in Nozzle Wall	A2	20.110	20.808	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	9.372	9.372	NA
TOTAL AREA AVAILABLE	Atot	31.927	48.025	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (82.2 \cdot 23.3029 \cdot 1.0 + 2 \cdot 27.0 \cdot 23.3029 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 19.155 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 85.148(1.0 \cdot 25.0 - 1.0 \cdot 23.303) - 2 \cdot 27.0$$

$$(1.0 \cdot 25.0 - 1.0 \cdot 23.3029) \cdot (1 - 1.0)$$

$$= 1.445 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 60.26)(17.57 - 0.89)1.0$$

$$= 20.110 \text{ cm}^2$$



Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi\text{-}can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(60.3, 471.6, 45.0)) \cdot (30.0 - 20.6) \cdot 1.0$$

$$= 9.372 \text{ cm}^2 \text{ Includes Bevel Area of } 0.888 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.8885 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 26.3029 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 26.3029 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 8.7300 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.73, \max(26.3029, 4.5)]$$

$$= 8.7300 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.8885, 8.73)$$

$$= 8.7300 \text{ mm.}$$

Available Nozzle Neck Thickness = 20.5740 mm. -> OK

Weld Size Calculations, Description: K3-3in

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm.
---	------	-------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (19.155 - 1.445 + 2 \cdot 27.0 \cdot 1.0 \cdot (1.0 \cdot 25.0 - 23.3029)) \cdot 138)$$

$$= 26210.18 \text{ Kg}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(Wi\text{-}Can/.707)^2 \cdot fr2) \cdot Sv$$

$$= (29.4818 + 1. - 0.0 \cdot 1.0) \cdot 138$$

$$= 42892.46 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (29.4818 + 0.0 + 1. + (13.5)) \cdot 138$$

$$= 61889.00 \text{ Kg}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (29.4818 + 0.0 + 1. + 0.0 + (13.5)) \cdot 138$$

$$= 61889.00 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 136.2 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 14751. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 54.6) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 45619. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 136.2 \cdot (24.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 53466. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (14751 + 45619) = 60370 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (14751 + 0 + 53466 + 0) = 68218 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (14751 + 53466 + 0) = 68218 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 60370	Kgf, must exceed W = 26210 Kg	or W1 = 42892 Kg
Path 2-2 = 68217	Kgf, must exceed W = 26210 Kg	or W2 = 61889 Kg
Path 3-3 = 68217	Kgf, must exceed W = 26210 Kg	or W3 = 61889 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 31 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 bars

The Drop for this Nozzle is : 24.1465 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 525.0358 mm.

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Input, Nozzle Desc: K8-2in

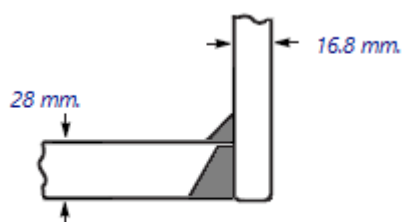
From: 20

Pressure for Reinforcement Calculations	P	29.400	bars
Temperature for Internal Pressure	Temp	65	°C
Design External Pressure	Pext	0.15	bars
Temperature for External Pressure	Tempex	0	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2150.00	mm.
Design Length of Section	L	6833.3330	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2000.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	477.6480	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.





Reinforcement CALCULATION, Description: K8-2in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 29.4 \cdot 1078.0 / (138 \cdot 1.0 - 0.6 \cdot 29.4)$
 $= 23.2653 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 29.4 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 29.4)$
 $= 0.6129 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.2565 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	134.4000 mm.
Parallel to Vessel Wall	Rn+tn+t	67.2000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K8-2in.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.6129 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 26.2653 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 26.2653 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 7.8000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(26.2653, 4.5)]$
 $= 7.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.6129, 7.8)$
 $= 7.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: A

Govrn. thk, tg = 16.8, tr = 0.613, c = 3.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.044$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -42 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 29.40/51.00 = 0.576

Weld Size Calculations, Description: K8-2in

Intermediate Calc. for nozzle/shell Welds Tmin 13.8000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

[Skipping the nozzle attachment weld strength calculations. Per UW-15\(b\)\(2\) the nozzles exempted by UG-36\(c\)\(3\)\(a\) \(small nozzles\) do not require a weld strength check.](#)

The Drop for this Nozzle is : 0.8286 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 506.4767 mm.

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 67.20 mm.

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Input, Nozzle Desc: K5-2in

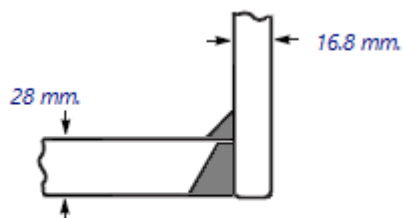
From: 20

Pressure for Reinforcement Calculations	P	29.422	bars
Temperature for Internal Pressure	Temp	65	°C
Design External Pressure	Pext	0.15	bars
Temperature for External Pressure	Tempex	0	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2150.00	mm.
Design Length of Section	L	6833.3330	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Cylinder/Cone Centerline	L1	735.0000	mm.
Distance from Bottom/Left Tangent		2300.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		48.21	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	477.6480	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K5-2in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 29.42 \cdot 1078.0 / (138 \cdot 1.0 - 0.6 \cdot 29.42)$
 $= 23.2831 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 29.42 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 29.42)$
 $= 0.6134 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.2565 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	154.5956 mm.
Parallel to Vessel Wall	Rn+tn+t	77.2978 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000 mm.

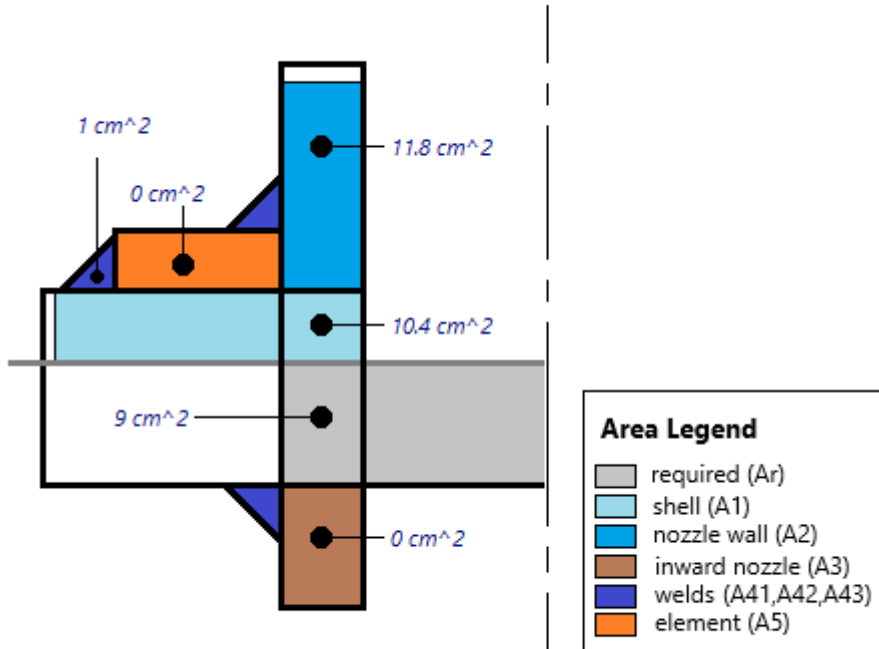


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	8.963	2.009	NA
Area in Shell	A1	10.366	15.351	NA
Area in Nozzle Wall	A2	11.849	12.170	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	23.216	28.521	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

47.54 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (76.9956 \cdot 23.2831 \cdot 0.5 + 2 \cdot 13.8 \cdot 23.2831 \cdot 0.5 \cdot (1 - 1.0)) \\
 &= 8.963 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 77.6(1.0 \cdot 25.0 - 0.5 \cdot 23.283) - 2 \cdot 13.8 \\
 &\quad (1.0 \cdot 25.0 - 0.5 \cdot 23.283) \cdot (1 - 1.0) \\
 &= 10.366 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot tnp)(tn - trn)fr2/\sin(\alpha3) \\
 &= (2 \cdot 34.5)(13.8 - 0.61)1.0/\sin(50.2) \\
 &= 11.849 \text{ cm}^2
 \end{aligned}$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.



Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: A

Govern. thk, $t_g = 16.8$, $t_r = 0.613$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.044$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	7 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-104 $^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 $^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 $^\circ\text{C}$
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-42 $^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $29.42/51.00 = 0.577$

Weld Size Calculations, Description: K5-2in

Intermediate Calc. for nozzle/shell Welds $T_{min} \quad 13.8000 \text{ mm.}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (8.9635 - 10.3662 + 2 \cdot 13.8 \cdot 1.0 \cdot (1.0 \cdot 25.0 - 11.6416)) \cdot 138)$$

$$= 3214.30 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_5 + A_4 - (W_i - Can / 0.707)^2 \cdot fr_2) \cdot S_v$$

$$= (11.8495 + 0.0 + 1. - 0.0 \cdot 1.0) \cdot 138$$

$$= 18081.18 \text{ Kg}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$$

$$= (11.8495 + 0.0 + 1. + (6.9)) \cdot 138$$

$$= 27790.52 \text{ Kg}$$

Weld Load [W3]:

$$= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$$

$$= (11.8495 + 0.0 + 1. + 0.0 + (6.9)) \cdot 138$$

$$= 27790.52 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 114.4089 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 12391. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 47.8511) \cdot (16.8 - 3.0) \cdot 0.7 \cdot 138$$

$$= 20434. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 114.4089 \cdot (24.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 44912. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (12391 + 20434) = 32826 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (12391 + 0 + 44912 + 0) = 57304 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (12391 + 44912 + 0) = 57304 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 32825	Kgf, must exceed W = 3214 Kgf	or W1 = 18081 Kgf
Path 2-2 = 57303	Kgf, must exceed W = 3214 Kgf	or W2 = 27790 Kgf
Path 3-3 = 57303	Kgf, must exceed W = 3214 Kgf	or W3 = 27790 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 31 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 bars

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K5-2in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (Sv \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 29.42 \cdot 1078.0 / (138 \cdot 1.0 - 0.6 \cdot 29.42)$$

$$= 23.2831 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (Sn \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 29.42 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 29.42)$$

$$= 0.6134 \text{ mm.}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.2565 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	134.4000 mm.
Parallel to Vessel Wall	Rn+tn+t	67.2000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000 mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K5-2in.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles



that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.6134 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 26.2831 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 26.2831 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 7.8000 mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
 = min[7.8, max(26.2831, 4.5)]
 = 7.8000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
 = max(3.6134, 7.8)
 = 7.8000 mm.

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Weld Size Calculations, Description: K5-2in

Intermediate Calc. for nozzle/shell Welds Tmin 13.8000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Nozzle is O.K. for the External Pressure: 0 bars

The Drop for this Nozzle is : 41.7866 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 557.3860 mm.

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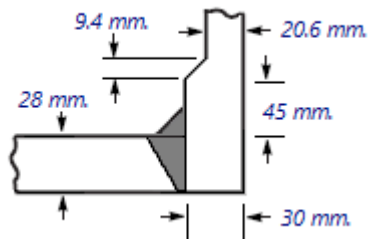
**Input, Nozzle Desc: K2-3in****From: 20**

Pressure for Reinforcement Calculations	P	29.484	bars
Temperature for Internal Pressure	Temp	65	°C
Design External Pressure	Pext	0.15	bars
Temperature for External Pressure	Tempex	0	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2150.00	mm.
Design Length of Section	L	6833.3330	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Cylinder/Cone Centerline	L1	290.0000	mm.
Distance from Bottom/Left Tangent		1200.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		105.24	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm.
Flange Material		SA-105	+IT
Hub Height of Integral Nozzle	h	45.0000	mm.
Height of Beveled Transition	L'	9.4260	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	221.5520	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K2-3in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.000 in.
Actual Thickness Used in Calculation	0.810 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 29.48 \cdot 1078.0 / (138 \cdot 1.0 - 0.6 \cdot 29.48)$
 $= 23.3326 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 29.48 \cdot 41.1 / (138 \cdot 1.0 - 0.6 \cdot 29.48)$
 $= 0.8896 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.2262 mm.

Intermediate Hub Nozzle Calculations:

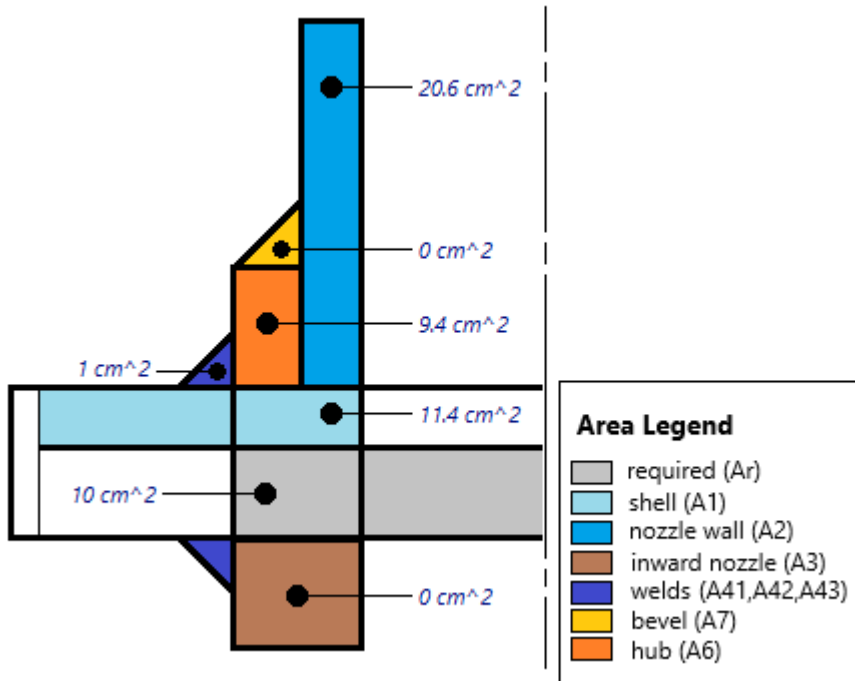
Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 9.426 / 0.5773$
 $= 16.3263 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	170.6044 mm.
Parallel to Vessel Wall, opening length	d	85.3022 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	60.2613 mm.



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	9.952	2.225	NA
Area in Shell	A1	11.374	16.875	NA
Area in Nozzle Wall	A2	20.641	21.462	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	9.372	9.372	NA
TOTAL AREA AVAILABLE	Atot	42.387	48.708	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

74.50 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (85.3022 \cdot 23.3326 \cdot 0.5 + 2 \cdot 27.0 \cdot 23.3326 \cdot 0.5 \cdot (1 - 1.0))$$

$$= 9.952 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 85.302 (1.0 \cdot 25.0 - 0.5 \cdot 23.333) - 2 \cdot 27.0$$

$$(1.0 \cdot 25.0 - 0.5 \cdot 23.3326) \cdot (1 - 1.0)$$

$$= 11.374 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2 / \sin(\alpha3)$$

$$= (2 \cdot 60.26) (17.57 - 0.89) 1.0 / \sin(77.0)$$

$$= 20.641 \text{ cm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.



Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(60.3, 221.6, 45.0)) \cdot (30.0 - 20.6) \cdot 1.0$$

$$= 9.372 \text{ cm}^2 \text{ Includes Bevel Area of } 0.888 \text{ cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -42 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 29.48/51.00 = 0.578$$

Weld Size Calculations, Description: K2-3in

Intermediate Calc. for nozzle/shell Welds T_{min} 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (9.9516 - 11.3739 + 2 \cdot 27.0 \cdot 1.0 \cdot (1.0 \cdot 25.0 - 11.6663)) \cdot 138)$$

$$= 8130.38 \text{ Kg}$$

For hub type nozzles, A₂ includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i - Can / 0.707)^2 \cdot fr_2) \cdot S_v$$

$$= (30.0128 + 1. - 0.0 \cdot 1.0) \cdot 138$$

$$= 43639.70 \text{ Kg}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$$

$$= (30.0128 + 0.0 + 1. + (13.5)) \cdot 138$$

$$= 62636.25 \text{ Kg}$$

Weld Load [W3]:

$$= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$$

$$= (30.0128 + 0.0 + 1. + 0.0 + (13.5)) \cdot 138$$

$$= 62636.25 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 141.3401 \cdot 10.0 \cdot 0.49 \cdot 138$$

= 15308. Kgf

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 56.6606) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 47341. Kgf$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 141.3401 \cdot (24.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 55484. Kgf$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (15308 + 47341) = 62649 Kgf$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (15308 + 0 + 55484 + 0) = 70792 Kgf$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (15308 + 55484 + 0) = 70792 Kgf$$

Summary of Failure Path Calculations:

Path 1-1 = 62648 Kgf, must exceed W = 8130 Kgf	or W1 = 43639 Kgf
Path 2-2 = 70792 Kgf, must exceed W = 8130 Kgf	or W2 = 62636 Kgf
Path 3-3 = 70792 Kgf, must exceed W = 8130 Kgf	or W3 = 62636 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 31 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 bars

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K2-3in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.000 in.
Actual Thickness Used in Calculation	0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (Sv \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 29.48 \cdot 1078.0 / (138 \cdot 1.0 - 0.6 \cdot 29.48)$$

$$= 23.3326 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (Sn \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 29.48 \cdot 41.1 / (138 \cdot 1.0 - 0.6 \cdot 29.48)$$

$$= 0.8896 \text{ mm.}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.2262 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:



$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

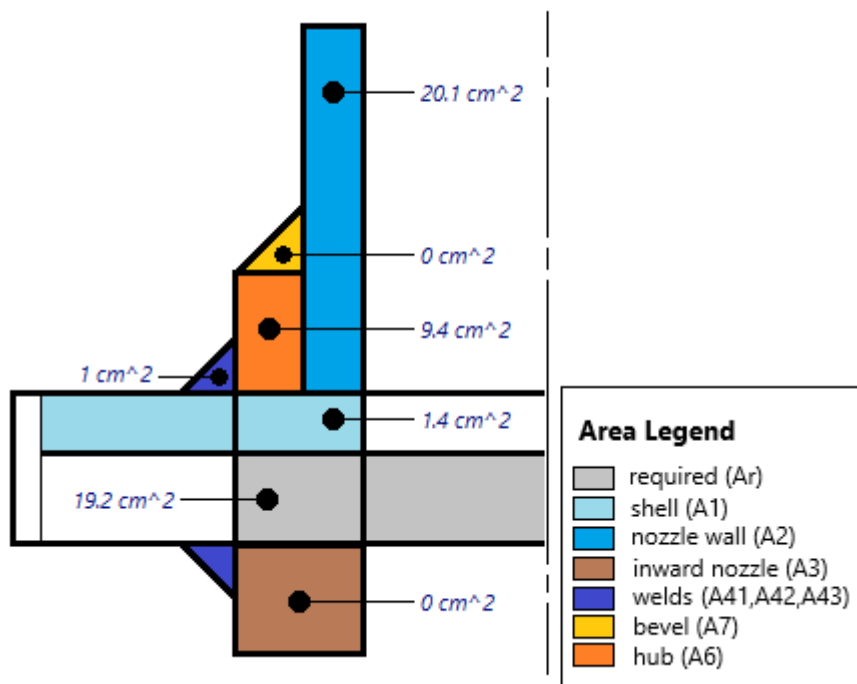
$$= 9.426 / 0.5773$$

$$= 16.3263 \text{ mm.}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	167.3480	mm.
Parallel to Vessel Wall	Rn+tn+t	83.6740	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	60.2613	mm.



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
Area Required	Ar	19.179	2.144
Area in Shell	A1	1.420	16.844
Area in Nozzle Wall	A2	20.108	20.908
Area in Inward Nozzle	A3	0.000	0.000
Area in Welds	A41+A42+A43	1.000	1.000
Area in Element	A5	0.000	0.000
Area in Hub	A6	9.372	9.372
TOTAL AREA AVAILABLE	Atot	31.900	48.124

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (82.2 \cdot 23.3326 \cdot 1.0 + 2 \cdot 27.0 \cdot 23.3326 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 19.179 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$



$$= 85.148(1.0 \cdot 25.0 - 1.0 \cdot 23.333) - 2 \cdot 27.0$$

$$(1.0 \cdot 25.0 - 1.0 \cdot 23.3326) \cdot (1 - 1.0)$$

$$= 1.420 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np})(t_n - t_r)fr_2$$

$$= (2 \cdot 60.26)(17.57 - 0.89)1.0$$

$$= 20.108 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(60.3, 221.6, 45.0)) \cdot (30.0 - 20.6) \cdot 1.0$$

$$= 9.372 \text{ cm}^2 \text{ Includes Bevel Area of } 0.888 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.8896 \text{ mm.}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 26.3326 \text{ mm.}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 26.3326 \text{ mm.}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 4.5000 \text{ mm.}$
Wall Thickness per table UG-45	$tb_3 = 8.7300 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[8.73, \max(26.3326, 4.5)]$$

$$= 8.7300 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.8896, 8.73)$$

$$= 8.7300 \text{ mm.}$$

Available Nozzle Neck Thickness = 20.5740 mm. -> OK

Weld Size Calculations, Description: K2-3in

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm.}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr))S_v)$$

$$= \max(0, (19.1794 - 1.4197 + 2 \cdot 27.0 \cdot 1.0 \cdot (1.0 \cdot 25.0 - 23.3326))138)$$

$$= 26257.58 \text{ Kg}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i - Can/0.707)^2 \cdot fr_2) \cdot S_v$$

$$= (29.4804 + 1. - 0.0 \cdot 1.0) \cdot 138$$

$$= 42890.54 \text{ Kg}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$$

$$= (29.4804 + 0.0 + 1. + (13.5)) \cdot 138$$

= 61887.09 Kgf

Weld Load [W3]:

= $(A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$

= $(29.4804 + 0.0 + 1. + 0.0 + (13.5)) \cdot 138$

= 61887.09 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

= $(\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$

= $(3.1416/2.0) \cdot 136.2 \cdot 10.0 \cdot 0.49 \cdot 138$

= 14751. Kgf

Shear, Nozzle Wall [Snw]:

= $(\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$

= $(3.1416 \cdot 54.6) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 138$

= 45619. Kgf

Tension, Shell Groove Weld [Tngw]:

= $(\pi/2) \cdot D_{lo} \cdot W_{g n v i} \cdot 0.74 \cdot S_{ng}$

= $(3.1416/2.0) \cdot 136.2 \cdot (24.0 - 0.0) \cdot 0.74 \cdot 138$

= 53466. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

= $(SONW + SNW) = (14751 + 45619) = 60370 \text{ Kgf}$

Failure Path [Path 2-2]:

= $(Sonw + Tpgw + Tngw + Sinw)$

= $(14751 + 0 + 53466 + 0) = 68218 \text{ Kgf}$

Failure Path [Path 3-3]:

= $(Sonw + Tngw + Sinw)$

= $(14751 + 53466 + 0) = 68218 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 60370 Kgf, must exceed W = 26257 Kgf or W1 = 42890 Kgf

Path 2-2 = 68217 Kgf, must exceed W = 26257 Kgf or W2 = 61887 Kgf

Path 3-3 = 68217 Kgf, must exceed W = 26257 Kgf or W3 = 61887 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 31 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 bars

The Drop for this Nozzle is : 21.5428 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 272.1439 mm.

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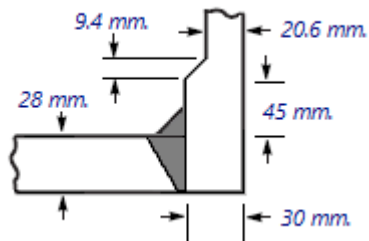
**Input, Nozzle Desc: K4-3in****From: 20**

Pressure for Reinforcement Calculations	P	29.484	bars
Temperature for Internal Pressure	Temp	65	°C
Design External Pressure	Pext	0.15	bars
Temperature for External Pressure	Tempex	0	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2150.00	mm.
Design Length of Section	L	6833.3330	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Cylinder/Cone Centerline	L1	290.0000	mm.
Distance from Bottom/Left Tangent		1750.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		105.24	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm.
Flange Material		SA-105	+IT
Hub Height of Integral Nozzle	h	45.0000	mm.
Height of Beveled Transition	L'	9.4260	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	221.5520	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K4-3in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.000 in.
Actual Thickness Used in Calculation	0.810 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 29.48 \cdot 1078.0 / (138 \cdot 1.0 - 0.6 \cdot 29.48)$
 $= 23.3326 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 29.48 \cdot 41.1 / (138 \cdot 1.0 - 0.6 \cdot 29.48)$
 $= 0.8896 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.2262 mm.

Intermediate Hub Nozzle Calculations:

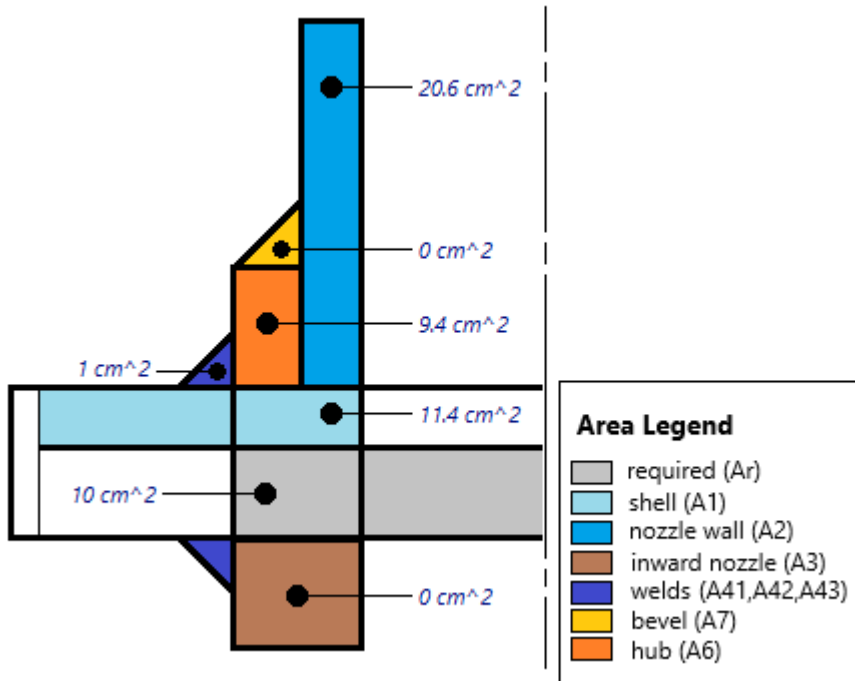
Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 9.426 / 0.5773$
 $= 16.3263 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	170.6044 mm.
Parallel to Vessel Wall, opening length	d	85.3022 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	60.2613 mm.



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	9.952	2.225	NA
Area in Shell	A1	11.374	16.875	NA
Area in Nozzle Wall	A2	20.641	21.462	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	9.372	9.372	NA
TOTAL AREA AVAILABLE	Atot	42.387	48.708	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

74.50 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (85.3022 \cdot 23.3326 \cdot 0.5 + 2 \cdot 27.0 \cdot 23.3326 \cdot 0.5 \cdot (1 - 1.0))$$

$$= 9.952 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 85.302 (1.0 \cdot 25.0 - 0.5 \cdot 23.333) - 2 \cdot 27.0$$

$$(1.0 \cdot 25.0 - 0.5 \cdot 23.3326) \cdot (1 - 1.0)$$

$$= 11.374 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp) (tn - trn) fr2 / \sin(\alpha3)$$

$$= (2 \cdot 60.26) (17.57 - 0.89) 1.0 / \sin(77.0)$$

$$= 20.641 \text{ cm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.



Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(60.3, 221.6, 45.0)) \cdot (30.0 - 20.6) \cdot 1.0$$

$$= 9.372 \text{ cm}^2 \text{ Includes Bevel Area of } 0.888 \text{ cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -42 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 29.48/51.00 = 0.578$$

Weld Size Calculations, Description: K4-3in

Intermediate Calc. for nozzle/shell Welds T_{min} 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (9.9516 - 11.3739 + 2 \cdot 27.0 \cdot 1.0 \cdot (1.0 \cdot 25.0 - 11.6663)) \cdot 138)$$

$$= 8130.38 \text{ Kg}$$

For hub type nozzles, A₂ includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i - Can / 0.707)^2 \cdot fr_2) \cdot S_v$$

$$= (30.0128 + 1. - 0.0 \cdot 1.0) \cdot 138$$

$$= 43639.70 \text{ Kg}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$$

$$= (30.0128 + 0.0 + 1. + (13.5)) \cdot 138$$

$$= 62636.25 \text{ Kg}$$

Weld Load [W3]:

$$= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$$

$$= (30.0128 + 0.0 + 1. + 0.0 + (13.5)) \cdot 138$$

$$= 62636.25 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 141.3401 \cdot 10.0 \cdot 0.49 \cdot 138$$

= 15308. Kgf

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 56.6606) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 47341. Kgf$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 141.3401 \cdot (24.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 55484. Kgf$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (15308 + 47341) = 62649 Kgf$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (15308 + 0 + 55484 + 0) = 70792 Kgf$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (15308 + 55484 + 0) = 70792 Kgf$$

Summary of Failure Path Calculations:

Path 1-1 = 62648 Kgf, must exceed W = 8130 Kgf	or W1 = 43639 Kgf
Path 2-2 = 70792 Kgf, must exceed W = 8130 Kgf	or W2 = 62636 Kgf
Path 3-3 = 70792 Kgf, must exceed W = 8130 Kgf	or W3 = 62636 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 31 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 bars

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K4-3in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.000 in.
Actual Thickness Used in Calculation	0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (Sv \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 29.48 \cdot 1078.0 / (138 \cdot 1.0 - 0.6 \cdot 29.48)$$

$$= 23.3326 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (Sn \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 29.48 \cdot 41.1 / (138 \cdot 1.0 - 0.6 \cdot 29.48)$$

$$= 0.8896 \text{ mm.}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.2262 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:



$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

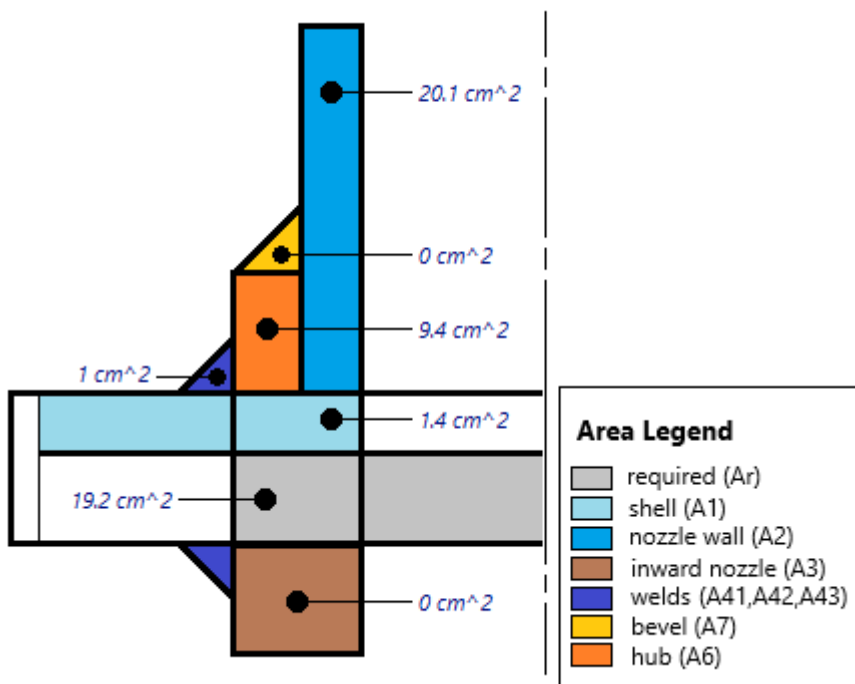
$$= 9.426 / 0.5773$$

$$= 16.3263 \text{ mm.}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	167.3480	mm.
Parallel to Vessel Wall	Rn+tn+t	83.6740	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	60.2613	mm.



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	19.179	2.144	NA
Area in Shell	A1	1.420	16.844	NA
Area in Nozzle Wall	A2	20.108	20.908	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	9.372	9.372	NA
TOTAL AREA AVAILABLE	Atot	31.900	48.124	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (82.2 \cdot 23.3326 \cdot 1.0 + 2 \cdot 27.0 \cdot 23.3326 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 19.179 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$



$$= 85.148(1.0 \cdot 25.0 - 1.0 \cdot 23.333) - 2 \cdot 27.0$$

$$(1.0 \cdot 25.0 - 1.0 \cdot 23.3326) \cdot (1 - 1.0)$$

$$= 1.420 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np})(t_n - t_{rn})fr_2$$

$$= (2 \cdot 60.26)(17.57 - 0.89)1.0$$

$$= 20.108 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(60.3, 221.6, 45.0)) \cdot (30.0 - 20.6) \cdot 1.0$$

$$= 9.372 \text{ cm}^2 \text{ Includes Bevel Area of } 0.888 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.8896 \text{ mm.}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 26.3326 \text{ mm.}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 26.3326 \text{ mm.}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 4.5000 \text{ mm.}$
Wall Thickness per table UG-45	$tb_3 = 8.7300 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[8.73, \max(26.3326, 4.5)]$$

$$= 8.7300 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.8896, 8.73)$$

$$= 8.7300 \text{ mm.}$$

Available Nozzle Neck Thickness = 20.5740 mm. -> OK

Weld Size Calculations, Description: K4-3in

Intermediate Calc. for nozzle/shell Welds	T_{min}	19.0000 mm.
---	-----------	-----------------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr))S_v)$$

$$= \max(0, (19.1794 - 1.4197 + 2 \cdot 27.0 \cdot 1.0 \cdot (1.0 \cdot 25.0 - 23.3326))138)$$

$$= 26257.58 \text{ Kg}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i - Can/0.707)^2 \cdot fr_2) \cdot S_v$$

$$= (29.4804 + 1. - 0.0 \cdot 1.0) \cdot 138$$

$$= 42890.54 \text{ Kg}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$$

$$= (29.4804 + 0.0 + 1. + (13.5)) \cdot 138$$



= 61887.09 Kgf

Weld Load [W3]:

= $(A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$

= $(29.4804 + 0.0 + 1. + 0.0 + (13.5)) \cdot 138$

= 61887.09 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

= $(\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$

= $(3.1416/2.0) \cdot 136.2 \cdot 10.0 \cdot 0.49 \cdot 138$

= 14751. Kgf

Shear, Nozzle Wall [Snw]:

= $(\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$

= $(3.1416 \cdot 54.6) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 138$

= 45619. Kgf

Tension, Shell Groove Weld [Tngw]:

= $(\pi/2) \cdot D_{lo} \cdot W_{gnv} \cdot 0.74 \cdot S_{ng}$

= $(3.1416/2.0) \cdot 136.2 \cdot (24.0 - 0.0) \cdot 0.74 \cdot 138$

= 53466. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

= $(SONW + SNW) = (14751 + 45619) = 60370 \text{ Kgf}$

Failure Path [Path 2-2]:

= $(Sonw + Tpgw + Tngw + Sinw)$

= $(14751 + 0 + 53466 + 0) = 68218 \text{ Kgf}$

Failure Path [Path 3-3]:

= $(Sonw + Tngw + Sinw)$

= $(14751 + 53466 + 0) = 68218 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 60370 Kgf, must exceed W = 26257 Kgf or W1 = 42890 Kgf

Path 2-2 = 68217 Kgf, must exceed W = 26257 Kgf or W2 = 61887 Kgf

Path 3-3 = 68217 Kgf, must exceed W = 26257 Kgf or W3 = 61887 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 31 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 bars

The Drop for this Nozzle is : 21.5428 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 272.1439 mm.

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Input, Nozzle Desc: K6-2in

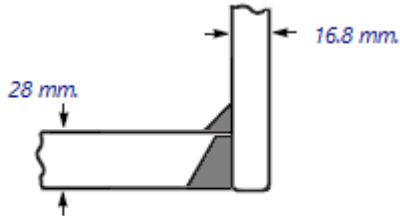
From: 20

Pressure for Reinforcement Calculations	P	29.484	bars
Temperature for Internal Pressure	Temp	65	°C
Design External Pressure	Pext	0.15	bars
Temperature for External Pressure	Tempex	0	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2150.00	mm.
Design Length of Section	L	6833.3330	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Cylinder/Cone Centerline	L1	290.0000	mm.
Distance from Bottom/Left Tangent		2300.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		105.24	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	227.6480	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K6-2in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 29.48 \cdot 1078.0 / (138 \cdot 1.0 - 0.6 \cdot 29.48)$
 $= 23.3326 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 29.48 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 29.48)$
 $= 0.6147 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.1881 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	136.5338 mm.
Parallel to Vessel Wall	Rn+tn+t	68.2669 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K6-2in.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: A

Govern. thk, tg = 16.8, tr = 0.615, c = 3.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.045$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-42 °C



Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 29.48/51.00 = 0.578

Weld Size Calculations, Description: K6-2in

Intermediate Calc. for nozzle/shell Welds Tmin 13.8000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K6-2in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 = $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $29.48 \cdot 1078.0 / (138 \cdot 1.0 - 0.6 \cdot 29.48)$
 = 23.3326 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 = $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $29.48 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 29.48)$
 = 0.6147 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.1881 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	134.4000 mm.
Parallel to Vessel Wall	Rn+tn+t	67.2000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K6-2in.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.6147 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 26.3326 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 26.3326 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 7.8000 mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
 = min[7.8, max(26.3326, 4.5)]
 = 7.8000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
 = max(3.6147, 7.8)
 = 7.8000 mm.

147-D

Ammonia LetDown Drum

QUEP

PV Elite 28 U1

Licensee: SL User

FileName : 147-D_Ammonia Letdown Drum_7%

Nozzle Calcs.: K6-2in

Nozl: 35 11:57am Sep 16,2026

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Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Weld Size Calculations, Description: K6-2in

Intermediate Calc. for nozzle/shell Welds Tmin 13.8000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

[Skipping the nozzle attachment weld strength calculations. Per UW-15\(b\)\(2\) the nozzles exempted by UG-36\(c\)\(3\)\(a\) \(small nozzles\) do not require a weld strength check.](#)

The Drop for this Nozzle is : 12.7614 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 269.4585 mm.

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Input, Nozzle Desc: D1-2in

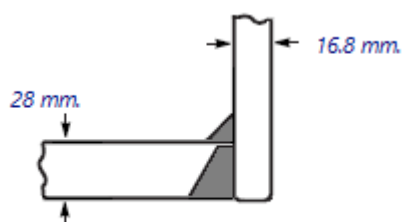
From: 20

Pressure for Reinforcement Calculations	P	29.533	bars
Temperature for Internal Pressure	Temp	65	°C
Design External Pressure	Pext	0.15	bars
Temperature for External Pressure	Tempex	0	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2150.00	mm.
Design Length of Section	L	6833.3330	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2050.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	227.6480	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.





Reinforcement CALCULATION, Description: D1-2in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 29.53 \cdot 1078.0 / (138 \cdot 1.0 - 0.6 \cdot 29.53)$
 $= 23.3719$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 29.53 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 29.53)$
 $= 0.6157$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.1881 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	134.4000 mm.
Parallel to Vessel Wall	Rn+tn+t	67.2000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: D1-2in.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.6157 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 26.3719 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 26.3719 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 7.8000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(26.3719, 4.5)]$
 $= 7.8000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.6157, 7.8)$
 $= 7.8000$ mm.

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: A

Govrn. thk, tg = 16.8, tr = 0.616, c = 3.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.045$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

147-D

Ammonia LetDown Drum

QUEP

PV Elite 28 U1

Licensee: SL User

FileName : 147-D_Ammonia Letdown Drum_7%

Nozzle Calcs.: D1-2in

Nozl: 36 11:57am Sep 16,2026

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Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -42 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 29.53/51.00 = 0.579

Weld Size Calculations, Description: D1-2in

Intermediate Calc. for nozzle/shell Welds Tmin 13.8000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

[Skipping the nozzle attachment weld strength calculations. Per UW-15\(b\)\(2\) the nozzles exempted by UG-36\(c\)\(3\)\(a\) \(small nozzles\) do not require a weld strength check.](#)

The Drop for this Nozzle is : 0.8286 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 256.4767 mm.

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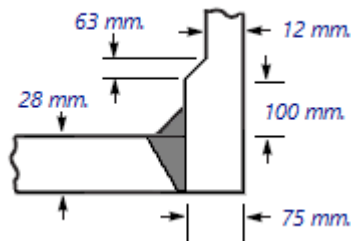
**Input, Nozzle Desc: N2-16in****From: 40**

Pressure for Reinforcement Calculations	P	29.400	bars
Temperature for Internal Pressure	Temp	65	°C
Design External Pressure	Pext	0.15	bars
Temperature for External Pressure	Tempex	0	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2150.00	mm.
Design Length of Section	L	6833.3330	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		4850.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	+IT
Hub Height of Integral Nozzle	h	100.0000	mm.
Height of Beveled Transition	L'	63.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		75.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	449.2000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N2-16in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	16.000 in.
Actual Thickness Used in Calculation	0.472 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 29.4 \cdot 1078.0 / (138 \cdot 1.0 - 0.6 \cdot 29.4)$
 $= 23.2653 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 29.4 \cdot 203.2 / (138 \cdot 1.0 + 0.4 \cdot 29.4)$
 $= 4.2928 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.6274 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 63.0 / 0.5773$
 $= 109.1193 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

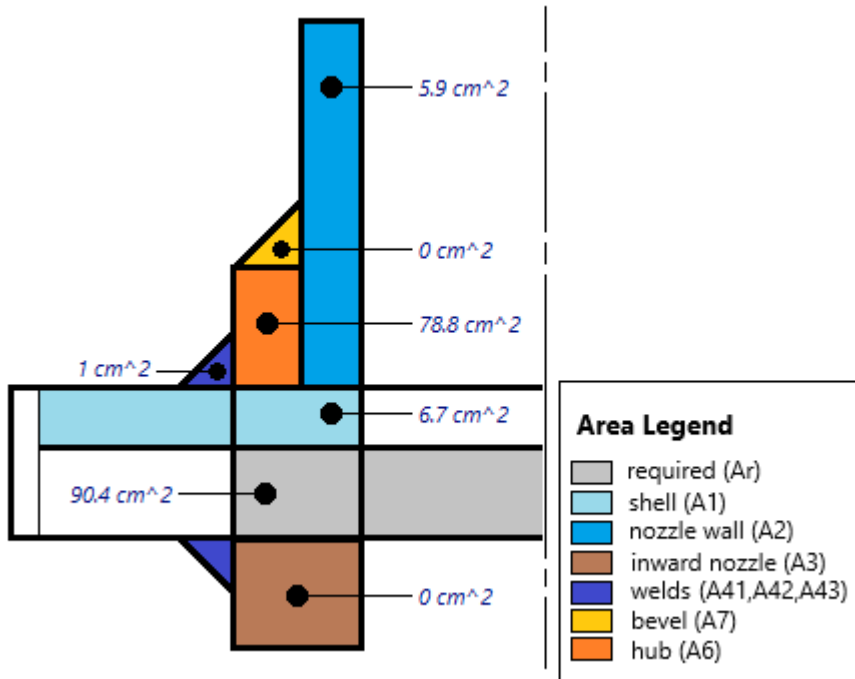
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	776.8000 mm.
Parallel to Vessel Wall, opening length	d	388.4000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	62.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	90.363	10.132	NA
Area in Shell	A1	6.737	76.835	NA
Area in Nozzle Wall	A2	5.884	10.466	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	78.750	78.750	NA
TOTAL AREA AVAILABLE	Atot	92.371	167.051	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (388.4 \cdot 23.2653 \cdot 1.0 + 2 \cdot 72.0 \cdot 23.2653 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 90.363 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 388.4(1.0 \cdot 25.0 - 1.0 \cdot 23.265) - 2 \cdot 72.0$$

$$(1.0 \cdot 25.0 - 1.0 \cdot 23.2653) \cdot (1 - 1.0)$$

$$= 6.737 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 62.5)(9.0 - 4.29)1.0$$

$$= 5.884 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

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$$\begin{aligned}
 &= W_o^2 \cdot fr2 + (W_i\text{-can}/0.707)^2 \cdot fr2 \\
 &= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 \\
 &= 1.000 \text{ cm}^2
 \end{aligned}$$

Area Available in the Hub Section [A6]:

$$\begin{aligned}
 &= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr2 \\
 &= (2 \cdot \min(62.5, 449.2, 100.0)) \cdot (75.0 - 12.0) \cdot 1.0 \\
 &= 78.750 \text{ cm}^2
 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 7.2928 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 26.2653 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 26.2653 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 11.3312 mm.

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned}
 &= \min[tb3, \max(tb1, tb2)] \\
 &= \min[11.331, \max(26.2653, 4.5)] \\
 &= 11.3312 \text{ mm.}
 \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned}
 &= \max(ta, tb) \\
 &= \max(7.2928, 11.3312) \\
 &= 11.3312 \text{ mm.}
 \end{aligned}$$

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 62.7,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 282.3 N./mm ²	Passed
Occasional	: 31.0,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 20.2,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-18 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-42 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 29.40/51.00 = 0.576$$

Weld Size Calculations, Description: N2-16in

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

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Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (90.3626 - 6.7374 + 2 \cdot 72.0 \cdot 1.0 \cdot (1.0 \cdot 25.0 - 23.2653)) \cdot 138)$$

$$= 121188.20 \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (84.634 + 1. - 0.0 \cdot 1.0) \cdot 138$$

$$= 120500.04 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (84.634 + 0.0 + 1. + (36.0)) \cdot 138$$

$$= 171157.52 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (84.634 + 0.0 + 1. + 0.0 + (36.0)) \cdot 138$$

$$= 171157.52 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 406.4 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 44016. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 198.7) \cdot (75.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 442710. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnv} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 406.4 \cdot (24.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 159536. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (44016 + 442710) = 486726 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (44016 + 0 + 159536 + 0) = 203552 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (44016 + 159536 + 0) = 203552 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 486725 Kgf, must exceed W = 121188 Kgf or W1 = 120500 Kgf
 Path 2-2 = 203551 Kgf, must exceed W = 121188 Kgf or W2 = 171157 Kgf
 Path 3-3 = 203551 Kgf, must exceed W = 121188 Kgf or W3 = 171157 Kgf

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Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 30 bars

Nozzle is O.K. for the External Pressure: 0 bars

The Drop for this Nozzle is : 19.3794 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 496.5795 mm.

Input Echo, WRC297 Item 1, Description: N2-16in :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2150.000	mm.
Shell Thickness	28.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	532.400	mm.
Nozzle Thickness	75.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:*External Forces and Moments in WRC 107/537 Convention:**These loads are assumed to be SUstained loads.*

Design Internal Pressure	Dp	29.40	bars
Radial Load	P	3058.18	Kgf
Circumferential Shear	Vc	2038.80	Kgf
Longitudinal Shear	Vl	2038.80	Kgf
Circumferential Moment	Mc	2283.54	Kg-m.
Longitudinal Moment	Ml	2283.54	Kg-m.
Torsional Moment	Mt	3262.22	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (7.099) is < 10
check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 532.4 + 2 \cdot 1.65 \cdot \sqrt{(1090.5(28.0 - 3.0))}$$

$$= 1077.275 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2181.000	mm.
Nozzle Outside Diameter	(d) =	532.400	mm.
Vessel Thickness used	(T) =	25.000	mm.
Nozzle Thickness used	(t) =	72.000	mm.

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T / t = 0.347
d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 2.280$$

Nr/P = 0.065
Mr/P = 0.130
M0/P = 0.039
N0/P = 0.083
MrD/Mc = 0.280
NrDL/Mc = 0.096
M0d/Mc = 0.086
N0DL/Mc = 0.079
MrD/Ml = 0.103
NrDL/Ml = 0.070
M0D/Ml = 0.031
N0DL/Ml = 0.105

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	-3	-3	-3	-3
Outplane Bending (P)	-11	11	-11	11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-7	-7	7	7
Outplane Bending (ML)	-12	12	12	-12
Normal Pressure Stress	125	128	125	128
Outplane Stress Summary	90	141	129	129

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	-3	-3	-3	-3
Inplane Bending (P)	-37	37	-37	37
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-41	41	41	-41
Inplane Pressure Stress	62	62	62	62
Inplane Stress Summary	-24	134	68	59

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	115	142	129	129

Vessel Stresses (N./mm²):

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CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	-3	-3	-3	-3
Outplane Bending (P)	-11	11	-11	11
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-34	34	34	-34
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	62	62	62	62
Outplane Stress Summary	7	99	87	40

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	-3	-3	-3	-3
Inplane Bending (P)	-37	37	-37	37
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-113	113	113	-113
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	125	128	125	128
Inplane Stress Summary	-34	269	204	55

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	42	269	204	56

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	-3	-3	-3	-3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-7	-7	7	7
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	8	11	8	11
Hoop Stress Summary	-2	0	11	14

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-3	3	-3	3
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	4	4	4	4

Axial Stress Summary	-5	8	5	5
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE
(Shear stress)

	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	1	1	0	0
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE
(Stress Intensities)

	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	5	8	11	14

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
(Stresses in the
hoop direction)

	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	-3	-3	-3	-3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-5	-5	5	5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	8	11	8	11

Hoop Stress Summary	-1	2	9	12
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
(Stresses Normal to
pipe cross-section)

	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-3	3	-3	3
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-11	11	11	-11
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4	4	4	4

Axial Stress Summary	-13	16	13	-2
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
(Shear stress)

	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	1	1
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Nozzle Stresses (N./mm²):

147-D

Ammonia LetDown Drum

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Nozzle Calcs.: N2-16in

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CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	13	16	14	15

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	125	128	125	128	125	128	125	128
Circ. Pl (SUS)	-10	-10	4	4	-9	-9	3	3
Circ. Q (SUS)	-23	23	1	-1	-150	150	76	-76
Long. Pm (SUS)	62	62	62	62	62	62	62	62
Long. Pl (SUS)	-7	-7	1	1	-8	-8	2	2
Long. Q (SUS)	-78	78	4	-4	-45	45	23	-23
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	125.0	128.0	125.0	128.0	125.0	128.0	125.0	128.0
Pm+Pl (SUS)	115.0	118.0	129.0	132.0	116.0	119.0	128.0	131.0
Pm+Pl+Q (Total)	115.1	141.5	130.1	131.1	43.2	269.0	204.0	55.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	128.00	138.00	Passed
Pm+Pl (SUS)	132.00	207.00	Passed
Pm+Pl+Q (TOTAL)	269.02	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	8	11	8	11	8	11	8	11
Circ. Pl (SUS)	-10	-10	4	4	-8	-8	2	2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	-6	6	0	0	-14	14	8	-8
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	8.0	11.0	8.0	11.0	8.0	11.0	8.0	11.0
Pm+Pl (SUS)	4.0	2.0	12.0	15.0	2.0	3.0	10.0	13.0
Pm+Pl+Q (Total)	4.0	8.0	12.0	15.0	12.0	16.0	14.0	15.0



Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	11.00	138.00	Passed
Pm+Pl (SUS)	15.00	207.00	Passed
Pm+Pl+Q (TOTAL)	16.00	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: N3-10in

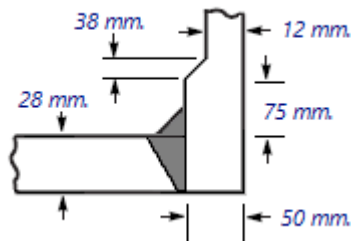
From: 40

Pressure for Reinforcement Calculations	P	29.533	bars
Temperature for Internal Pressure	Temp	65	°C
Design External Pressure	Pext	0.15	bars
Temperature for External Pressure	Tempex	0	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2150.00	mm.
Design Length of Section	L	6833.3330	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		4650.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	+IT
Hub Height of Integral Nozzle	h	75.0000	mm.
Height of Beveled Transition	L'	38.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	232.6520	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N3-10in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	10.000 in.
Actual Thickness Used in Calculation	0.472 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 29.53 \cdot 1078.0 / (138 \cdot 1.0 - 0.6 \cdot 29.53)$
 $= 23.3719 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 29.53 \cdot 127.0 / (138 \cdot 1.0 + 0.4 \cdot 29.53)$
 $= 2.6950 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3624 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 38.0 / 0.5773$
 $= 65.8180 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

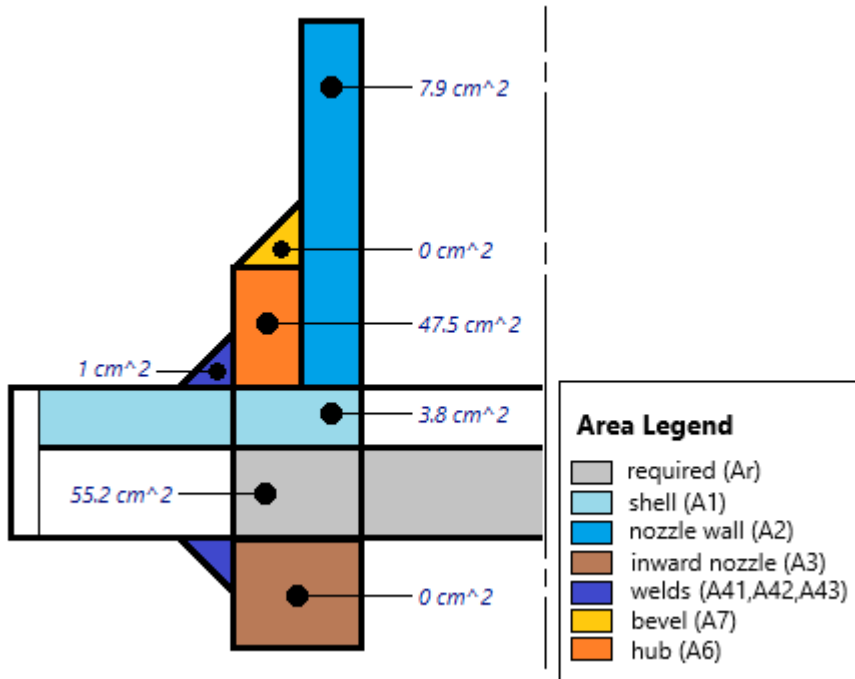
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	472.0000 mm.
Parallel to Vessel Wall, opening length	d	236.0000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	62.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	55.158	6.157	NA
Area in Shell	A1	3.842	46.687	NA
Area in Nozzle Wall	A2	7.881	10.797	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	47.500	47.500	NA
TOTAL AREA AVAILABLE	Atot	60.223	105.984	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (236.0 \cdot 23.3719 \cdot 1.0 + 2 \cdot 47.0 \cdot 23.3719 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 55.158 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 236.0(1.0 \cdot 25.0 - 1.0 \cdot 23.372) - 2 \cdot 47.0$$

$$(1.0 \cdot 25.0 - 1.0 \cdot 23.3719) \cdot (1 - 1.0)$$

$$= 3.842 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 62.5)(9.0 - 2.7)1.0$$

$$= 7.881 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

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Nozzle Calcs.: N3-10in

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$$\begin{aligned}
 &= W_o^2 \cdot fr2 + (W_i - can/0.707)^2 \cdot fr2 \\
 &= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 \\
 &= 1.000 \text{ cm}^2
 \end{aligned}$$

Area Available in the Hub Section [A6]:

$$\begin{aligned}
 &= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2 \\
 &= (2 \cdot \min(62.5, 232.7, 75.0)) \cdot (50.0 - 12.0) \cdot 1.0 \\
 &= 47.500 \text{ cm}^2
 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.6950 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 26.3719 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 26.3719 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 11.1026 mm.

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned}
 &= \min[tb3, \max(tb1, tb2)] \\
 &= \min[11.103, \max(26.3719, 4.5)] \\
 &= 11.1026 \text{ mm.}
 \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned}
 &= \max(ta, tb) \\
 &= \max(5.695, 11.1026) \\
 &= 11.1026 \text{ mm.}
 \end{aligned}$$

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 51.5,	Allowable	: 138.0 N./mm2	Passed
Expansion	: 0.0,	Allowable	: 293.4 N./mm2	Passed
Occasional	: 18.7,	Allowable	: 183.5 N./mm2	Passed
Shear	: 21.2,	Allowable	: 96.6 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-18 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-42 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 29.53/51.00 = 0.579$$

Weld Size Calculations, Description: N3-10in

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:



Nozzle Weld Required Thickness Actual Thickness
 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (55.1578 - 3.8422 + 2 \cdot 47.0 \cdot 1.0 \cdot (1.0 \cdot 25.0 - 23.3719)) \cdot 138)$$

$$= 74362.23 \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (55.3812 + 1. - 0.0 \cdot 1.0) \cdot 138$$

$$= 79336.92 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (55.3812 + 0.0 + 1. + (23.5)) \cdot 138$$

$$= 112404.99 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (55.3812 + 0.0 + 1. + 0.0 + (23.5)) \cdot 138$$

$$= 112404.99 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 254.0 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 27510. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 122.5) \cdot (50.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 178165. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnv} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 254.0 \cdot (24.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 99710. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (27510 + 178165) = 205675 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (27510 + 0 + 99710 + 0) = 127220 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (27510 + 99710 + 0) = 127220 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 205675 Kgf, must exceed W = 74362 Kgf	or W1 = 79336 Kgf
Path 2-2 = 127219 Kgf, must exceed W = 74362 Kgf	or W2 = 112404 Kgf
Path 3-3 = 127219 Kgf, must exceed W = 74362 Kgf	or W3 = 112404 Kgf

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Nozzle Calcs.: N3-10in

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Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 31 bars

Nozzle is O.K. for the External Pressure: 0 bars

The Drop for this Nozzle is : 7.5282 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 268.1803 mm.

Input Echo, WRC297 Item 1, Description: N3-10in :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2150.000	mm.
Shell Thickness	28.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	330.000	mm.
Nozzle Thickness	50.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:*External Forces and Moments in WRC 107/537 Convention:**These loads are assumed to be SUstained loads.*

Design Internal Pressure	Dp	29.53	bars
Radial Load	P	1911.36	Kgf
Circumferential Shear	Vc	1274.24	Kgf
Longitudinal Shear	Vl	1274.24	Kgf
Circumferential Moment	Mc	892.01	Kg-m.
Longitudinal Moment	Ml	892.01	Kg-m.
Torsional Moment	Mt	1274.31	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (6.600) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 330.0 + 2 \cdot 1.65 \cdot \sqrt{(1090.5(28.0 - 3.0))}$$

$$= 874.875 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2181.000 mm.
Nozzle Outside Diameter	(d) =	330.000 mm.
Vessel Thickness used	(T) =	25.000 mm.
Nozzle Thickness used	(t) =	47.000 mm.

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T / t = 0.532
d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 1.413$$

Nr/P = 0.086
Mr/P = 0.177
M0/P = 0.052
N0/P = 0.115
MrD/Mc = 0.297
NrDL/Mc = 0.100
M0d/Mc = 0.093
N0DL/Mc = 0.055
MrD/Ml = 0.152
NrDL/Ml = 0.080
M0D/Ml = 0.050
N0DL/Ml = 0.132

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-3	-3	-3	-3
Outplane Bending (P)	-9	9	-9	9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-12	12	12	-12
Normal Pressure Stress	125	128	125	128
Outplane Stress Summary	94	141	131	127

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-31	31	-31	31
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-38	38	38	-38
Inplane Pressure Stress	62	62	62	62
Inplane Stress Summary	-13	127	70	56

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	108	142	131	127

Vessel Stresses (N./mm²):

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CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	-3	-3	-3	-3
Outplane Bending (P)	-9	9	-9	9
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-23	23	23	-23
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	62	62	62	62
Outplane Stress Summary	24	90	76	47

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-31	31	-31	31
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-75	75	75	-75
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	125	128	125	128
Inplane Stress Summary	11	229	171	86

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	24	229	171	87

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	-3	-3	-3	-3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-5	-5	5	5
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	10	7	10
Hoop Stress Summary	-1	1	9	13

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

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Nozzle Calcs.: N3-10in

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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-6	6	-6	6
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-8	8	8	-8
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-15	14	7	5
----------------------	-----	----	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	2	2	1	1
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	15	15	10	13
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	-3	-3	-3	-3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-2	-2	2	2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	10	7	10

Hoop Stress Summary	1	5	6	9
---------------------	---	---	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-6	6	-6	6
Axial Membrane (Mc)	-3	-3	3	3
Axial Bending (Mc)	-17	17	17	-17
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-24	24	17	-4
----------------------	-----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	1	1	2	2
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

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Nozzle Calcs.: N3-10in

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CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	C1 Inside	Du Outside	D1 Inside
Two * Max Shear Stress	26	24	17	14

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	C1	Du	D1
Circ. Pm (SUS)	125	128	125	128	125	128	125	128
Circ. Pl (SUS)	-8	-8	2	2	-6	-6	2	2
Circ. Q (SUS)	-21	21	3	-3	-106	106	44	-44
Long. Pm (SUS)	62	62	62	62	62	62	62	62
Long. Pl (SUS)	-5	-5	1	1	-5	-5	-1	-1
Long. Q (SUS)	-69	69	7	-7	-32	32	14	-14
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	125.0	128.0	125.0	128.0	125.0	128.0	125.0	128.0
Pm+Pl (SUS)	117.0	120.0	127.0	130.0	119.0	122.0	127.0	130.0
Pm+Pl+Q (Total)	108.1	141.3	130.1	127.1	25.3	228.0	171.0	86.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	128.00	138.00	Passed
Pm+Pl (SUS)	130.00	207.00	Passed
Pm+Pl+Q (TOTAL)	228.03	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	C1	Du	D1
Circ. Pm (SUS)	7	10	7	10	7	10	7	10
Circ. Pl (SUS)	-8	-8	2	2	-5	-5	-1	-1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-3	-3	3	3	-3	-3	3	3
Long. Q (SUS)	-14	14	2	-2	-23	23	11	-11
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	7.0	10.0	7.0	10.0	7.0	10.0	7.0	10.0
Pm+Pl (SUS)	1.0	2.0	9.0	12.0	2.0	5.0	6.0	9.0
Pm+Pl+Q (Total)	14.1	14.1	9.6	12.1	25.1	23.1	17.1	14.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	10.00	138.00	Passed
Pm+Pl (SUS)	12.00	207.00	Passed
Pm+Pl+Q (TOTAL)	25.08	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Nozzle Calcs.: M2_24in

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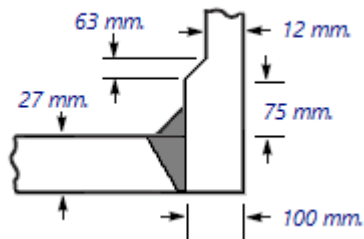
Input, Nozzle Desc: M224in**From: 50**

Pressure for Reinforcement Calculations	P	29.466	bars
Temperature for Internal Pressure	Temp	65	°C
Design External Pressure	Pext	0.15	bars
Temperature for External Pressure	Tempex	0	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	2150.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	27.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	+IT
Hub Height of Integral Nozzle	h	75.0000	mm.
Height of Beveled Transition	L'	63.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		100.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	23.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: M224in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 24.000 in.
Actual Thickness Used in Calculation 0.472 in.

Note:
Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $29.47 \cdot 2156.0 \cdot 0.8976 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 29.47)$
= 20.7072 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $29.47 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 29.47)$
= 6.4536 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.6290 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 88.0/0.5773
= 152.4205 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1183.2001	mm.
Parallel to Vessel Wall, opening length	d	591.6000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	60.0000	mm.

Weld Strength Reduction Factor [fr1]:
= min(1, Sn/Sv)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, Sn/Sv)
= min(1, 138.0/138.0)



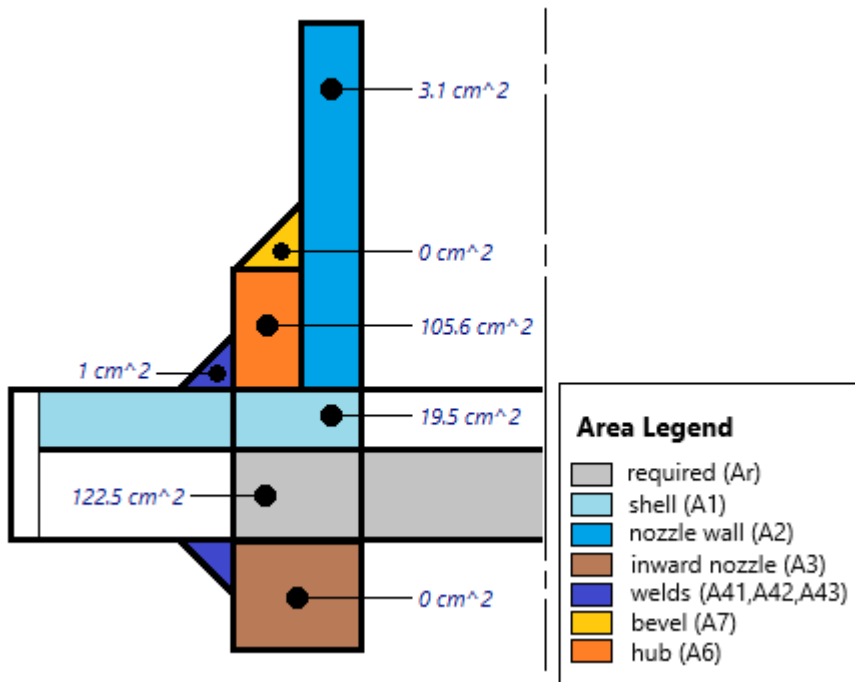
= 1.000

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)

= min(1.0, 1.0)

= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	122.504	6.429	NA
Area in Shell	A1	19.480	129.127	NA
Area in Nozzle Wall	A2	3.056	10.045	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	105.600	105.600	NA
TOTAL AREA AVAILABLE	Atot	129.136	245.772	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

= (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)

= (591.6•20.7072•1.0+2•97.0•20.7072•1.0•(1-1.0))

= 122.504 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

= d(E1•t - F•tr) - 2 • tn (E1•t - F•tr) • (1 - fr1)

= 591.6(1.0 • 24.0 - 1.0 • 20.707) - 2 • 97.0

(1.0 • 24.0 - 1.0 • 20.7072) • (1 - 1.0)

= 19.480 cm²

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Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_{r2}$$

$$= (2 \cdot 60.0) (9.0 - 6.45) 1.0$$

$$= 3.056 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot f_{r2} + (W_i - c_{an}/0.707)^2 \cdot f_{r2}$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(60.0, 250.0, 75.0)) \cdot (100.0 - 12.0) \cdot 1.0$$

$$= 105.600 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:Wall Thickness for Internal/External pressures $t_a = 9.4536 \text{ mm}$.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 9.4536 \text{ mm}.$$

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld (Impact tested) :**

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : $-41 \text{ }^\circ\text{C}$ **Flange MDMT including Temperature reduction per UCS-66.1:**MDMT of ASME B16.5/47 flange per Matl. Specification $-18 \text{ }^\circ\text{C}$
Flange MDMT with Temp reduction per UCS-66(i)(2) $-42 \text{ }^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 29.47/51.00 = 0.578$$

Weld Size Calculations, Description: M224inIntermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm}$.**Results Per UW-16.1:**

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) S_v)$$

$$= \max(0, (122.5037 - 19.4803 + 2 \cdot 97.0 \cdot 1.0 \cdot (1.0 \cdot 24.0 - 20.7072)) 138)$$

$$= 153958.50 \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$$

$$= (108.6556 + 1. - 0.0 \cdot 1.0) \cdot 138$$

$$= 154302.09 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (108.6556 + 0.0 + 1. + (46.56)) \cdot 138$$

$$= 219819.09 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (108.6556 + 0.0 + 1. + 0.0 + (46.56)) \cdot 138$$

$$= 219819.09 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 66024. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 300.3) \cdot (100.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 901397. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 609.6 \cdot (23.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 229332. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (66024 + 901397) = 967421 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (66024 + 0 + 229332 + 0) = 295356 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (66024 + 229332 + 0) = 295356 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 967420 Kgf, must exceed W = 153958 Kgf or W1 = 154302 Kgf
 Path 2-2 = 295356 Kgf, must exceed W = 153958 Kgf or W2 = 219819 Kgf
 Path 3-3 = 295356 Kgf, must exceed W = 153958 Kgf or W3 = 219819 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 30 bars

Nozzle is O.K. for the External Pressure: 0 bars

The Drop for this Nozzle is : 23.8867 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 300.8866 mm.

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: